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MARCIA CRISTIANE ROSSI

EFFECTS OF GOVERNANCE STRUCTURE ON MITIGATION OF CYBER RISKS IN SERVICES

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MARCIA CRISTIANE ROSSI

EFFECTS OF GOVERNANCE STRUCTURE ON MITIGATION OF CYBER RISKS IN SERVICES

Doctoral Thesis presented to the Graduate Program in Business Administration at Universidade Presbiteriana Mackenzie, of the requirements for the degree of Ph.D. of Business Administration.

Advisor: Prof. Dr. Gilberto Perez
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Doctoral Thesis presented to the Graduate Program in Business Administration at Universidade Presbiteriana Mackenzie, in partial fulfillment of the requirements for the degree of Ph.D. of Business Administration.

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RESUMO

A tese aborda a exploração de riscos cibernéticos em serviços, com foco na estrutura de governança e seu impacto na mitigação desses riscos. Os objetivos específicos incluem a investigação das particularidades dos serviços no ambiente cibernético, a identificação das medidas adotadas pelas empresas, a análise do papel da tecnologia nas estruturas de governança em empresas que enfrentam riscos cibernéticos e, ainda, o desenvolvimento de um framework de riscos cibernéticos em serviços. Inicialmente, conduziu-se um estudo bibliométrico para a obtenção de informações sobre o estado da arte das publicações relacionadas aos riscos cibernéticos em serviços. Com base nessa análise, os objetivos da pesquisa foram delineados. Para atingir esses objetivos, foi desenvolvido o referencial teórico. Sua principal base é a Teoria dos Custos de Transação, que investiga como as empresas lidam com a complexidade das transações para coordenar atividades, com referências relevantes em Coase (1937), Williamson (1985, 1991) e Ménard (2004). Além disso, a pesquisa concentra-se nas abordagens de Faïz Gallouj e Faridah Djellal (2011, 2018, 2023), cujas análises são pertinentes para compreender a Inovação em Serviços, especialmente em um contexto cibernético em constante evolução. A análise e discussão dos Contratos Econômicos foram baseadas principalmente no trabalho dos pesquisadores Brousseau e Glachant (2002). Adotou-se uma estratégia de pesquisa mista, composta por duas etapas distintas, utilizando uma abordagem mista (Bryman, 2021; Creswell, 2010): na primeira etapa quantitativa, questionários foram aplicados a empresas responsáveis pelo gerenciamento de riscos cibernéticos em serviços, visando a coleta de dados e a definição de categorias fundamentais. Na sequência, realizou-se uma fase qualitativa em que as entrevistas foram exploradas por meio da análise de conteúdo (Bardin, 2020), permitindo uma análise mais detalhada das práticas, políticas e desafios associados ao gerenciamento de riscos cibernéticos em serviços. Foram discutidos os resultados da pesquisa nestas duas fases. Na primeira fase (quantitativa), 175 profissionais participaram de um questionário, que foi analisado por meio de agrupamento, cuja análise descritiva permitiu desenvolver um framework de mitigação de riscos cibernéticos em serviços, posteriormente organizada em eixos temáticos. Na segunda fase da pesquisa (qualitativa), foram realizadas entrevistas com 20 participantes, e a análise qualitativa foi conduzida com base no roteiro de entrevistas desenvolvido a partir dos eixos temáticos obtidos na fase quantitativa do estudo. As categorias foram desenvolvidas *a posteriori* por meio da análise de conteúdo, seguindo as premissas de Bardin (2020). Como resultados, a análise integrada dos dados levou à construção do Modelo Teórico dos Efeitos da Estrutura de Governança nas práticas de mitigação de riscos cibernéticos em serviços. Além disso, é apresentada uma proposta de contribuição e expansão do conhecimento, destacando o ambiente cibernético como uma especificidade de ativos na Teoria dos Custos de Transação, levando à proposta da Taxonomia de Serviços Cibernéticos: Serviços Digitais e Plataformas, atualizando o modelo proposto por Bordoloi et al. (2019) e alinhando-o também com os 'Setores Núcleos Típicos' da Taxonomia Tecnológica Setorial de Serviços de Miozzo & Soete (2001), proporcionando informações sobre a caracterização dos Serviços Digitais na economia. Os modelos teóricos e fluxos apresentados nesta tese encontraram suporte e evidência empírica, sugerindo futuras investigações empíricas.

Palavras-chave: Estruturas de governança. Mitigação dos riscos cibernéticos. Especificidade dos serviços. Serviços cibernéticos. Governança contratual.

ABSTRACT

The thesis addresses the exploration of cyber risks in services, focusing on the governance structure and its impact on mitigating these risks. The objectives include investigating the particularities of services in the cyber environment, identifying the measures adopted by companies, analyzing the role of technology in governance structures in companies facing cyber risks, and developing a framework of cyber risks in services. Initially, a bibliometric study was carried out to obtain information on state-of-the-art publications related to cyber risks in services. Based on this analysis, the research objectives were outlined. To achieve these objectives, the theoretical framework was developed. Its primary basis is Transaction Cost Theory, which investigates how companies deal with the complexity of transactions in order to coordinate activities, with relevant references in Coase (1937), Williamson (1985, 1991), and Ménard (2004). In addition, the research focuses on the approaches of Faïz Gallouj and Faridah Djellal (2011, 2018, 2023), whose analyses are pertinent to understanding Innovation in Services, especially in a constantly evolving cyber context. The analysis and discussion of Economic Contracts were based mainly on the work of researchers Brousseau and Glachant (2002). A mixed research strategy was adopted, consisting of two distinct stages, using a mixed approach (Bryman, 2021; Creswell, 2010). In the first quantitative stage, questionnaires were applied to companies responsible for managing cyber risks in services in order to collect data and define fundamental categories. This was followed by a qualitative phase in which the interviews were explored using content analysis (Bardin, 2020). This allowed for a more detailed analysis of the practices, policies, and challenges associated with cyber risk management in services. The results of the research were discussed in these two phases. In the first (quantitative) phase, 175 professionals took part in a questionnaire, which was analyzed utilizing clustering, whose descriptive analysis made it possible to develop a framework of cyber risk mitigation in services, later organized into thematic axes. In the second phase of the research (qualitative), interviews were conducted with 20 participants, and the qualitative analysis was conducted based on the interview script developed from the thematic axes obtained in the quantitative phase of the study. The categories were developed *a posteriori* using content analysis, following the premises of Bardin (2020). As a result, the integrated data analysis led to the construction of the Theoretical Model of the Effects of Governance Structure on cyber risk mitigation practices in services. In addition, a proposal to contribute to and expand knowledge is presented, highlighting the cyber environment as an asset specificity in Transaction Cost Theory, leading to the proposal of the Cyber Services Taxonomy: Digital Services and Platforms, updating the model proposed by Bordoloi et al. (2019) and also aligning it with the 'Typical Core Sectors' of Miozzo & Soete's (2001) Technological Sectoral Taxonomy of Services, providing information on the characterization of Digital Services in the economy. This thesis's theoretical models and flows have found support and empirical evidence, suggesting future empirical research.

Keywords: Governance structures. Cyber risk mitigation. Services specificities. Cyber services. Contractual governance.

RESUMÉ

La thèse traite de l'exploration des cyber-risques dans les services, en mettant l'accent sur la structure de gouvernance et son impact sur l'atténuation de ces risques. Les objectifs spécifiques incluent l'étude des particularités des services dans l'environnement cybernétique, l'identification des mesures adoptées par les entreprises, l'analyse du rôle de la technologie dans les structures de gouvernance des entreprises confrontées aux cyber-risques et l'élaboration d'une structuredes cyber-risques dans les services. Dans un premier temps, une étude bibliométrique a été réalisée pour obtenir des informations sur l'état de l'art des publications relatives aux cyber-risques dans les services. Sur la base de cette analyse, les objectifs de la recherche ont été définis. Pour atteindre ces objectifs, le cadre théorique a été développé. Il repose principalement sur la théorie des coûts de transaction, qui étudie la manière dont les entreprises gèrent la complexité des transactions afin de coordonner leurs activités, avec des références pertinentes à Coase (1937), Williamson (1985, 1991) et Ménard (2004). De plus, la recherche se concentre sur les approches de Faïz Gallouj et Faridah Djellal (2011, 2018, 2023), dont les analyses sont pertinentes pour comprendre l'innovation dans les services, en particulier dans un contexte cybernétique en constante évolution. L'analyse et la discussion des contrats économiques ont été principalement basées sur les travaux des chercheurs Brousseau et Glachant (2002). Une stratégie de recherche mixte a été adoptée, consistant en deux étapes distinctes, utilisant une approche mixte (Bryman, 2021; Creswell, 2010): dans la première étape quantitative, des questionnaires ont été appliqués aux entreprises responsables de la gestion des cyber-risques dans les services, afin de collecter des données et de définir des catégories clés. Cette étape a été suivie d'une phase qualitative au cours de laquelle les entretiens ont été explorés à l'aide d'une analyse de contenu (Bardin, 2020), ce qui a permis une analyse plus détaillée des pratiques, des politiques et des défis associés à la gestion des cyberrisques dans les services. Les résultats de la recherche ont été examinés au cours de ces deux phases. Dans la première phase (quantitative), 175 professionnels ont répondu à un questionnaire, qui a été analysé au moyen d'une classification, dont l'analyse descriptive a permis d'élaborer une structure de l'atténuation des cyberrisques dans les services, organisée par la suite en axes thématiques. Dans la deuxième phase de la recherche (qualitative), des entretiens ont été menés avec 20 participants, et l'analyse qualitative a été réalisée sur la base du script d'entretien élaboré à partir des axes thématiques obtenus dans la phase quantitative de l'étude. Les catégories ont été développées *a posteriori* à l'aide de l'analyse de contenu, en suivant les prémisses de Bardin (2020). En conséquence, l'analyse intégrée des données a conduit à la construction du modèle théorique des effets de la structure de gouvernance sur les pratiques d'atténuation des cyberrisques dans les services. De plus, une proposition est faite pour contribuer et élargir les connaissances, en soulignant l'environnement cybernétique comme une spécificité des actifs dans la théorie des coûts de transaction, ce qui conduit à la proposition de la Taxonomie des services cybernétiques : services et plateformes numériques, en mettant à jour le modèle proposé par Bordoloi et al. (2019) et en l'alignant également sur les "secteurs de base typiques" de la taxonomie sectorielle technologique des services de Miozzo & Soete (2001), en fournissant des informations sur la caractérisation des services numériques dans l'économie. Les modèles théoriques et les flux présentés dans cette thèse ont trouvé un soutien et des preuves empiriques, suggérant une recherche empirique future.

Mots-clés: Structures de gouvernance. Mitigation des cyber-risques. Spécificité des services. Cyber-services. Gouvernance contractuelle.

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1 INTRODUCTION

In an increasingly digital and interconnected society, where the speed of change is becoming increasingly noticeable, the importance of companies reducing uncertainty and mitigating risks in their business has become even more pressing. The impact of digitalization on business processes has encouraged transformation markets to create or improve products and services (Barile et al., 2020). At the same time, another active front has progressively explored new business models, driven by companies seeking replacements and adopting strategies to meet their customers' needs (Porter, 1996). These strategies can involve internal development or the exploitation of external (March & Simon, 1993) since, historically, in a pendulum swing, strategy theory has varied in its focus between the external and internal aspects of companies (Hoskisson et al. 1999).

In the context of how companies structure and coordinate their operations based on considerations of cost, uncertainty, and complexity, Transaction Cost Theory was initially developed by Ronald Coase in 1937 and later expanded by Oliver Williamson, gaining prominence and recognition through the Nobel Prize in Economics awarded to Williamson in 2009 (Nobel, 2009). This theory addresses how economic transactions are carried out between parties and how the costs associated with these transactions can influence organizations' governance structure and functioning. These costs go beyond monetary values, encompassing the time, effort, and information needed to conduct business (Macher & Richman, 2008; Ménard, 2004; Williamson, 1985).

In this respect, the governance structure is an essential element within this theoretical context. It refers to the organizational arrangement and decisions about how a company manages its activities and transactions (Williamson, 1985, 1991; Zylbersztajn, 2018). The analysis of transaction costs supports the understanding of how companies decide between internal production (hierarchy), external acquisition (market), or a combination of both (hybrid mode) to acquire goods and services. In this sense, understanding the governance structures adopted and implemented in practice becomes challenging since business models and forms of transactions have evolved rapidly.

In addition, depending on the choice of governance structure, transaction costs can arise due to the lack of perfect information, asymmetry of information between the parties, the need to establish contracts and guarantees, and the search for reliable suppliers and partners (Böhme et al. 2018; Macher & Richman, 2008; Simon, 1980). The occurrence of uncertainty, opportunism, and limited rationality can have an impact on compliance with business and

regulatory objectives, including agency costs between regulatory bodies, bureaucracy, and the organizations themselves (Dixit, 2002; Frant, 1996; Miller, 1992).

Many institutions have the purpose of reducing asymmetry and informational uncertainty - these institutions include certification, reputation, and, especially, contractual guarantees, contingent contracts and various quality signals (Holmstrom, 1985). Therefore, based on the concepts of bounded rationality and opportunism, the establishment of contracts, which is the base of transaction cost theory, to regulate transactions will always tend towards complexity and incompleteness (Brousseau & Glachant, 2002; Coase, 2022 Tadelis & Williamson, 2012; Williamson, 1985).

From this perspective, operational risk management, historically focused on the organizational and financial aspects of the company, is now embracing cyber risk management as an essential component for protecting corporate assets (Fox, 2018; King, 2001; Lam, 2014; Strupczewsk, 2021). The spectrum of cyber threats has grown exponentially, becoming a challenge that crosses geographical boundaries and business sectors (Amanowicz & Kamola, 2022; ChainAnalysis, 2020). This is because, with the advancement of technology and the increasing digitalization of business operations, cyber risks have emerged as a significant sub-risk of operational risk, presenting unique challenges for organizations.

The services sector has played a crucial role in this discussion due to its growing relevance in the global economy (Rubalcaba & Solano, 2023; Stoshikj et al., 2016). Transactions in the service sector are changing, and their constantly evolving nature has contributed to an increase in complexity and uncertainty, making the business environment increasingly unpredictable (Barile et al., 2020). However, scholars express concern about the innovation gap in services research, pointing to factors such as limited investment in research and development initiatives (Gallouj & Djellal, 2011; Rajapathirana & Hui, 2018) as significant factors for this issue.

Although the economy is thriving in some sectors, such as services, there has also been an increase in digital threats in the form of cyber risks, a concern that has remained in the background of CEOs worldwide. This finding has been reinforced by studies, including a survey carried out by Marsh in partnership with Microsoft involving 600 global companies, which revealed that 61% of these companies do not have cyber risk insurance coverage, while 22% were unable to answer whether their company invests in this type of insurance (Funke, 2021).

The impacts go beyond the financial, such as the exposure of customer information and exposure to commercial or political espionage, causing organizations to need to repair their

image in the market due to a lack of confidence in their operational and cyber security processes (Norton, 2020).

Schneier (2022) states that cybersecurity has become crucial because cyberspace is crucial. Cyber risk can be considered a broad concept and used in a multidisciplinary way, with the complexity involved in its operational process being one of the most accepted reasons (Strupczewski, 2021). Although there is an investment in creating control processes, standards, internal policies, training, and technologies to support the fight against cyber risks, these mechanisms prove to be ineffective or neglected, allowing third parties to appropriate confidential information and divert financial resources (ChainAnalysis, 2023).

1.1 RESEARCH QUESTION

Considering that the problem presented involves how operations have been conducted and yet present vulnerabilities in transactions carried out in the cyber environment, this study aims to explore the existing processes in the management of cyber risks in services through the research question: How does the governance structure affect cyber risk mitigation in services?

The response to this question will allow to identify the specificities of the assets, the behavioral assumptions (bounded rationality and opportunism), and how governance has been constituted and applied in companies that must deal with cyber risks.

Furthermore, this exploration seeks greater relevance when considering the role of technology in the service economy. The aim is to investigate how the companies surveyed use and mediate technology in their transactions to boost business growth while managing risks.

1.2 RESEARCH OBJECTIVES

This research explores how the governance structure affects the mitigation of cyber risks in services from the perspective of professionals who work directly or indirectly in this area. To achieve this objective, the following specific objectives were established as goals to be achieved.

a) Explore the specificities of services transacted in the cyber environment.
b) Identify the measures companies use to deal with the challenges associated with transactions that permeate cyberspace.
c) Identify the role of technology in governance structures in companies facing cyber risks.

d) Develop a Framework of Cyber Risk in Services.

In order to achieve the objectives and answer the research question, a research strategy consisting of two distinct stages was adopted through mixed research (Bryman, 2021; Creswell, 2010; Creswell & Creswell, 2021): an initial quantitative approach in which questionnaires were applied to companies that manage cyber risks in services, to collect data and establish fundamental categories; followed by a qualitative phase that explored the interviews through content analysis (Bardin, 2020), enabling a more detailed analysis of the practices, policies and challenges associated with managing cyber risks in services.

1.3 JUSTIFICATIONS FOR ACADEMIC AND MANAGERIAL RESEARCH

Over two decades of experience in controllership and taxation, I have acquired a deep understanding of the challenges of managing operational risks in various business settings. During this period, it became clear that tax and financial management requires a meticulous analysis of the risks that permeate business operations, including the complex facets of cyber risk. This awareness intensified when I realized the centrality of the service sector, which plays a fundamental and transversal role in business activities and the economy as a whole. This professional path naturally led me to research cyber risk management, recognizing its indispensability in an increasingly digital and interconnected business environment.

Considering technological evolution and the digitization of processes, the service economy has undergone significant changes, and with this comes new challenges and risks related to security, data protection, and digital infrastructure (Norton, 2020) and the reliability of the services provided.

The service economy represents a significant portion of the global economy (Gallouj, 2002), and companies in the sector are increasingly adopting technological solutions to improve efficiency, meet customer expectations, and gain a competitive advantage (Barile et al., 2020). However, these technological initiatives also bring risks related to cybersecurity, data privacy, and digital resilience (Nordin et al., 2011; Raddats et al., 2019; Rajapathirana & Hui, 2018). It should be noted that the risk reverberating from institutions and organizations to society has made the discussion of combating fraud and cybercrime a frequent one.

Cyber threats constantly evolve, and new vulnerabilities are discovered regularly (Biener et al., 2015; Brewer, 2000; Liu et al., 2022). The theories and concepts developed by Williamson shed light on how institutions coordinate and establish governance structures to

deal with uncertainties and risks. However, since Williamson was awarded the Nobel Prize in Economics, technological advances, changes in markets and business models, and the evolution of the cyber environment, have played a significant role in the way transactions are conducted.

In addition to the specific objectives, this study seeks to go beyond the conventional limits of academic research by proposing to make contributions to both the area of Transaction Cost Theory and Service Economics. Regardless of any field of activity, the scientific output developed by a researcher must have "a social commitment and be known as useful to the academic community and society in general" (Barrancos & Duarte, 2013, p. 108). With this objective, this research explores the complexities inherent in governance structures in the face of cyber risks. It aims to reveal new perspectives that can enrich the field of Transaction Cost Theory, providing a framework of Cyber Risk in Services for understanding coordination and transaction costs in diversified economic environments since services permeate other industries.

In addition, by focusing on the Economics of Services, the research aims to explore the nature of services in activities carried out in the physical-digital environment. In this way, the research aims to contribute to the understanding of services as fundamental elements in the modern economy, providing subsidies for developing more effective and informed policies and strategies in sectors increasingly dependent on this economic dimension.

1.4 ASSUMPTIONS AND LIMITATIONS

The choice to analyze the services transacted by companies, rather than focusing on a specific type of service, is a strategic decision for the research. By addressing cyber risks across the entire service economy, the study seeks a broader scope and a more complete view of the challenges faced by companies in different sectors. This approach seeks to be relevant to a broad audience, including academics, practitioners, and policymakers.

In addition, by considering the interconnectedness of services, it is possible to identify patterns and interconnections between them, demonstrating how cyber vulnerabilities in one sector can affect other areas of the Economy.

2 THEORETICAL BACKGROUND

This theoretical framework will be a conceptual basis for analyzing cyber risk in services transacted by companies. Its main foundation is the Transaction Cost Theory, which investigates how companies deal with the complexity of transactions in order to coordinate activities, with relevant references in Coase (1937), Williamson (1985, 1991, 1996), and Ménard (2004). The analysis and discussion of Economic Contracts will be based mainly on the work of researchers Brousseau and Glachant (2002).

In addition, the research focuses on the approaches of Faïz Gallouj and Faridah Djellal (2011, 2018, 2023), whose analyses are relevant to understanding Service Innovation, especially in a constantly evolving cyber context. As these innovations are explored, it is necessary to recognize the emerging challenges related to data security, digital infrastructure robustness, and the services' reliability. Digital technology has primarily permeated business, and companies face new challenges and opportunities. One of these challenges is the growing complexity of cyber risk in transactions that arises from the interdependence of operations between companies, customers, and governments. In order to understand the dynamics underlying this phenomenon, we seek to explore and integrate theoretical concepts that shed light on decision-making mechanisms and the nature of the services transacted by companies.

2.1 THE NEW INSTITUTIONAL ECONOMICS

Understanding the organizational structures and dynamics of companies and explaining their limits has been a central objective in the search for answers that can contribute to a cohesive understanding of organizations in the economy. In this context, the New Institutional Economics (NEI) originated from the pioneering ideas of Oliver Williamson in the early 1970s. These roots go back to Coase's analysis of the firm (1937) and Hayek's work (1937 and 1945) on knowledge in the history of industrial companies, complemented by the contributions of notable thinkers such as Simon (1961), Arrow (1963), North (1990), Davis and North (1971) and Williamson himself (1971, 1975 and 1985).

The NEI represents an approach beyond the purely economic, integrating interdisciplinary contributions from fields such as Law (Brousseau & Glachant, 2002; Klein, 2002) and Strategy (Alchian & Demsetz, 1972; Porter, 1996). Its central aim is to shed light on the various elements that shape not only structure and operations but also the development and evolution of institutions. A relevant aspect of its scope is the investigation of the impacts of

these factors on the way organizations conduct their transactions and the costs associated with these activities (Williamson, 1996).

Transaction costs arise due to factors such as uncertainty, information asymmetry, and coordination difficulties, directly affecting the functioning of companies (Akerlof, 1978; Stiglitz, 2002; Williamson, 1993a). Understanding how organizations deal with these costs and make decisions amid uncertainty and how this affects organizational efficiency is essential for analyzing economic processes and interactions.

Institutions play a crucial role in determining transaction costs. For example, transparent rules and effective judicial systems can reduce the costs associated with enforcing contracts (North, 1991). On the other hand, transaction costs can shape institutions as parties seek to create contractual structures that minimize risks and uncertainties (Macher & Richman, 2008).

Institutional logic is based on the structure of rules, norms, procedures, and practices that guide the functioning of an institution (Klein, 1999; North, 1990). In the context of companies, each one is responsible for establishing and implementing specific internal rules for its operation to create value, whether through its internal processes or its interactions with other companies.

Macro-institutions strongly impact the functioning and performance of markets (Aoki, 2007). They establish the game's rules, determine the degree of competition, influence participants' confidence, and define the environment in which companies operate (North, 1990; Williamson, 1991, 1993b and 2000). Governments and regulatory bodies play a significant role in this context. The research carried out by the World Bank, known as 'Doing Business' (World Bank, 2020), explores precisely how these institutions shape the business environment and, consequently, impact the performance of markets in different countries.

In this scenario, meso-institutions play an essential intermediary role by connecting and facilitating the effective implementation of the rules established by macro-institutions. They ensure that these rules are adequately transmitted, monitored, and observed at the microlayer, where practical implementation takes place. Ultimately, meso-institutions act as essential elements in the regulatory and governance chain, promoting coordination between policies and their practical execution (Vinholis et al., 2021; Ménard, 2017; Ménard et al., 2022; Williamson, 2000;).

Therefore, institutions establish the context in which agents operate, while transaction costs influence economic coordination choices. These complementary approaches show that the economy is a complex interaction between rules, behaviors, and decisions within an institutional environment and transaction costs.

2.2 TRANSACTION COST THEORY

The Transaction Cost Theory (TCT) originated with Ronald Coase in 1937 through the publication of "The Nature of the Firm." However, the theory gained notoriety with Williamson's studies by expanding the understanding that transaction costs were fundamental in negotiations and contracts.

The central purpose of Transaction Cost Theory is to maximize results through efficiency, considering the behavior and coordination of the participants involved in transactions. In developing the theory between the early 1970s and the mid-2000s, Williamson transformed it into an influential theoretical framework[1], enabling an understanding of the particularities of organizations and the costs of coordination, monitoring, control, and management inherent in the transactions carried out.

Williamson (1979, p. 234) even ironized the idea of insignificant transaction costs, arguing that this would make economic activity irrelevant since gratuitous contracting would eliminate any advantages. In this way, the trade-off between specialization and transaction costs is recognized as the starting point for understanding the institutional environment, in which economic activity is driven by production that uses resources, labor, and capital (Farina et al., 1997).

The optimal organizational structure seeks to achieve economic efficiency by mitigating the costs of exchange but also recognizes the inevitability of the incidence of these costs, such as the costs of research, information, negotiation, decision-making, monitoring, and enforcement, which are essential to mitigate risks and achieve higher levels of efficiency in transactions (Williamson, 1979). In comparison with other approaches to the study of transaction cost economics, Williamson defined it as:

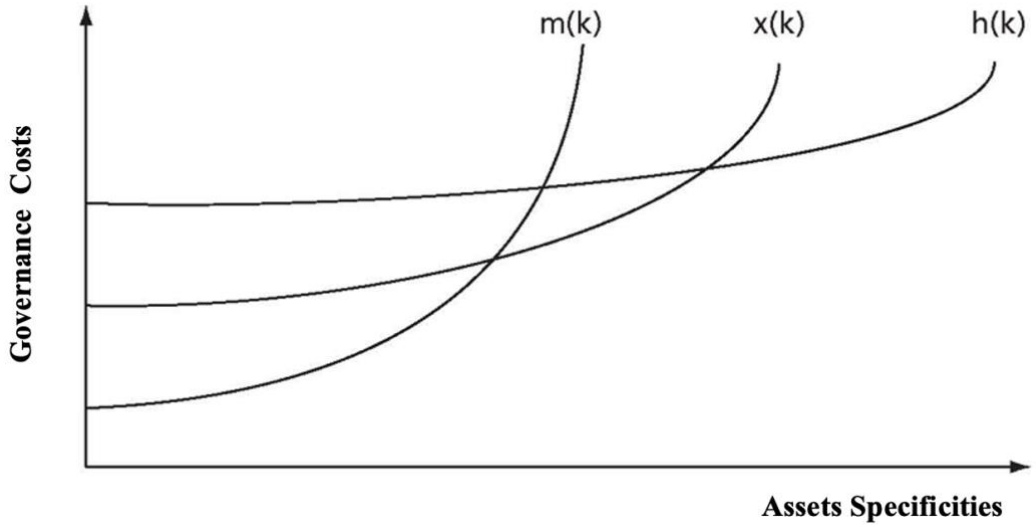
> The transaction cost economics (1) is more microanalytic, (2) is more conscious about its behavioral assumptions, (3) introduces and develops the economic importance of asset specificity, (4) relies more on comparative institutional analysis, (5) regards the business firm as a governance structure rather than a production function, and (6) place greater weight on the *ex post* institutions of contract, with special emphasis on private ordering, as compared with court ordering (Williamson, 1985, pp.17-18).

[1] Until September 2023, 19.558 scientific articles related to the topic of "transaction costs" had been identified on the Web of Science platform.

Transaction Cost Economics (TCE) is a micro-analytical approach that delves into the costs inherent in economic transactions; it focuses its attention on the details, especially the costs that emerge due to the lack of perfect information and the need for coordination between the parties involved. In addition, ECT stands out for its awareness of behavioral assumptions. Unlike theories that assume perfect rationality, ECT recognizes and considers the nuances of individuals' behaviors and decisions in economic transactions.

One of the dimensions that ECT explores is the economic importance of asset specificity. This dimension highlights that highly specialized assets, which have little value outside a given transaction, can increase the risks and costs. In addition, Williamson highlights the comparative analysis of institutions as paramount to ECT: this analysis explores the organizational structures, rules, contracts, and other institutional factors that affect economic transactions. Through these comparisons, ECT seeks to understand the implications of different institutional arrangements, as shown in Figure 1.

Figure 1 – Comparative Governance Costs



Source: Williamson, 1991, p. 83

Figure 1 represents a relationship between asset specificity and the cost associated with organizations' governance mode. Asset specificity refers to the extent to which an organization's assets are highly specialized or specific to a particular task or use. The figure shows that there are three distinct categories of governance structures. The first line (m) represents market governance, which implies transactions based on market contracts. The second line refers to the

intermediate form of governance, known as the hybrid mode (x), which combines hierarchical (h) and market (m) governance elements. Moreover, the third line (h) is governance through hierarchy, which involves internal organization with strong centralized control. The (k) represents a measure of the transaction specificity of the assets involved. The notion of coordination, in turn, is based on the synergy between the governance structure adopted and the specificities of the assets that comprise it. Thus, in principle, no governance structure is intrinsically superior to the others.

The analysis highlights that the more specific an asset becomes, i.e., the more specialized it is for a specific task, the higher the cost of monitoring that asset becomes when used in the market (Farina et al., 1997; Williamson, 1991). Highly specific assets become more challenging and expensive to control and monitor through market transactions. Therefore, when the costs of market monitoring become excessively high, it is strategically advantageous to consider changing the mode of governance. This may involve transitioning to a hybrid mode, where market monitoring costs are reduced compared to the pure market mode (Ménard, 2004; Williamson, 1991). In the hybrid mode, specific contracts can be used to create a form of quasi-vertical coordination, where the organization retains some control over the assets. However, if monitoring costs rise to the point of making the hybrid mode unviable, analysis suggests that the transaction will force the organization to adopt the hierarchical mode (Williamson, 1991). In this scenario, the organization uses its internal structure to coordinate and control the assets, centralizing decision-making, and control (Klein et al., 1978). This analysis proposed by Williamson addresses essential points about how organizations make strategic choices regarding the governance of their assets.

In addition, Williamson exposes, as a unique characteristic of ECT, the company as a governance structure, not just a production function (Williamson, 1991). The theory recognizes that companies play an essential role in coordinating and facilitating transactions, which can reduce the costs involved. In addition, the emphasis on the importance of institutional contracts is a highly relevant attribute in ECT. The theory places significant value on contracts and formal agreements that outline the rights and responsibilities of the parties involved, prioritizing specific contractual arrangements, also discussed by Brousseau & Glachant (2002).

Based on the following dimensions, Williamson (1996) postulated that these costs are influenced by behavioral assumptions (limited rationality and opportunism) and by the dimensions of the transaction (frequency, asset specificity, and uncertainty), which are combined in the governance structures (market, hierarchy, and hybrid) that make up a transaction as a unit of analysis.

Faced with governance structure options, such as internal hierarchy, the external market (Williamson, 1985), or a hybrid approach that combines elements of both (Ménard, 2004; Williamson, 1991), the organization will choose the one that best aligns with the transaction costs associated with each option. The theory emphasizes the importance of defining the organization's boundaries and deciding which activities should be carried out internally or outsourced. In addition, it is important to note that transactions are not restricted to monetary agreements; factors such as technological changes, automation, innovation, new products, services, and the exploration of growing markets also profoundly impact transactions (Mulligan & Levi, 2020).

Therefore, organizations seek to reduce transaction costs and achieve growth and profit maximization through their transactions. In order to stand out in the market, differentiating the specifics of the assets with which they transact plays a significant role.

2.2.1 Behavioral Assumptions

Although neoclassical economics defends the application of full rationality to production costs, in addition, or even in opposition to this premise, the analysis of transaction costs is based on two behavioral assumptions regarding the economic agents involved in the transactions: the first is bounded rationality, and the second is opportunism (according to Williamson, 1981 and 1985). From an economic point of view, Williamson (1985, p. 45) emphasized Coase's statement that 'modern institutional economics must study man as he acts within the limits imposed by real institutions.'

Williamson (1985) affirms that although human behavior is rational, the limits and obstacles to this rationality cannot be overlooked. In this sense, Simon (1959) argued that individuals' very cognitive and behavioral limits can interfere with how organizational processes develop. This is because organizational knowledge and knowledge management are complex and multifaceted (Alavi & Leidner, 2001).

Bounded rationality stems from the need to consider uncertainties relating to the evolution of the economic environment, while opportunism requires a strategic assessment of the possible conduct of transaction participants in the face of unforeseen events (North, 1990; Sarto & Almeida, 2015; Williamson, 1985).

Therefore, based on the concepts of limited rationality and opportunism, establishing contracts to regulate transactions will always tend toward complexity and incompleteness (Brousseau & Glachant, 2002; Tadelis & Williamson, 2012; Williamson, 1985).

2.2.2 Bounded Rationality

Simon (1959) pioneered conceptualizing bounded rationality, allowing other concepts to be subsequently developed and explored in behavioral economics (Kahneman, 2003, 2012). By interpreting bounded rationality as a cognitive assumption on which transaction cost economics is dependent, Williamson (1985, p. 45) rejected the idea that "economists who oppose bounded rationality misinterpret this opposition as irrationality."

Simon (1990) argued that individuals and companies have limitations when dealing with complex problems. When this limitation affects managers, it hinders more assertive decision-making since managing the organization's problems and strategies rationally is challenging. Simon (1990) supports Smith's (1776) understanding, that people hardly behave like utility maximizers; they do not maximize their objectives, and there are a variety of human desires without a utility function. This is because the limited rationality of agents in itself does not pose any obstacles, but when added to scenarios of uncertainty and complexity, it negatively impacts the circumstances of the transaction to be carried out and makes it difficult to make contracts (Williamson, 1981).

Bounded rationality refers to a disparity between the information possessed by the parties involved in a contract. This can happen when one of the parties possesses information that the other party does not, creating information asymmetry, and this can occur in internal and external contracts, although the dynamics can vary (Barzel, 1982; Dequech, 2001; Stiglitz, 2002; Williamson, 1985).

In addition to rational limitation, there are situations in which actors overestimate their own knowledge and the resources available due to overconfidence or optimism (Gervais & Odean, 2001; Jordão, 2018; Rubisteins, 2001). This position can also be applied to the probability of errors not uncommon in managerial decision-making (Kaplan & Mikes, 2012; Teece, 2009), whose deliberations can underestimate the risks inherent in the organization's operations and systems.

Human action will constantly be exposed to mistakes, whether intentional or not. Feldman and Pentland (2003) contribute to this position by postulating that repetitive patterns of action tend to emerge as members of the organization opt for easier actions and avoid more complex ones. So much so that the conceptualization attributed by Dequech (2001, p. 913) can reinforce the understanding that "bounded rationality challenges economic transactions when complexity and uncertainty are present," whether in workflows or decision-making. In addition, Nelson and Winter (2002, p. 27) revealed that the evolution of "company behavior is an

essential element concerning market results and long-term survival." However, there are limitations to the rationality of individual agents, in which it is unlikely that all alternatives or solutions are known by the agents (March & Simon, 2005).

Therefore, the behavioral assumptions literature highlights the importance of recognizing human rationality's limitations and human behavior's influence on economic transactions and organizational management. Ignoring these behavioral nuances can result in simplistic approaches that fail to deal with the multifaceted reality of business and economic interactions (Williamson, 1993a). Furthermore, considering opportunism adds another layer of complexity to the this equation.

2.2.3 Opportunism

Ignoring or underestimating the role of opportunism can have significant implications for understanding economic systems and formulating effective strategies and policies (Williamson, 1993a). According to the behavioral assumption of transaction cost analysis, the opportunism can be understood as a calculated and premeditated effort that seeks the incomplete or distorted distribution of information that leads to error. Opportunism is a less technical term compared to "adverse selection" and "moral hazard" (Williamson, 1993a, p. 100).

Simon (1985, p. 303) presented two views of human nature concerning economics: 'homo economicus,' in line with the classical view that considers human beings to be perfectly rational agents capable of processing all available information quickly and accurately, making choices that maximize their economic interests. Moreover, Simon's second view would be that of 'homo psychologicus,' in which human beings are not as perfectly rational as 'homo economicus' suggests. Instead, economic decisions are influenced by psychological, cognitive, and emotional factors - people have limitations in their ability to process information, deal with complexity, and make consistent decisions in all circumstances. This can result in choices that do not necessarily maximize long-term economic interest (Ferrreira, 2023).

Years after developing this same understanding, Williamson expanded the concept of opportunism in several works, admitting that, in relationships between economic agents, not everyone acts and negotiates with integrity. He recognized that there could be situations in which “economic agents can lie, steal, and cheat flagrantly” (Williamson, 1985, p. 47), acting out of “self-interest” and with intentional malice (Williamson, 1988, p. 88). He also pointed out that the aim is not to describe human agents in a favorable light but rather to understand complex economic organization (Williamson, 1993, p. 99). In this context, Pauleen et al. (2017, p. 26)

provided a model of failure in an economic organization, exemplifying an "arrogant and unethical corporate culture in which employees manipulated accounting and earnings to generate bonuses for senior executives."

Opportunistic behavior, according to Santos et al. (2019, p.115), "is not predictable, and does not even assume conventional patterns found in the organizational environment" and can occur *ex ante* or *ex post* to the drafting of contracts. In this sense, Williamson (2000, p. 611) introduced the idea of a deeper level of interaction, which he called the 'game of the game'. At this stage, the dynamics go beyond the boundaries of the basic rules and embrace the complexity of the governance of contracts and relationships. In other words, the author pointed out that many transactions are not limited to the direct exchange of goods and services but also include the intricate management of contractual agreements and the maintenance of long-lasting relationships between the parties involved.

Furthermore, ECT recognizes that trust and safeguards are equally relevant to efficient transactions. In this context, the definition of trust converges on the idea that it is the subjective probability that an agent or group will carry out a specific action, in which trust implies cooperation and the expectation of mutual benefit (Williamson, 1993a). When there are no contractual safeguards in place, the parties may choose to invest in a less specific or less specialized way to avoid possible future problems or costs (Langlois & Foss, 1997).

Additionally, ignoring or suppressing opportunism in transactions can result in a loss of understanding of many problems of economic organization (Williamson, 1993a). In this sense, opportunism is related to contractual incompleteness caused by bounded rationality and suppressing it can lead to simpler contracts. However, the ability to deal with complex situations could not be lost.

2.2.4 Asset Specificities

Asset specificity has been widely discussed as a dominant theory for explaining decisions related to organizational boundaries (Geyskens et al., 2006). Asset specificity implies that assets are considered specific because they cannot be easily reused or exchanged without incurring a loss of value. This distinction of specificity has led to the categorization of assets into six distinct types (Williamson, 1991, pp. 281-282):

(i) Local. These assets are physically close to each other, such as side-by-side workstations. This saves inventory and transportation costs, as physical proximity reduces the need to move resources and materials, resulting in lower transaction costs.

(ii) Physical. Indicates that a particular component or resource is essential for producing a product or service. This specificity refers to the need to possess a specific physical asset to perform a task or process, which may imply additional transaction costs if this asset is not available or needs to be acquired.

(iii) Human. This relates to the knowledge, skills, and experience of the people or teams who play a role in certain activities. This type of specificity manifests itself in 'learning-by-doing' (Schank et al., 2013).

(iv) Brand. This refers to the value and reputation associated with a brand. The specificity of this asset lies in the trust and recognition that customers have concerning the brand.

(v) Dedicated. Involves the production of customized or specific products or services for a client or order. The specificity of this asset is related to the need to adapt processes and resources to meet the specific requirements of a client.

(vi) Temporal. These are assets that are sensitive to time and timely response. This is related to technological non-separability, where the execution of specific tasks or activities requires precise coordination in time.

Williamson also points out that "asset specificity, especially in its first five forms, creates bilateral dependence and presents additional contracting risks" (Williamson, 1991, p. 282). In addition, they are linked to the uniqueness or lack of substitutes for certain asset. Furthermore, increasing specificity generally results in more significant adaptation challenges, translating into higher transaction costs, as Williamson (1985) and Farina et al. (1997) observed. This means that as the specificity of an asset increases, so do the associated risks. In the context, the relationship between assets, technology, and organization, it is important to highlight Williamson's (1975) observation:

> While the relation of technology to organization remains important, it is scarcely determinative […] Rather I contend that transactional considerations, not technology, are typically decisive in determining which mode of organization will obtain in what circumstances and why. Central to the analysis is what I refer to as the "organizational failures framework." (Williamson, 1975, p. 2)

Williamson is arguing that, when considering how an organization should be structured in a specific context, it is more important to take into account transactional factors such as

transaction costs and the structure of organizational failures, rather than simply relying on technology.

Incidentally, Williamson (1996) did not address including innovative practices in commercial relations, as Grassi (2002) observed. However, Williamson (2000, p. 599) acknowledged that "technical and organizational innovations have been overlooked and ways must be sought to operationalize them in a combined manner." This suggests the need to take advantage of technological and organizational innovations to achieve results or progress.

2.2.5 Uncertainty

Ambiguity is present when distinguishing between uncertainty and risk (Williamson, 1985, 1991). However, the work entitled "Risk, Uncertainty and Profit" is a seminal work by Knight (1921), whose concepts distinguish risk as a measurable probability, while uncertainty is characterized by immeasurability.

Knight's uncertainty points to three categories of probabilities to distinguish phenomena, both of which are measurable: the first two are "a priori probability" and "statistical probability", inserted in the "logical space of mathematical propositions" (Knight, 1921, p. 224). While the third is called "estimation", whose result is not measurable, in which the author explains that it is a judgment by "induction from experience" (Knight, 1921, p. 225).

In 1937, Keynes made a significant contribution to the issue of uncertainty by postulating that future knowledge does not exist but is uncertain. The author argues that certain events, such as the outcome of a roulette game, life expectancy, and the weather, can be considered only moderately uncertain. However, the author is more interested in the deep and intractable uncertainty related to events such as "the prospect of a European war, the price of copper and the rate of interest twenty years hence, as well as the obsolescence of a new invention, or the position of private wealth" (Keynes, 1937, pp.113-114).

In light of Knight and Keynes' conceptualizations of "uncertainty," which permeates the competitive market and any operations carried out by organizations, it is worth highlighting information design, which has emerged as a mechanism for detecting weak signals to anticipate risks. So much so that since the 1970s, research has specialized in the relationship between the graphic representation of information and its interpretation (Sidhom & Lambert, 2011). Ansoff (1975) argued that although strategic planning is a successful program, there is still a dependence on input information that will enable decision-making to be carried out in a way that allows uncertain events to be predicted.

Under Williamson's analysis, uncertainty would emerge on two significant fronts: from acquiring and accessing information that demonstrates what skills are needed from potential partners and from ensuring that this information is reliable. That said, Williamson portrays uncertainty as a "disturbance" that permeates any transaction:

> Transactions conducted under certainty are relatively uninteresting. Except as they differ in the time required to reach an equilibrium-exchange configuration, any governance structure will do. More relevant are transactions where uncertainty is present to an intermediate or high degree. […] The question here is how the governance of transactions is affected by increasing the degree of uncertainty (Williamson, 1979, pp. 253-254)

Although Williamson (1985) considers asset specificity to be the primary dimension that distinguishes transactions, the presence of uncertainty is relevant to the choice of an efficient governance structure. Moreover, bounded rationality tends to become more critical as uncertainty sets in.

The elements that characterize an organization, together with the changes that occur in its environment, are seen as sources of uncertainty and are more pronounced in dynamic and complex environments. Silva & Brito (2013) brought a relevant perspective to the topic by analyzing that "uncertainty" within the field of organizational theory has been interpreted as a measure of the unpredictability of changes and the level of difference between the elements that make it up.

A high-performance economy is one that is able to adapt efficiently in uncertain environments. Anticipation plays a central role in this adaptation, making it possible to foresee future changes. Governance structures that convert inputs into outputs also deal with incomplete information. In short, the ability to adapt, anticipate, and manage imperfect knowledge are critical characteristics of high-performance economies (Hayek, 1945).

Considering the positions of Knight, Keynes, and Williamson, it is possible to identify a point shared by the authors: the concept of uncertainty is directly linked to vulnerability or lack of knowledge and is constantly confronted with the idea of knowledge about measurable risk. Therefore, this study will admit the existence of two conditions: measurability (risks) and immeasurability (uncertainty), as well as the frequency with which transactions occur, which are essential elements for analyzing transaction costs, asset specificity, and uncertainty.

2.2.6 Frequency

Of the three attributes mentioned by Williamson, frequency is less discussed than asset specificity and uncertainty. The author emphasizes that frequency refers to the regularity or frequency of a buyer's transactions in the market, which can be classified as "unique, occasional, or recurrent" (Williamson, 1979, pp. 246-247). However, the author does not offer a more in-depth discussion of this attribute.

In fact, the author points out that there will rarely be an interaction between organizations in situations that occur on a one-off or occasional basis. In any of these scenarios, the complexity of the contractual relationship may be obvious. Although the frequency is an important attribute for choosing the appropriate structure for this transaction, Farina et al. (Farina et al., 1997, p. 88) highlight it in two ways, stating that "complex transaction mechanisms can bring benefits," such as the equitable distribution of costs and reputation building, especially concerning the dimensions of asset specificity and uncertainty.

According to Williamson (1985), the frequency attribute refers to the number of times economic agents meet to carry out a transaction, considering that the greater the frequency, the greater the likelihood of the partners developing a reputation, limiting opportunistic behavior (Augusto et al., 2013).

Although the literature does not deal extensively with the subject, it is crucial to highlight frequency as a significant element in cyber threats and attacks, even in contractual contexts, and the restrictions associated with exchange activity in the market. Indicators show a real increase in cyber-attacks, with human entities being the most frequently targeted by malicious cyber actors (Ani et al., 2019; Instituto Brasileiro de Segurança, Proteção e Privacidade de Dados [IBRASPD], 2023).

2.3 GOVERNANCE STRUCTURE

Governance structures should be distinct from corporate governance. The former is grounded in the literature of transaction cost economics, drawing mainly on organizational, economic literature, and contractual theory. It focuses on selecting the most efficient form of governance for a specific economic transaction (Williamson, 1975 and 1991). On the other hand, Corporate Governance involves Agency Theory; it focuses predominantly on the exchange relationship between corporate owners and the individual employee, usually

represented by the CEO, who manages the business on behalf of the owners (Jensen & Meckling, 1976).

However, Williamson (1996) dedicated himself to examining Corporate Finance through the lens of Transaction Cost Theory under the aspects of Agency Theory. The author noted the similarities in both theories because "they operate from a configuration of managerial discretion, adopting an efficient contracting orientation for the economic organization." For example, both theories deal with a board of directors and start from an endogenous format (Williamson, 1996, p. 193), i.e., internal cost and expense policies.

Governance formats from the perspectives of hierarchical governance (internalization), hybrid governance, and market governance (outsourcing) would be reconciled with the other aspects conditioned to the costs of this governance (Williamson, 1991), precisely to identify what would stimulate agents to seek better economic achievement. Commons (1932, p. 4) postulated that a transaction unit must incorporate three essential principles: "conflict, mutuality and order." He argued that while the transaction is the fundamental element of the unit, governance plays a key role in creating order, mitigating conflict, and seeking mutual benefits. As Williamson (2000, p. 599) pointed out, 'transaction is the foundation of unity, but governance is a crucial element in establishing order, resolving conflict and achieving shared gains.'

So much so that Williamson (1996, p. 12) argued that "the feasibility and effectiveness with which governance structures serve to mitigate risk are directly related to bounded rationality and opportunism." This statement reinforces the understanding that the effectiveness of governance structures in dealing with risks is linked to the capacity of these structures to accommodate the limited rationality of the people involved and to avoid opportunism.

Furthermore, defining the boundaries of organizations has been a significant challenge, considering the exponential growth and indistinct frontiers of technology (Hitt et al., 2015). The importance of a company's positioning in the market is a constant concern for its leaders, investors, and employees. This reinforces the idea that governance structures in an organization or economic system are not all the same.

Understanding the governance structures applied in practice, including the contractual elements, can be a considerable challenge. This becomes even more relevant due to the evolution of business models and how transactions are carried out, especially in a scenario where activities demand customized information and knowledge.

2.3.1 Contractual Governance

Contracts cannot foresee all possible situations that may arise over time, making them incomplete by nature (Hart & Moore, 1988; Williamson, 1985). This intrinsic limitation of contracts has led to countless disputes and litigation, where the parties involved can find themselves caught in a complex web of unspecified interpretations and intentions. However, incomplete contracts are a rational choice, as they offer a more efficient way of structuring commercial agreements, minimizing excessive bureaucracy, reducing search and negotiation costs, and avoiding the significant 'inking costs' that would occur if all details were specified in complete contracts (Klein, 2002).

The flow of transaction costs in contracts covers two aligned moments: the first, *ex-ante*, during the negotiation and formalization of contracts, and the second, *ex-post*, seeking to adjust the rules through contractual additives in new circumstances, including costs of monitoring the established rules (Galanter, 1981; Grossman & Hart, 1986).

The gap between what a contract covers and the actual circumstances that may unfold is a constant source of challenge for both individuals and companies. Even with comprehensive clauses and meticulously crafted legal language, it is impossible to foresee all the nuances and variations that real life may present (Brousseau & Glachant, 2002). This becomes even more evident in constantly evolving sectors, such as online services, where new scenarios emerge almost daily.

In the context of approaching and analyzing contracts, Brousseau and Glachant (2002) presented three fundamental theoretical frameworks that support the exploration of how contracts work and the decisions made by the parties involved when faced with the need for agreements in line with Transaction Cost Theory:

a) **Incentives**: The core of the investigation is influenced by rewards and penalties in the parties' behavior. ECT emphasizes the need to "organize a structure of incentives that will lead the actors in a coordinated manner towards their objectives" in contractual relationships of an incomplete nature (Siffert Filho, 1995, p. 104).
b) **Incomplete Contracts**: In contracts, it is not possible to foresee all the eventualities that may arise during their execution, i.e., they may not cover all possible conditions or situations. The parties face these gaps and make decisions when encountering circumstances not foreseen in the contracts.
c) **Transaction Costs**: The analysis includes the costs associated with drafting, negotiating, and executing contracts. This is also intrinsically related to the choice

> between conducting transactions internally within an organization or resorting to the external market. This interaction of the organization with other market agents results in the formulation of contracts and makes up the so-called 'contractual nexus' (Williamson, 1996, p. 98). This approach reinforces the concept of the firm and the importance of contractual agreements for cooperation and efficient coordination in the business environment.

These theoretical approaches not only broaden the understanding of the dynamics of contracts but also have ramifications in wider spheres, covering both microeconomics, with its individual decisions and interactions, and macroeconomics, addressing the overall functioning of the economy and the institutional setting.

2.3.2 Market Governance (Outsourcing)

In the face of uncertainty, Williamson (1985 and 1996) understood that market governance is the preferred supply method due to internal organizations' incentives and bureaucratic deficiencies, especially concerning controlling production costs. This is why this structure is recognized for its high-powered incentives, resulting in reduced costs and an operation focused on efficiency. Although "market arrangements offer the most incentives, the possibility of control will depend on the existence of substitute partners to discipline deviations from the agreement" (Mizumoto & Zylbersztajn, 2006, p. 150) - which may lead some companies to adopt internalization so that activities are coordinated, even if the incentives are reduced (Augusto et al., 2013).

In the market mode, high asset specificity promotes a bilateral monopoly situation, commitment, and dependence between supplier and customer, even in a scenario of uncertainty and incomplete contracts (Williamson, 1979). Contracts are crucial elements, especially for organizations that hold market power, as imperfect competition gives them a greater concentration of privileged information, making the other party vulnerable to opportunistic conditions (Williamson, 1975).

It is worth noting that when the product or service involved in the exchange is accessible and traded abundantly and easily on the market, and there is still a need to adapt, this will be done unilaterally, and the contract to be used is the classic one (Quevedo, 2016; Williamson, 1991). In classical contract law, "classical contract law applies to the ideal transaction in law and economics, in a clear and clear agreement for a clear performance" (Williamson, 1991, p. 271).

Ménard (2004) reinforces Williamson's understanding (1996, p. 14) by stating that legal contracts are one of the parties' most important safeguards in market economies. The importance of these contracts is based on five causes: (i) bilateral dependence, (ii) measurement problems, (iii) changes in terms over time, (iv) inability to respond in real-time, prolonged latency and abuse of strategies, and (v) risks that can lead to weaknesses in the institutional environment.

Specifically, the market mode, through outsourcing, is considered superior for dealing with transactions of a generic nature in which autonomous adaptations are sufficient. This is illustrated by the concept of the "wonder of the market" described by Hayek in 1945. On the other hand, transactions in which the parties are dependent on each other benefit from the coordinated adaptation provided by hierarchy, is called the "wonder of hierarchy," presented by Barnard in 1938.

2.3.3 Hierarchical Governance (Internalization)

The hierarchical governance structure is adequate for transacting specific assets, considering that this structure offers broad administrative control (Williamson, 1975; Zylbersztajn, 1995). The presence of incentives is scarce, although it can be inferred that target contracts and bonuses for performance can be considered a form of incentive to comply with the internal contract.

Due to the specificity of the assets, the high transaction costs demand an organization with a hierarchical structure, in which the market is small for this type of structure. It also adds that competitiveness between players is incapable of inhibiting opportunism on the part of the players. In this respect, the hierarchical governance structure is expanded within limits set by the organizational environment and develops through the behavioral assumptions of internal agents (Williamson, 1991).

The internal, or hierarchical, organization is qualified as an even more elastic and adaptive mode of organization, characterized by "tolerance." The rationale for the law of tolerance is based on two logical lines: (i) the parties to an internal dispute are aware of the circumstances surrounding a conflict and would demand costs if taken to court, and (ii) taking internal disputes to court would reveal a scenario of fragility in the effectiveness and integrity of the hierarchy. In other words, the hierarchy is characterized by its "greater reliance on instrumental reasoning and the result of mutual accommodation" (Williamson, 1991, p. 274 and 1996, p. 97).

Considering that the market model uses the classic contractual instrument, hierarchical governance resembles this modality due to the difference that "the hierarchy is its own court" (Williamson, 1991, p. 274) when it manages disputes and conflicts, while at the same time adjusting them to reduce differences internally, through the limits and zones of acceptance by the organization's tolerance. However, a perspective presented by Granovetter (1985) goes against the presumption of effectiveness of the institutional view, as proposed by Williamson, regarding the lower transaction costs in hierarchies. Granovetter challenges the idea that hierarchies always result in lower transaction costs, suggesting that this view overestimates hierarchical power, known as *fiat*[2] power. In a way, Williamson's approach leads us to infer an emphasis on the importance of internal organization through the more active participation of employees seeking to reduce costs linked to negotiation, administrative activities, inefficiencies in operational processes, and monitoring linked to contractual compliance.

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2.3.4 Hybrid Governance

Considering that, as Williamson (1996, p. 104) stated, "markets and hierarchies are polar modes," hybrid governance represents a combination of market and hierarchical structures. The grouping of market incentives characterizes it, focused on reducing transaction costs, together with the use of administrative controls as coordination mechanisms in the hierarchical structure (Williamson, 1991).

Williamson (1991 and 1996) points out that in the hybrid mode, the parties can give up some individual incentives in favor of more efficient and effective coordination of activities or transactions. This suggests that, in certain situations, the ability to coordinate and control interactions may be more valuable than maintaining high individual incentives, especially when achieving specific objectives more efficiently. Ménard (2004) states that adopting a hybrid governance structure reduces transaction costs and allows for non-hierarchical interdependence while avoiding the risks of opportunism. This differs from a purely market-based structure, where such risks are more likely to occur.

[2] *Fiat* is a Latin term that means "let it be done". Williamson (1991) used this term to describe situations that require conflict resolution through the intervention of a higher authority that can interfere with the parties involved.

Although there is a quest to preserve market incentives, hybrid governance is based on complex contracts that serve as coordination mechanisms. External incentives can arise through partnerships, open innovation, and public policies cooperating with regulatory bodies. This justifies the analysis of hybrid contracting forms, including contracts, joint ventures, subcontracting, networks, franchises, partnerships, cooperatives, and alliances (Ménard, 2004; Silva, 2020; Williamson, 1991; Zylbersztajn, 1995). Williamson (1996, p. 314) states that the predominant form of organization is based on partnerships in the service industry, such as law, accounting, investment banking, and management consulting. The logic of transaction costs explains why these industries opt for partnership-based organizational structures rather than other forms of organization. This suggests that transaction cost analysis helps understand why these industries choose this organization model.

In order to better detail how hybrid governance is adapted, it will require a complete approach given the evolution that contracts undergo over time (Grassi, 2002). Also noteworthy are the interactions characterized by competition and cooperation practiced between organizations, which are subject to analysis, considering the new business models emerging in the service sector.

Ménard (2004) characterized hybrid governance by its variety of forms, including trust, networks of relationships, leadership, and governance. Based on the graphic representation originally presented by Williamson (Fig. 1), Ménard (2004) presented the diversity of hybrid arrangements, which are promoted through contracts and agreements concluded close to the market structure. In addition, Ménard (2004) warned of the particularly challenging risks for hybrid organizations, where reducing them through contracts requires selecting partners and defining clauses that can efficiently restrict opportunistic behavior. In large markets with a diversity of offers and organizations, administering effective rules proves to be a challenge for the regulatory body, which will need to act so that market participants can transact (Ménard, 2004, p. 22).

However, in situations when asset specificity is relatively high, hierarchies (internal management) tend to prevail over hybrid forms. This is because managing conflicts and implementing adaptations through direct management decisions prove to be effective and less costly than resorting to third parties or getting involved in legal disputes (Williamson, 1996). Concerning the adoption of outsourcing services, whether in the market structure or in the hybrid governance structure, it will be related to low to medium specificity assets. High specificity assets involve high transaction costs, making hierarchical governance more

advantageous, although transaction costs are not eliminated, and interactions between individuals are not immune to opportunism (Williamson, 1991).

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2.4 CYBER RISKS

In order to identify the state of the art of scientific production that addresses cyber risks in services, a bibliometric study was carried out study was carried out by the author, that explored the progress of scientific production, identifying relevant researchers and institutions on the subject, measuring impact and identifying trends. The result of this bibliometric study, initially carried out in 2021 and updated for publication in 2023 (Rossi & Perez, 2023).

Cyber risk is a relatively recent scientific debate, while its plurality and complexity have marked a colossal change in the cybersecurity and cyber threat scenarios. Under a broad, multidisciplinary concept, cyber risk encompasses areas such as computer science, engineering, business management, economics, and social sciences (Strupczewski, 2021).

Risk is the effect that uncertainty has on an organization's objectives. "It is the possibility of the occurrence of events that affect the achievement or attainment of objectives, combined with the impact and occurrence on the intended results" (Tribunal de Contas da União [TCU], 2018, p. 8). In addition, risk is related to the results of the chosen strategies, in which companies have been operating in an environment known for imperfect information, information asymmetry, negative externalities, as well as common risk factors in operationalization and non-predictability (Riek & Böhme, 2019). However, authors such as Knight (1921) and Mousavi and Gigerenzer (2014) argue that although risk and uncertainty seem synonymous, each has its own heuristics.

Risks can manifest in two distinct ways: first, in the static or pure form, which is characterized by risk involving only the possibility of loss; second, in the speculative or

dynamic form, which implies the possibility of loss for one of the parties while the other obtains some gain (Durak, 2020; Powers, 2006). It is important to note that uncertainty, on the other hand, arises due to complexity and requires a regulatory approach based on simplicity (Mousavi & Gigerenzer, 2014, p. 5)

In this sense, the quest for competitiveness has driven constant changes in organizations' processes, intending to improve their positioning in the market (Hermundsdottir et al. 2021), even in the face of risks and uncertainties. The nature of the risks is determined by the threats that organizations face, whether intentional or not. In this context, companies adopt business strategies and strategic decisions to mitigate these risks (Egan et al., 2019).

Amit and Schoemaker (1993) highlight the importance of managers anticipating the future, assessing competitive dynamics, and overcoming organizational inertia through strategic decisions. Changes in environments characterized by growing concern about sustainability, social responsibility, and corporate citizenship are accelerating, but they also bring uncertainties (Hays, 2008) and multifaceted risks.

In this context, financial institutions have been pioneers in risk management since the 1980s; this was driven by a series of occurrences that resulted in substantial losses, leading them to focus their efforts on controlling and reducing the variability of operations through financial risk management (Rossi, 2017). However, it is essential to note that risk is not exclusively related to the organization's financial results but also to operational risk, which is intrinsically linked to how the company operates, makes decisions, and manages its internal control processes. Operational risks stem from numerous operational failures and have grown exponentially with the advent of global connectivity and its expansion through value chains (Fox, 2018; King, 2001; Lam, 2014). In addition, operational risk does not fit easily into classic risk management approaches since most cases of loss associated with it involve complex and non-linear interactions between risks and business processes (Van Grinsven & Grinsven, 2009).

Considering these complex interactions and challenges of operational risk, it is also important to consider its specific sub-risk, cyber risk, which can compromise the confidentiality, availability, and integrity of the information or systems of organizations (Cebula & Young, 2010).

Organizations worldwide have been challenged with confronting situations and demanding knowledge about cyber security. Cyber risk has been cited as one of the most important sources of operational risk in organizations. Consequently, public awareness of events related to this type of risk has steadily increased (Egan et al., 2019).

Cyber risk cannot be limited to the internet, hardware, software, and information systems (Silva & Nogueira, 2019); integrating people and their social interactions in computer networks must be considered (Klimburg, 2012). As it gains attention in the market due to the rapid digitalization of the economy and social relations, cyber risk is amplified due to the combination of probabilities that exist within an organization, including "assets, physical resources and communication between them, which trigger and impact an organization's processes" (Bank of International Settlements [BIS], 2016, p. 24).

High competitiveness in the market has forced changes in organizations to improve their positioning in the face of innovation and emerging technologies. To this end, researchers have explored, analyzed, and conceptualized cyber risks (Figure 2).

Figure 2 – Cyber Risk Research *Locus*

Cyber Risk Concepts	Locus	Source
By multiplying the threat by the asset value, it obtained a measure reflecting the vulnerability level. The greater the result of this multiplication, the greater the perceived vulnerability, indicating a greater risk that threats may cause significant damage to the organization's assets.	Measurement, threat and asset value	Brewer (2000)
The negative impact on the organization's operation is influenced by non-formal factors, such as its mission, image, and reputation. In this context, resources and intellectual capital are the primary assets used in information systems.	Potential, organizational operations and assets	National Institute of Standards and Technology (2006)
This subtype of operational risk applies to information and technology-related assets and directly impacts the confidentiality, availability, and integrity of information and systems.	subrisk, operational risk, impacts	Cebula and Young (2010)
It is a combination of the probability of an event in the area of network information systems and the effects of this event on the assets and reputation of an organization.	combination of probability, event, assets	World Economic Forum (2012)
Cyber damage to the organization occurs when potential threats successfully exploit vulnerabilities present in the organization's information assets.	threat, vulnerabilities and assets	ISO/IEC 27000 (2014)
Any risk of financial loss, interruption, and damage to an organization's image or reputation caused by information and systems failure.	financial loss, reputation and informational failure	The Institute of Risk Management (2014)
Cyber risk is based on three parameters: impact, which represents the potential for damage; the threat, which indicates the probability of occurrence; and vulnerability, which evaluates the effectiveness of security measures.	level of damage, expressed threat and vulnerability	Biener et al. (2015)
Risk related to a malicious electronic event that could cause business interruption and financial loss. It covers all other risks associated with online activity, such as storing personal data and transactions that can result in image damage, financial loss, and adversities in life and business.	business discontinuity, malicious event, adversities in life and business	National Association of Insurance Commissioners (2018)

To be continued...

Continuation of Figure 2

Cyber Risk Concepts	Locus	Source
Two aspects can highlight it: (i) Technical: process flow, reprogrammable behavior, and global dynamic threat and (ii) Economic: information asymmetry, externalities, and common risk factors in operationalization.	technical and economic aspects.	Böhme et al. (2018)
Risk depends on whether the organization faces malicious or non-malicious threats and how organizations mitigate risks through business and strategic decisions.	threat, business decisions and strategies	Egan et al (2019)
Cyber risk is based on the risk of financial loss, interruption, or damage to reputation due to the failure of information technology systems.	financial loss, reputational risk, technology failures	Northbridge Insurance (2019)
Cyber risks are linked to company security: remote work, teleworking, access to strategic and sensitive information, and careless use of equipment such as notebooks and smartphones.	remote work, resources synchronized with company systems and carelessness	Strupczewski (2021)
Cybersecurity has become crucial because cyberspace is crucial. Cyber risk can be considered a broad concept and used in a multidisciplinary way.	cyberspace, multidisciplinary, operational processes	Schneier (2022)

Based on a synthesis of Figure 2, the locus of cyber risk research covers a wide range of factors, including threats, vulnerabilities, and organizations' assets in the cyber environment (ISO/IEC, 2014; NIST, 2006; World Economic Forum, 2012).

Cyber risk is characterized as potential damage caused by threats exploiting vulnerabilities in information assets. This definition highlights the importance of understanding the interaction between threats, vulnerabilities, and assets in order to effectively protect organizations. The CSIS Center for Strategic & International Studies (CSIS)[3] is known for keeping a comprehensive record of significant cyber incidents since 2006. This detailed history focuses mainly on cyber-attacks targeting government agencies, defense contractors, and high-tech companies, as well as economic crimes that resulted in losses exceeding one million dollars. This initiative is a valuable source of information for understanding the growing trends and challenges in the cybersecurity landscape.

Both public and private organizations have faced attacks from sophisticated groups using various types of malware designed to harm or exploit programmable devices, services, and networks, posing a threat to systems and individuals (McAfee, 2021). These attacks have had a significant impact on the market.

[3] The CSIS Center for Strategic & International Studies (CSIS). Retrieved from https://www.csis.org/programs/strategic-technologies-program/significant-cyber-incidents

The digitalization of the economy and society has become an essential element with a potentially significant impact on global prosperity. From the perspective of science and data processing, essential for communication technologies such as artificial intelligence, cloud computing, and more, Younan et al. (2020) offered a comprehensive analysis of the future of the Internet of Things, exploring the challenges. Their analysis covered both the public and private spheres.

2.4.1 Cyber Incidents and Financial Impact

Information and communication technologies have exponentially boosted personal information sharing, storage, and processing (Conger et al., 2012; Saridakis et al., 2016). This evolution has resulted in incidents with significant economic and social implications (Mantha & Soto, 2020), making cybersecurity an urgent concern for companies, regardless of whether they operate in regional or global ecosystems.

Companies invest in various resources to ensure information and cyber security in their business models. However, in addition to these companies, third parties also allocate considerable resources in this regard. These third parties are engaging in cybercrime in an increasingly sophisticated way, turning it into a highly lucrative business model (Chainalysis, 2021, 2023). This illicit enterprise involves the collaboration of specialists from various fields, including individuals who can convert the amounts obtained in cryptocurrencies into tangible physical goods (Center for Strategic & International Studies [CSIS], 2023; IBM, 2023).

Considering that "human risk is considered the weakest link in information security" (Cybersecurity Guide, 2021; Dodel & Mesch, 2019, p. 75), opportunism can be revealed by actions aimed at circumventing the system through malicious behavior and limited rationality by atomized skills and knowledge when it comes to protection against cyber threats. So much so that the perception of uncertainty in a transaction can increase opportunistic behavior, making it difficult to predict future environmental conditions (Silva & Brito, 2013; Williamson, 1993).

An economic exchange relationship can entail significant transaction costs, which involve the time, material, and financial resources a focal individual must allocate before the exchange occurs (Williamson, 1981). However, allocating resources to mitigate operational and digital vulnerabilities is essential since no company wants these weaknesses to be part of its economic transactions. This is because assessing the damage caused by cyber-attacks has presented colossal challenges on a global scale for companies and governments. Such attacks

are known to cause catastrophic and cascading service disruptions, resulting in billion-dollar financial losses to organizations and critical infrastructures worldwide (Pal et al., 2021). In 2020, the amounts sent and received from illicit activities through digital currencies totaled US$10 billion, representing 0.34% of all global transactions with digital currencies (ChainAnalysis, 2021).

Attacks occur through social engineering, in which the attacker uses human interactions to obtain or compromise information about an organization or its systems. Cybercriminals try to exploit the individual's lack of knowledge by presenting themselves as trustworthy and posing as new employees. They even offer credentials to support this identity. By asking questions, they can collect enough information to infiltrate the network and do so in relation to other people within the same organization (CISA, 2021; IBM Security, 2023; SERPRO, 2021). Another IBM report (2021) revealed that vulnerabilities in the cyber environment had exceeded phishing[4] as the "most common infection vector," being the most applied way to access victims' environments in 2020.

Cyber risks are emerging as a growing topic in the current scenario. A study by the Ponemon Institute, in collaboration with IBM Security (2023), analyzed 553 companies in 16 countries and regions, covering 17 different sectors. The research focused on companies that faced data breaches between March 2022 and March 2023. The report's main objective was to identify "how much a data breach would cost," as the global average cost between 2022 and 2023 reached the significant mark of US$4,45 million, representing a 15% increase over the last three years. These figures highlight the seriousness of the threat that organizations face in the cyber environment. Another relevant aspect highlighted by the report is that 51% of organizations plan to increase their investments in cyber security. This increase in investment includes measures such as incident response (IR) planning and testing, employee training, and the adoption of advanced threat detection and response tools. This trend reflects the growing awareness of cyber risks and the pressing need to protect companies.

Another point that deserves to be highlighted is the substantial savings achieved by organizations that have adopted cutting-edge technologies such as Artificial Intelligence (AI) and security automation. These organizations manage to save an average of $1,76 million compared to those that do not adopt these innovations. In addition, the report highlighted the

[4] *Phishing* is an attack using email or malicious websites to request personal information by posing as a trustworthy organization. The attack is through an email from a trusted company or institution that requests information from the target person. They also pass themselves off as charities, taking advantage of current events such as natural disasters, epidemics, and elections.

importance of involving law enforcement agencies in ransomware[5] related incidents. This is because the figures indicate that organizations that did not involve these agencies faced significantly higher costs and a longer breach lifecycle.

Another highlight is the 53.3% increase in data breach costs in the healthcare sector since 2020. This highly regulated sector has seen a substantial increase in the costs associated with data breaches, highlighting the vulnerability even in sensitive sectors.

Given this data, the study of cyber risk is becoming more critical. Mitigating cyber risks is no longer an option but a necessity for any organization's security in the cyber environment. In addition, it corroborates the understanding that dealing with situations requires investment in knowledge and action in cyber security, going beyond just providing training for teams (Ng et al., 2009). However, in this scenario, regulatory agencies and the government have established crucial roles in administrating and regulating data security and cybersecurity.

2.4.2 Agencies and Regulatory Authorities

As cyber-attacks have grown, regulators have taken notice and taken steps to increase the pressure on organizations to find solutions (Bowcut, 2021). Regulatory and compliance requirements are, at the same time, a significant challenge for organizations to coordinate and manage their structures and processes, which has caused movement on the part of legislators on the issue of cybercrime. The General Data Protection Regulation (GDPR) is a rule of European law on privacy and personal data protection, applicable to all individuals in the European Union. The General Data Protection Regulation (LGPD) acts in different sectors and services, seeking legal security in the data of individuals, companies, and governments (SERPRO, 2021), and both legislations were published in 2018.

Bringing light to the context of regulators and supervisory authorities whose premise is to create, execute, monitor, and supervise the interactions of agents involved in cyber risks, they have also acted as an incentive and even an accelerator of innovation, involving partners interested in sharing initiatives that will speed up solutions for society (OECD, 2018), bringing in financing agents such as entrepreneurial capital, crowdfunding, start-ups, laboratories, and angel investors.

[5] Ransomware is a type of malware in which social engineering is usually carried out before the data is encrypted. The data remains on the individual's or organization's computer, but without them being able to access it. Forms of attack include hijacking an organization's server or service platform and demanding a ransom. Another example is blackmailing victims by threatening to expose sensitive or strategic information and data.

The macro and meso institutions involved in cyber risk management can encompass various entities, including regulatory bodies in the technology sector, government cybersecurity agencies, cybersecurity standardization organizations, cybersecurity service providers, and other entities that play a crucial role in the cybersecurity sphere (Brazilian Ministry of Science, Technology and Innovation, 2023; European Union Agency for Cybersecurity, s.d.; Oversight FBI Cyber Divison, 2022).

Therefore, given the highly competitive nature of the market, forcing organizations to change in order to improve their positioning in the face of risks and uncertainties, as well as the growing complexity and expansion of the markets in which they require their processes to be compliant and sustainable, the social and economic impact of the services sector is also relevant to this study.

2.5 SERVICES

Understanding the importance of services in the contemporary economy is crucial to analyzing their nature and characteristics. The service economy is a diverse field encompassing a broad range of services, each with its characteristics regarding operations, competition, and regulations (Gallouj, 2023; Metters, 2023). In fact, the economy can be seen as a vast system of services comprising various “interconnected entities and subsystems” (Stoshikj et al., 2016, p. 212; Vargo & Lusch, 2004).

The relevance of the services sector is also evidenced by its significant contribution to countries' Gross Domestic Product (GDP). The research by Rubalcaba and Solano (2023) reveals indicators that highlight the solid presence of services in developing economies, with a contribution that reaches approximately 76% of the total added value in these regions. On a global scale, services play a vital role, accounting for around 51% of global GDP.

Furthermore, the service economy can be represented by traditional activities and associated with a modern, developed economy. In this sense, Rubalcaba (2023, p. 18) approaches this issue from two fronts: from the traditional perspective, services are defined as activities that offer "intangible, immaterial, non-transportable, non-cumulative, non-quantifiable, transitory and unpredictable products." Their value is intrinsically linked to their socio-economic function, often complementing the production of goods. However, in the modern view, services are considered to be "changes, actions, relationships, co-productions, interactions and dialogues that result in significant changes, benefits, value and effects" (Gadrey, 2000; Hill, 1977; Maglio & Lim, 2018; Vargo & Lush, 2004).

Aligned with the vision of a modern economy where services play a key role in economic and technological advancement, Gallouj's (2002) analysis was instrumental in examining some misconceptions, including the widespread myth that services are incapable of innovation. His research contributed to paving the way for an understanding of the dynamics of innovation in services, even in predominantly service-based economies, questioning the common idea that services only absorb technological innovations from the manufacturing industry or that they depend exclusively on labor intensity, a claim also questioned by Miozzo & Soete (2001).

Furthermore, the growing value of services has been so well recognized that the industrial manufacturing sector is actively seeking ways to diversify its offerings and improve customer service through servitization (Ostrom et al., 2015). However, when adopting the extended service strategy, it is important to recognize that, despite its positive benefits, it is essential not to neglect the risks and uncertainties associated with this new operation. Previous research, such as that by Fang et al. (2008), Nordin et al. (2011), Raddats et al. (2019) and Rajapathirana & Hui (2018), highlights the importance of a balanced approach that takes into account both the positive aspects and the risks involved in implementing this strategy.

In this perspective, concerning the coordination of organizational activities, Djellal and Gallouj (2023, p. 157) distinguish between core and peripheral activities as key elements in service strategy. Core activities refer to the organization's essential competencies related to the core service, defining its identity and core value in the market. In contrast, peripheral activities are complementary, responding to secondary customer needs or facilitating access to the core service.

The authors also emphasize that these activities have a central role to play in differentiating the company, often forming part of innovation and competitiveness strategies. This approach to managing organizational activities fits the context discussed by Feldman and Pentland (2003), who understand that organizational routines are challenging because of their variety. They counter the view that these routines foster the creation of inertia in the company, starting from the premise that the organization's structure can be seen as a function of the problem-solving process.

In addition, another strategy companies use to coordinate their operations and focus on their core competencies is outsourcing, which has been perceived as a complementary activity in organizations so that they can devote themselves to their core (Segate et al., 2019).

In this sense, business models have become increasingly dynamic and have shown a greater central focus on experience, whose services have provided value to customers, as

highlighted by Sundbo et al. (2022). It is possible to highlight the transformation of the role of customers and their skills in the socio-economic context, guided by co-creation, experiences, data generation, and initial experiments that lead to improving new products and services (Curley & Salmelin, 2018). However, it also capitalizes on digital technologies, opening doors to operational efficiency in companies while strengthening ties with customers, as Tabrizi et al. (2019) noted.

In addition, how companies manage their core and peripheral activities, as discussed by Djellal and Gallouj (2023), together with the considerations on the relationship between value co-creation and organizational learning, highlighted by Desmarchelier et al. (2020), lead to the consideration that how a company conducts its operations impacts not only its efficiency but also its ability to co-create value with customers and adapt to a constantly changing environment. Furthermore, this adaptation to the environment is highlighted by the exponential growth of data and the convergence of different technologies, which have come along with the definitive establishment of technology and information, significantly transforming all areas of the economy (Kristensson, 2019). This interconnection promotes complexity of economic systems, driving the creation of new activities and services (Desmarchelier, 2019).

In this context, it is clear that research into the generation of value in e-business during the transition to the online environment has attracted the attention of researchers in the fields of entrepreneurship and strategic management (Amit & Zott, 2001). So much so that Scupola (2023, p. 33) discusses the significant transformation in the economy since the rise of the World Wide Web (www) in recent decades, which has led to a transition of the main units of economic exchange from traditional goods and services to “electronic services”.

However, it is relevant to emphasize that the cyber environment goes beyond traditional business-to-business (B2B), direct-to-consumer (B2C), and e-government services, as listed by Scupola, 2008. Behind the scenes is a dynamic ecosystem of companies providing various services and resources in the cyber environment.

Services have become a cross-sector activity in the economy, and their expansion is intrinsically linked to other economic sectors, as stated by Miozzo & Soete (2001), whose authors contributed a taxonomy that represents a comprehensive revision in the categorization of service sectors concerning the sources of technological change previously proposed by Pavitt (1984). This taxonomy emphasizes the interaction between the manufacturing and service sectors in terms of the origin and application of technological innovations, as shown in Fig. 3.

Figure 3 – Services Sector Taxonomy: Determinants, Directions and Characteristic

Category firm	Typical core sectors	Sources of technology manuf./serv.	Type of user	Means of appropriation	Technological trajectory	Source of technology	Relative size of innovating firms
Supplier Dominated	Personal services Restaurants Laundry Beauty	Manuf.	Performance sensitive	Non-technical	Product design	Suppliers	Small
	Public and social services Health Education	Both	Quality sensitive	Not allowed, public	Improving performance	Suppliers	Large
Scale-intensive physical networks	Transport Wholesale	Manuf.	Price sensitive	Standards, Norms	Cost-cutting, networking	In-house, suppliers	Large
Information networks	Finance Insurance Communications	Both					
Specialized suppliers/ science based	Software Specialized Business services	Serv.	Performance sensitive	R & D, Know-how, Skills, Copyright, Product Differentiation	System design	In-house, customers, suppliers	Small

Source: Miozzo & Soete (2001)

In order to comprehend the content of the services taxonomy shown in Fig. 3, a brief analysis will be presented. This analysis will allow exploration of the distinct characteristics of each group of companies and sectors, understanding their specific innovation strategy that each represents and coordinates.

The "Supplier Dominated" services category concerns personal services sectors. These companies depend on suppliers to acquire products and services essential to their operations. Their emphasis is on manufacturing products or providing quality services, and innovation often focuses on improving the design and performance of their products or services. Although they may be important players in their respective industries, companies in this group tend to be smaller than other sectors and groups. A subcategory is also dedicated to critical sectors, including public and social services, such as health and education. These areas rely heavily on suppliers to operate and are sensitive to the services' quality. Innovation in this group tends to be aimed at improving the performance and quality of services, as they have a significant impact on society. Innovative companies within these sectors tend to be significant.

Companies in the second category, represented by "Scale-intensive physical networks," which cover sectors such as transportation and wholesale, emphasize large-scale operations and operational efficiency. These sectors are price-sensitive and need to adopt standards and norms to guarantee quality in their operations. Companies in this group are often large, seeking to maintain a competitive edge by striving for efficiency and cost control.

In the "Information Network" category are the finance, insurance, and communications sectors. These sectors are dependent on information technology to offer services. They are price-sensitive, justifying the prioritization of efficiency and cost reduction. Companies in these sectors adhere to standards and norms to guarantee the reliability and security of their operations. Sources of technology include internal and external resources. The size of innovative companies may vary, but they all share the need to adopt advanced technologies to maintain their competitiveness, possibly because it is a constantly evolving environment.

The "Specialized Suppliers/Science-Based" category comprises specialized service sectors, such as Specialized Software and Business Services. These sectors are also sensitive to performance and constantly seek improvements in research and development (R&D), know-how, skills, copyright, and product differentiation. The technology path is focused on designing advanced systems to meet specific customer needs. Sources of technology include in-house development and outsourcing. The size of companies in this group tends to be smaller.

In addition, the transformation of the role of agents and their competencies in the socio-economic context, guided by co-creation, experiences, data generation, and initial experiments

that lead to improvement or new products and services (Vargo & Lusch, 2004; Curley & Salmelin, 2018). This involves collaboration between companies and customers based on data and customer behavior, resulting in interactive and co-productive actions. In addition, competition would be at the center of competitiveness studies, but cooperation and competition can coexist through coopetition (Bouncken et al., 2018; Brandenburger & Nalebuff, 2020). These approaches that support growth and agility for companies can enable them to adapt quickly to changing customer needs and demands, as discussed by Nagle et al. (2020) and Rubalcaba (2023).

This leads to the understanding that technological innovations originate and diffuse in services, often driven by the interaction between the manufacturing and service sectors in response to the demands of the cyber environment. This dynamism has resulted in an ecosystem made up of various sectors and companies that range from network infrastructure and cybersecurity to e-commerce platforms, applications, data analysis, cloud services, virtual and augmented reality, artificial intelligence, and much more (Bryson, 2023; CISA, 2021; Wei & Blake, 2010; Younan et al., 2020).

Each group shown in Fig. 3, with its distinct characteristics and specific innovation strategies, has the choice of internalization (in-house) or outsourcing (supplier) in its technology source. In this context, Gallouj & Djellal (2011, p. 483) suggest that, as companies gain more experience in contract design and learning, contracts can evolve to become "repositories of knowledge" on how to manage collaborations effectively. The governance structure directly impacts the operating costs and contractual relations of each group (Williamson, 1991, 1996). For example, outsourcing can be a compelling choice for coordinating transaction costs in physically scale-intensive sectors such as transportation and wholesale, taking advantage of economies of scale (Saes, 2009).

Therefore, Gallouj and Djellal's (2011) argument reinforces the concept of transaction costs in the context of service innovation, which can be particularly significant due to the uncertainty and ambiguity associated with service innovations and the need to negotiate new and complex service contracts. This can have significant implications for companies seeking to innovate in their services and manage these associated costs.

3 METHODOLOGICAL PATHWAY

The mixed method, or mixed research, combines qualitative and quantitative techniques in the same research study. This approach allows for a complete understanding of the object of study, integrating subjective and objective aspects in the same analysis (Creswell, 2021).

Scientific research is a complex process that involves various methods and techniques to answer a research question or problem (Bryman, 2021). When choosing the most appropriate method for a study, it is essential to consider the research objectives, the nature of the phenomenon to be investigated, and the resources available. The mixed method has been increasingly used in academic research, especially when seeking a more complete and in-depth understanding of the phenomenon studied (Bryman, 2021; Creswell, 2021; Headley & Plano Cark, 2020). In the case of research that seeks to identify how governance structures affect the mitigation of cyber risks in services, using the mixed method can be particularly useful, as this is a complex and multifaceted phenomenon involving many variables and factors.

In this research, the mixed method strategy is justified by the need to obtain a broader and deeper view of the options for governance structures to mitigate cyber risk. While the quantitative approach focused on measuring objective variables and analyzing the data statistically, the qualitative approach sought to understand the subjective perspective of the individuals involved, exploring their experiences and meanings attributed to the research topic. Qualitative and quantitative techniques were combined and integrated to complement the information obtained in each approach and guarantee the validity and reliability of the results.

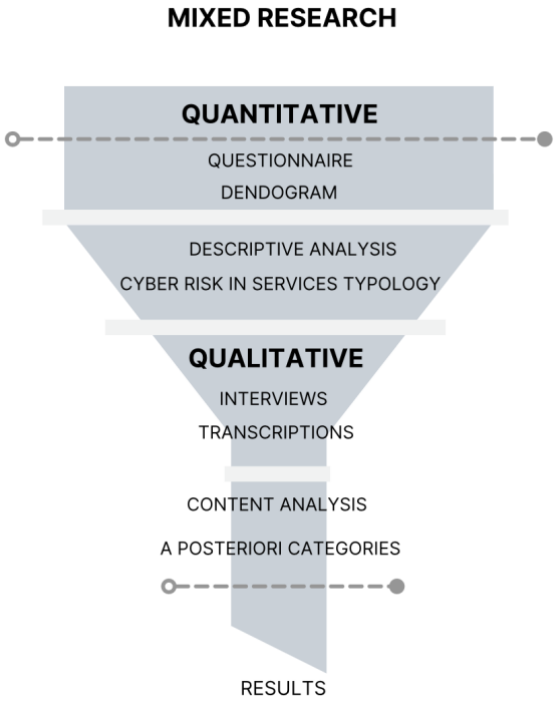
In conducting this research, an explanatory sequential strategy was adopted (Creswell, 2021), starting with quantitative analysis and followed by qualitative exploration.

The strategy of starting the research by collecting quantitative data aimed to provide an up-to-date view of cyber risk mitigation in the companies under study. The quantitative results provided input for understanding emerging trends in the characterization of the services transacted by the companies, strategic practices adopted for growth, and the issue of contractual governance in services.

The sequential explanatory strategy was advantageous for providing a broad understanding. It excels at explaining unexpected quantitative results and offers flexibility to adjust the research based on initial findings (Creswell, 2021). However, dealing with prioritization decisions between the phases was essential, and coordinating the two distinct stages was challenging.

During the quantitative phase, descriptive analysis of the data generated by dendrograms identified emerging categories or patterns. Subsequently, in the qualitative phase, a more in-depth and exploratory analysis was carried out in order to confirm and deepen the understanding of these categories identified in the quantitative phase, as shown in Figure 4.

Figure 4 – Mixed Research Design



Source: author (2023)

The funnel format represented the process stages in this mixed research flow. Although there are no direct arrows, it is essential to emphasize that the process was fluid, allowing for a holistic approach to developing findings during the analyses. After completing each research stage, the journey did not end; it gave way to a reflexive return to the previous stages. Revisiting was an aspect not intended to alter the results but to deepen understanding and find new particularities in the data collected and analyzed.

The methodological approach of this study was structured as follows: it began with a quantitative approach, followed by the definition of the variables and the construction of the questionnaire. The data from this questionnaire was analyzed using dendrogram analysis, allowing a descriptive analysis to be carried out. The categories resulting from this analysis facilitated the development of a framework of cyber risk mitigation in services and the definition of the thematic axes used to construct the interview script in the qualitative phase.

In the qualitative approach, the study was based on constructing an interview script based on the conclusions of the quantitative phase (thematic axes) and the questions considered relevant to deepen the understanding of the emerging themes. In the qualitative phase, interviews were conducted, and then transcripts of these interviews were analyzed using the content analysis method (Bardin, 2020). This resulted in organizing *a posteriori* categories, which were later analyzed and combined with the thematic axes.

It is essential to clarify that, although this study followed a sequential explanatory process (Creswell, 2021), for better clarity and organization, separate chapters were dedicated to describing the methodological path of the mixed research and presenting the results.

3.1 QUANTITATIVE APPROACH

The quantitative approach aimed to support achieving the objectives outlined in this research. To this end, this stage sought to create thematic axes as a foundation for the subsequent qualitative research phase.

3.1.1 Definition of Variables and Construction of the Questionnaire

The questionnaire on cyber risk mitigation in services was developed to assess companies' choices of governance structure to face the challenges related to cyber risk (Apendix A). In this context, the application of this questionnaire aims to explore the characterization of the services that permeate an organization (contracting and production), the means of mitigating cyber risks, and contractual governance.

The questionnaire was adapted from the "Harmonised Data Collection for the CIS 2018" (Community Innovation Survey [CIS], 2018), regarding data collection methodology and survey structure. Just as the CIS strived to understand innovation in companies, the questionnaire in this study aimed to understand how companies, considering all their specificities and contractual relationships, faced the challenges linked to cybersecurity. Using the CIS as a model, we sought to adapt its best practices to the research to gain an understanding of the characteristics of the companies involved in mitigating cyber risks in services and in their contractual relationships.

The questions were developed based on a literature review, which was an essential step in identifying the main issues related to Transaction Cost Theory - TCT (Coase, 1937, 1998; Williamson, 1985, 1991, 1996), Services Innovation Theory (Gallouj & Djellal, 2011, 2018)

and Economic Contracts (Brousseau & Glachant, 2002), as central references for the development of the research. This study considers the mentioned authors as central references for the research development.

A common object in the transaction cost literature is identifying the costs involved in an economic transaction. These costs include information seeking, negotiation, contract governance, and monitoring. Identifying these costs helps to understand how they affect transaction efficiency (Farina et al., 1997). Transaction Cost Theory also emphasizes the role of institutions in reducing transaction costs. Therefore, it can be applied to assess how legal, regulatory, and normative institutions (contracts) affect business decisions and the transaction costs associated with those decisions (Quevedo, 2016; Williamson, 1991).

TCT suggests that firms can reduce transaction costs by adopting efficient management practices, such as vertical integration, outsourcing, and forming strategic alliances. Also, TCT emphasizes that transaction costs can affect competition in each market and can be used to assess how transaction costs affect competition and whether competition is sufficiently robust to encourage efficiency and innovation.

Innovation is an essential factor in the competitiveness of companies, allowing them to develop new products, services, and processes that enable them to remain at the forefront of the market (Christensen & Raynor, 2013). However, innovation is not an isolated process and often involves co-creation to allow companies to work with other companies, customers, and stakeholders to develop innovative solutions (Djellal, Gallouj & Miles, 2013).

The literature review was a fundamental step in the questionnaire development process, as it allowed the identification of the most relevant variables for the research and the understanding how they are related. Therefore, twenty-four questions made up the questionnaire and were designed to ensure clarity, objectivity, and the ability to accurately measure the variables defined. The questionnaire is subdivided into topics related to the Organization of Activities and Knowledge Flow, Cyber Risk Mitigation, and Contractual Governance.

3.1.2 Respondent Selection

Before applying the questionnaire to the research participants, a pre-test was carried out with three professionals with characteristics similar to those of the future participants. At this stage, these professionals were asked to answer the questionnaire and give feedback on their understanding and ease of answering. After analyzing the results, some adjustments were made

to the questionnaire before its final application. These adjustments included incorporating additional concepts into specific questions to improve their comprehension. Finally, pre-testing proved an essential and practical stage in the questionnaire development process (Bryman, 2021). It allowed to assess the quality and clarity of the questions before collecting data on a large scale.

In this study, 618 professionals were approached after conducting a search on the LinkedIn platform, in which the following keywords were used: "cyber risk," "services," "risk mitigation," and "risk management."

The goal was to have a multidisciplinary and representative sample of professionals who work in different areas and share the common challenge of mitigating cyber risks in their respective companies. This multidisciplinary view was fundamental to understanding how risk mitigation is approached in the different activity segments and identifying the best practices and challenges professionals face.

For selecting participants, managerial positions and analysts linked to areas such as compliance, cybersecurity, IT development, controlling, finance, commercial and internal development, among others, were considered.

3.1.3 Questionnaire Application

An informed consent procedure was adopted before the questionnaire was sent out to ensure that respondents understood the nature of the research and were willing to participate. This involved an explanation of the purpose and content of the study, followed by a question about interest in participating. Only after confirmation of interest was the questionnaire link made available to the respondents. This informed consent approach is a practice that seeks to maintain research ethics (Bryman, 2021) and helps to ensure that participants are aware of what they are getting involved in and voluntarily agree to participate.

The questionnaire used in this study was developed and applied through the online platform Google Forms. Of the 618 professionals approached, 175 participated in the questionnaire, representing a response rate of approximately 28%. The data collection period in the field covered 21 days, from the beginning of January to February 2023. The researcher was available for any questions about the completion of the survey or any other inquiries. During the survey, some respondents informed that they could not continue answering the questionnaire, even though they were aware of the terms of confidentiality and secrecy of the survey. This was because the survey included questions that these respondents considered

strategic to their companies. In particular, 6,8% of the respondents, all with managerial positions, provided this feedback.

3.1.4 Methodology for Clustering Analysis and Data Exploration

Clustering is a statistical technique that groups individuals into homogeneous groups based on their characteristics (James et al., 2021). In this context, a questionnaire applied to 175 participants was analyzed to identify groups of individuals with similar characteristics regarding the research question: How does the governance structure affect cyber risk mitigation in services? To develop this, the process of performing the clustering in Python (Spyder v. 5.3.3) involved the following steps (Desmarchelier, 2023):

- Export the Google Forms spreadsheet to a CSV file;
- Import the Pandas library into Python;
- Load the CSV file into a Pandas data frame;
- Select the relevant variables for clustering and prepare the data accordingly;
- Use clustering libraries available in Python to perform the data clustering.

After processing the data, the next step was to perform the hierarchical grouping by creating dendrograms, which are graphical representations of the hierarchical structure of the clusters (James et al, 2021; Desmarchelier, 2023). After creating the dendrogram, it was possible to determine the number of clusters to be used in the analysis and the clustering method to be applied.

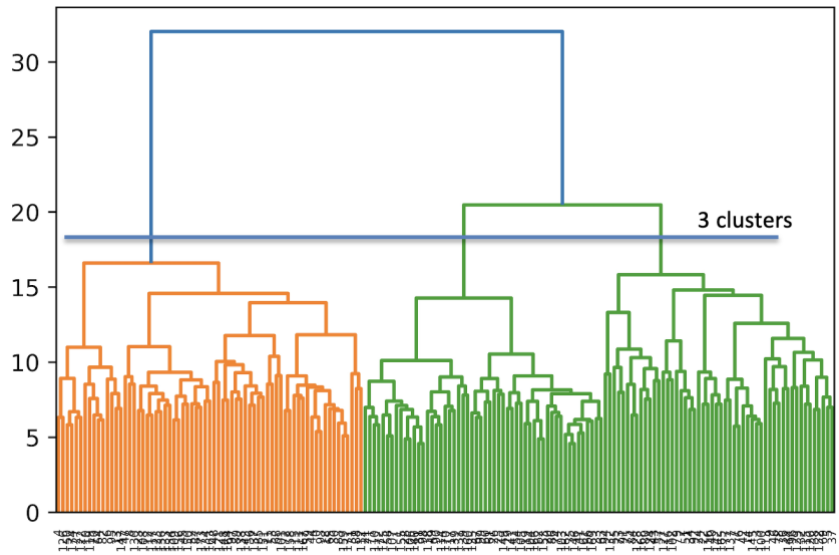
The choice of distance metric is a critical aspect of cluster analysis, as it directly influences the formation of clusters. In this study, we used Euclidean distance as the dissimilarity metric. Euclidean distance is a widely recognized metric that measures the distance between two points in an Euclidean space. It is appropriate when the units in the dimensions of the characteristics are comparable and when you want to consider the differences in magnitude between the traits.

The cluster analysis identified significant groups of firms, with more than 40% of them grouped into clusters. This suggests that each cluster is a relatively homogeneous group of companies with similar characteristics. Identifying specific patterns may be helpful to understand better the characteristics, behaviors, and needs of the companies in each cluster. However, only some questions had their results analyzed globally, including all clusters.

The cutting method, used to determine the number of clusters, involved selecting a point in the dendrogram based on specific criteria, which may include visual interpretation and the

height of the cut in the dendrogram, as shown in Figure 5. The analysis resulted in three distinct clusters, in line with the research objectives. At this study stage, the clusters were divided using the dendrogram method. The dendrogram, a graphic representation of the stages of the clustering process, was used to group the companies based on their similarities concerning the variables analyzed. This technique made it possible to visualize the interactions between the companies as they grouped and separated throughout the process.

Figure 5 – Dendrogram with Euclidean distance measure



Source: Research Data / Python (Spyder v. 5.3.3)

This process of hierarchical division by dendrogram provided a detailed view of the structure of the clusters and the relationships between the companies. The study comprised 175 companies, distributed as follows: 68 companies in Cluster 1, 62 companies in Cluster 2, and 45 companies in Cluster 3.

In order to develop the questionnaire analysis, 24 variables were measured in the questionnaire. The data was then analyzed using the agglomerative hierarchical clustering method with the complete linkage method, which is a clustering method that considers all observations at each stage of the cluster.

Although there is no specific reliability test for dendrograms, it is possible to perform robustness tests to assess the stability of the clusters obtained. Random permutation: In this test, clusters were randomly permuted several times, and the dendrogram was reconstructed with each permutation. The comparison of the dendrograms obtained with the original dendrogram allows the statistical significance of the associations between the objects to be assessed.

An essential step in the data analysis was coding categorical variables to make the answers more accessible to quantitative analysis. In order to code the answers to questions that included categories such as 'high,' 'medium,' 'low,' and 'not important,' it was decided to use the dummy variable coding method. This method creates separate binary variables for each category, transforming categorical answers into numerical values. For example, if a participant's original answer was 'high' or 'medium,' a value of 1 was assigned to the dummy variable corresponding to the category, indicating 'Relevant.' However, if the answer was 'low' or 'not important,' the dummy variable was assigned the value 0, representing 'Irrelevant.' The dummy variable coding method made a significant contribution to the data analysis in this study, allowing for a quantitative interpretation of the categorical responses and contributing to a comprehensive understanding of the results.

Based on the results of the exploratory analysis, further descriptive statistical analyses were conducted to assess the statistical significance of the relationships between the variables. Initially, each cluster was individually analyzed to comprehend its unique characteristics, which aided in identifying significant differences among the companies in each group. Subsequently, the clusters were compared to understand the overall trends.

3.2 QUALITATIVE APPROACH

The qualitative research was conducted based on the interview script developed from the thematic axes obtained in the quantitative phase of the study (Apendix B). The primary purpose of this stage was to explore, deepen, and contextualize the information collected through content analysis. This approach aimed to enrich the understanding of the specific objectives outlined for this study, thus providing a more comprehensive view by combining and analyzing the results obtained in both phases.

3.2.1 Constructing the Interview Script

In a competitive market, companies seek growth, innovation, and adaptation and have the crucial objective of remaining sustainable. They often face a complexity that transcends numbers since they are involved in various services, each with specificities. Moreover, these companies operate in an environment where cyber risks are a presence in the new businesses carried out in the digital world.

In this context, the quantitative results, analyzed in isolation, might only capture some of the complexity and richness this study requires. For this reason, the qualitative approach was adopted as an extension of the quantitative analysis, allowing for a deeper understanding of the processes and experiences of the participants. At this stage, interviews were conducted and applied, allowing for exploring the themes, contexts, and meanings underlying the quantitative results.

The advantage of the thematic axes (*a priori* categories) is that they provided a consistent and systematic structure for analyzing the interviews. This made it easier to compare participants' responses and to identify patterns, trends, and specific relationships. The creation of the thematic axes was not intended to restrict the qualitative analysis, so it was possible to reconcile the identification of categories *a posteriori* after the content analysis. Therefore, by using the framework of cyber risks as *a priori* categories, converted into thematic axes, the aim was to ensure that the categories were flexible enough to accommodate themes and new information that might emerge from analyzing the content of the interviews.

3.2.2 Thematic Axes and Research Parameters

The second stage of the research adopted a qualitative approach, using interviews to explore further themes aligned with the theoretical framework and the contextualization of the quantitative phase. The questions addressed in the interviews were related to the following thematic axes: (i) Organization of Activities and Knowledge Flow, (ii) Contractual Governance, and (iii) Mitigation of Cyber Risks in Services.

During the content analysis, the researcher maintained a flexible approach when analyzing the content of the interviews, which made it possible to code the interview responses and take ownership of the categories *a posteriori*.

3.2.3 Data Collection

During this research phase, 38 individual invitations were issued to the participants, who were selected based on their willingness to participate during the quantitative phase. These invitations aimed to ensure the professionals' continued participation in the subsequent stage of the interviews, seeking a more in-depth exploration of the topics covered. Of the invitations sent out, 20 interviews were scheduled and carried out, and the data collection period in the field covered 63 days, from March to May 2023. It is also important to note that all these

interviews were conducted using the Teams platform. This option was deliberate to optimize flexibility and overcome mobility and geographical limitations, allowing professionals to participate.

After explaining the objectives of the investigation, complying with the conditions established by the Ethics Council, and obtaining authorization for recording from the interviewees, it was possible to accumulate 22 hours of interview material. The average time spent on each interview was approximately 65 minutes. It is worth noting that all the interviews were transcribed manually and inserted into a Word document format, generating 453 pages.

The choice of manual transcription provided a prosperous space for new ideas and perceptions about the research to be generated during this process. In this way, the analysis was conducted manually, capturing the content expressed and encouraging the discovery of deeper layers of meaning, contributing significantly to the richness of the research.

In addition, the interviewees pointed out that the questions asked provided moments of reflection and opportunities for them to re-examine the process of mitigating cyber risks in their respective organizations and partnerships, as participant E18 said “even during the process, I ended up with some ideas to implement in my company”.

3.2.4 Profile of the Interviewees

At the beginning and end of the interviews, all the participants emphasized their interest in the results of this research. It is important to emphasize that the active contribution of these participants was fundamental to the successful conclusion of the interview stage, giving them a central role in the progress of this research. In order to safeguard the confidentiality of the interviewees, they have been coded as "E" followed by a number, as shown in Table 1.

Table 1 – Categorization of Participants

ID	Position/Function	Company Local	
E1	Cyber Security Manager	E-Commerce Technology	Multinational
E2	Tax Compliance Analyst	Technology in mobility platform	Multinational
E3	New Businesses Executive	Information Systems	Local
E4	Cyber Risk Officer – CRO	Insurance Company	Multinational
E5	Business Intelligence Director	Digital Entertainment	Multinational
E6	Internal Audit Manager	Financial Institution	Multinational
E7	Corporate Risk Executive	Broadcast television and other ventures	Local
E8	Cyber Security Executive	Digital Innovation Provider	Multinational

To be continued...

Continuation of Table 1

ID	Position/Function	Company Local	
E9	Data Engineering Manager	Consulting, Technology and Operations	Multinational
E10	Data Scientist	Financial Institution	Multinational
E11	Legal Risk Lawyer	Law Firm	Local
E12	Cyber Security Partner	Auditing and Consulting (Strategic/Tax)	Multinational
E13	Tax Associate	Auditing, Consulting Advisory Services	Multinational
E14	Financial Planning Specialist	Financial Services	Local
E15	Auditing Executive	Utilities - Water Service Provider	Local
E16	Risk and Compliance DPO	Logistics	Local
E17	Controllership Director	Industry and Sale of Personal Care Products	Multinational
E18	Technology System Admin	Reinsurance Company	Multinational
E19	Administrative Manager	Technology Outsourcing Firm	Multinational
E20	IT Coordinator	Technological Partner City Hall	Local

Although there was no active selection, the companies' participation in the interview phase contributed to the representativeness and diversity of the perspectives included in the study. The voluntary participation of the interviewees made it possible to explore different approaches and experiences, considering a variety of job profiles and segments.

3.2.5 Data Processing

This study adopted the methodological structure proposed by Bardin (2020), which was organized as a content analysis grid to categorize, code, and interpret the multiple layers of meaning in the data collected in the interviews.

This stage of the research required special attention to the organization of the content. This pre-analysis phase involved systematizing ideas and reading through all the interviews, getting to know the context and letting impressions and orientations flow (Bardin, 2020), which comprised the interview corpus.

Thematic analysis was used and made it possible to categorize all the significant elements of the interviews. The verbs, words, and excerpts from the interviews were extracted, organized, and interpreted manually in Excel, thus enabling the classification of deductive elements based on the literature and inductive elements representing emerging topics.

In addition, the "dangers of the illusion of transparency" that can result in a lack of clarity and objectivity in content analysis were noted by Bardin (2020, p. 30). In order to avoid these dangers, a detailed description of the categories of analysis, the coding process, and the

criteria used to interpret the data was carried out. This helped to avoid implicit assumptions and will allow other researchers or readers to understand the meaning and context of the analyzed data properly.

As a result, 2.648 codes were created and organized manually. Through multiple readings, it was possible to thoroughly analyze the information, facts, and data presented in the interviews. All the interviews complied with the rule of pertinence, the order of which, presented by Bardin (2020), establishes that the interviews must be suitable as a source of information in order to correspond to the objectives initially set and the research question.

Concomitantly, it was possible to make inferences that made it possible to identify characteristics of the messages and make careful interpretations. This approach enabled the identification of elements that might have needed to be more evident in the information and facts presented in the interviews.

Therefore, in the process of layers of meaning and analysis, the coding, added to the verbs, words, and passages, was organized into categories, subcategories, and elements of analysis (UR). Through multiple readings, a thorough analysis of the information, facts, and data provided by the interviews was sought.

3.2.6 Ethical Aspects

The study was submitted to the Human Research Ethics Committee (CEPSH) of Mackenzie Presbyterian University. Data collection of the qualitative phase began only after this Committee approved the project. Participants were contacted both in person and by telephone. Each participant received a Free and Informed Consent Form (FICF) detailing the purposes and procedures of the research. It is worth emphasizing that participation was entirely voluntary, at no cost, and no point during the study were the participants identified.

Furthermore, during the preparation of the survey, special care was taken to avoid any exposure of the identity of the respondents or their companies, as requested by them. It was possible to adopt this approach without compromising the quality of the information collected, guaranteeing confidentiality, and preserving the integrity of the data. The interviews were recorded, with the participants' authorization, and later transcribed. The researcher was available to answer any questions the participants might have.

4 MIXED RESEARCH RESULTS

4.1 PRESENTATION OF QUANTITATIVE RESULTS

In this section, the quantitative results will be presented and analyzed descriptively, with references to the relevant literature to support the analysis and interpretation of the data. The organization of the results encompasses different topics, covering aspects such as internal and externalized activities, motivations for internalizing and externalizing services, investments in technology, technological dependence, mediation, economic growth initiatives in the physical and digital environment, risk mitigation strategies, and contractual governance. In addition, the developed framework of cyber risks in services, called Cyber Risk Framework in Services - CyberFrame, will be presented and contextualized in the light of the findings and the available literature. This will support the analysis of the results, providing input for the subsequent qualitative stage.

4.1.1 Organization of Activities and Knowledge Flow in Services

At the beginning of the questionnaire, the professionals were asked to provide the annual turnover of their companies. This measure was adopted to enable a comprehensive analysis of governance structures and their relationship to cyber risk mitigation in a diverse business context. This information made it possible to understand how different companies with a wide range of resources face and respond to cyber challenges in their operations - especially when discussing the results of the qualitative analysis phase.

Table 2 – Annual Revenue

	%
< 5 millions	9,1%
< 10 millions	5,7%
10 millions – 100 millions	16,0%
100 millions – 500 millions	10,9%
500 millions – 1 billion	5,7%
1 billion – 10 billions	24,6%
> 10 billions	28,0%
	100,0%

Data analysis revealed a diverse distribution of participants' companies based on annual revenue. Companies were grouped into various revenue brackets, ranging from those with less than 5 million annual revenue to companies with 10 billion dollars in annual revenue. This diversity is crucial to the research since different sizes of companies can adopt different approaches to cyber risk management according to their specific resources and needs (Barney, 1991; Peteraf, 1993)

Most companies in the sample are large, with 52.6% reporting an annual revenue of over 1 billion dollars. This scenario suggests these organizations may have more complex governance structures and resources to mitigate digital risks. This predominance of large companies is a significant aspect to consider in the analysis.

Despite the predominance of large companies, it is essential to note the significant presence (14.8%) of companies with annual revenues of less than 10 million dollars in the sample. This creates an opportunity to explore how governance structures vary between different companies. sizes and how this can affect their approaches to cyber risk mitigation.

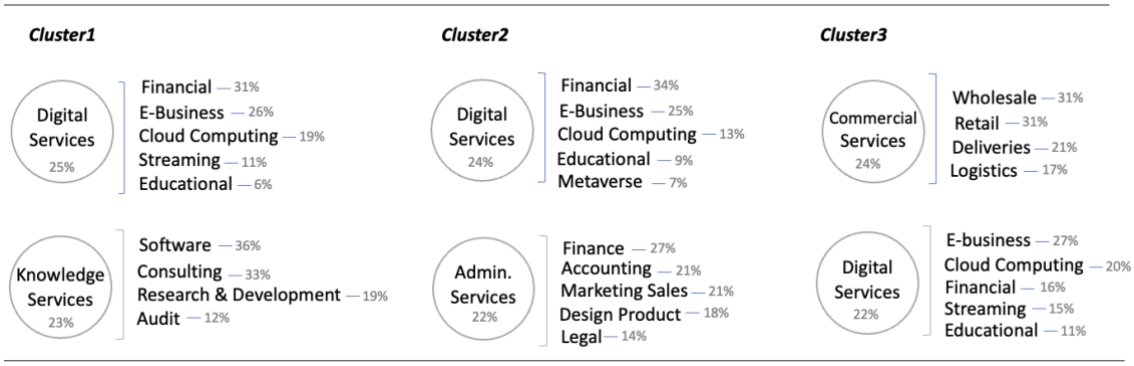
In addition, in order to obtain clarification about the types of services offered by the participating companies, they were asked to indicate which services are part of their core business and which are peripheral activities (Djellal & Gallouj, 2023). For this, a comprehensive list of services was presented, adapted from the ‘Role of Services in an Economy Model’ (Bordoloi, Fitzsimmons & Fitzsimmons, 2019, p. 5), which allowed the participants to indicate the categories of services and their respective activities, as shown in Figure 6. In addition to listing the service categories and their activities, a blank field was included so that the participants could inform another category of service or activity that was not mentioned there. As a result, the information provided has been added in Figure 6, with the category "Infrastructure," in addition to the activities highlighted in italics.

Figure 6 – Service Activities transacted by the participants' companies

	Activities
Commercial	Delivery, logistics, retail, *storage*, wholesale
Digital	Cloud Computing, e-business, educational, IT infrastructure, financial, metaverse, social networks, streaming
Financial	*Assets management*, financing, *brokerage*, *credit cards*, insurance, *leasing*, payments, *pension plan*, *private loan*
Governmental	Educational, judiciary, police, public systems
In Company (administrative)	Accounting, *consulting*, *compliance*, *data security*, design & product, finance, legal, marketing & sales
Infrastructure	*Airport*, *telecommunication*, supply systems, utilities
Knowledge	Auditing, *business solutions*, consulting, research & development, software, and *hardware* development
Personal	Entertainment, health & insurance, hotel, *parking*, restaurant, tourism

Understanding the activities of a company is fundamental to distinguish between those that generate revenue and are considered the core of the business, as well as those that complement these activities, the so-called peripheral activities (Djellal & Gallouj, 2023). These core and peripheral activities are crucial for the development of the business. They are relevant in understanding how they are related and interdependent, especially regarding the governance structures adopted. Therefore, we will start with the results that point to the core business of the participating companies, as shown in Figure 7.

Figure 7 – Core Business



In descending order within each cluster, Figure 7 describes the service activities provided by the companies. Regarding the companies' core business, Digital Services were the most common activity categories, ranking first in importance in Clusters 1 and 2, with 25% and 24% participation, respectively. Digital Services also stood out as the second position with a 22% presence in Cluster 3. However, it's important to note that the percentage differences between clusters are not statistically significant, indicating a similar presence of Digital

Services across all groups. Also, in Digital Services as core business, financial activities emerged in this category, considering that they are present and include financial institutions to companies that operate as intermediaries in online financial transactions, known as payment service providers.

Another point to note is cloud-based service. This type of activity has been enhanced by sharing and expanding the possibilities in service delivery, such as applications, platforms, network resources, and storage (Wei & Blake, 2010). Cloud computing and streaming activities were considered core businesses in Clusters 1 and 2, while in Cluster 3, they are classified as peripheral activities.

This result shows that companies are expanding their business activities into new areas as technology evolves and consumer needs change. The market for cloud services is overgrowing and has created opportunities for traditional companies to diversify their business and enter new markets.

As the second most offered type of service by Cluster 1 companies, Knowledge Services stand out with a 23% share, in which software development, consulting, research and development (R&D), and auditing constitute formal representations of explicit knowledge (Nonaka et al., 2000). These presented results reveal a strong synergy of Digital Services with Knowledge Services in Cluster 1.

A relevant result concerns Commercial Services. In Cluster 3, both wholesale and retail activities stand out, each contributing 31%. Suppose we perform a joint analysis with the e-business activity (Digital Services). In that case, it is noticeable that the primary orientation of these companies is to boost their sales of products and services. In terms of a target audience, wholesale and retail are distinguished by sales volume and prices. At the same time, e-business can encompass wholesale, retail, and sales in the online environment (Zhu et al., 2005; Chen et al., 2018).

In Cluster 2, in second place, Administrative Services were considered a core business by 22% of the companies in this group. This result proves relevant and deserves to be explored, considering that much of the literature does not consider administrative activities, the so-called back-office activities, as an integral part of the company's core operation that aims to generate revenue in a direct manner. Generally, back-office activities are performed remotely and are not visible to customers. However, in some specific cases, there may be some interaction between back-office activities and customers (Metters & Vargas, 1999; Voss & Zomerdijk, 2007).

Concerning government and personal services sectors, the survey revealed a limited presence of firms. Although the number of firms belonging to these sectors was relatively low, it is essential to note that these firms played a significant role in analyzing the characteristics, similarities, and divergences among the clusters of firms examined.

Therefore, describing the core business activity categories in descending order of result, we have Digital, Commercial, Knowledge, and Administrative. Next, we introduce the results related to peripheral activities.

Figure 8 – Peripheral activities

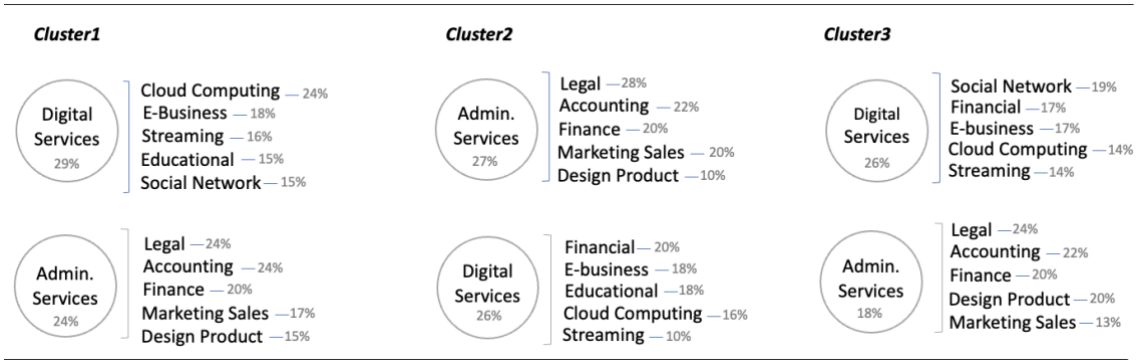


Figure 8 reveals two dominant service categories: Digital Services and Administrative Services. Digital Services activities comprise 81% of the activities, while Administrative Services activities total 69%. Administrative activities generally only generate revenues indirectly for the business. However, how they are coordinated can affect the company's profitability through cost savings and operational improvements, which may explain this result.

Another remarkable point is that the percentage of activities developed as peripheral is higher than the core business activities. It may be related to the diversification of services that companies have sought to expand their offerings to customers.

Nevertheless, following the example of the results obtained from the companies' core business activities in Cluster 1, cloud computing and streaming are also highlighted as peripheral activities, demonstrating that the cyber environment has constituted an active activity that complements the core business of the companies.

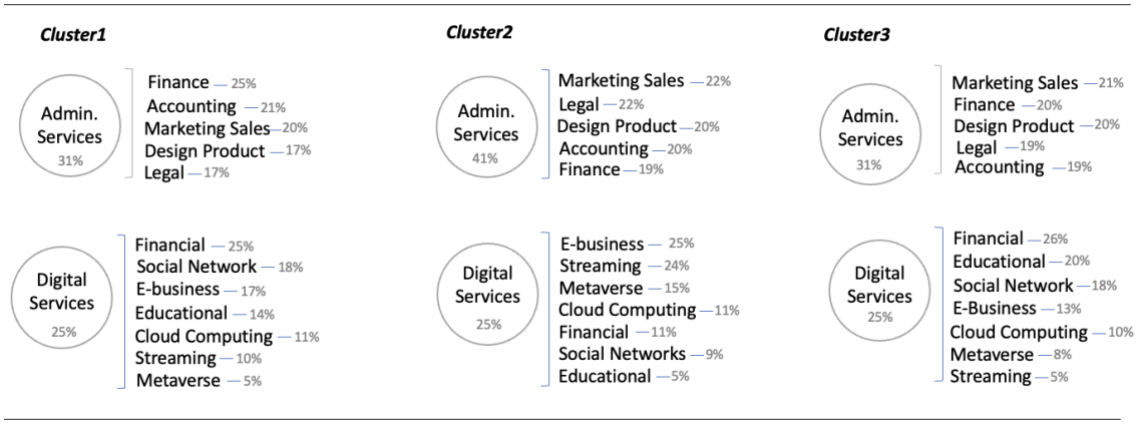
It is relevant to recognize that although these administrative activities in Cluster 2 have been classified as peripheral concerning the core business, there may be interdependence between back-office and front-office functions (Zomerdijk & Vries, 2007). Regardless of the company's governance structure, even those whose primary focus is not one of these administrative activities rely on the effective coordination of both.

In Cluster 3, financial activities and the social network are the most present and, considering that the dominant core business of this same cluster is commercial wholesale and retail services lead to the perception that the activities are linked to brand building, customer engagement, and advertising, mainly through social networks. Concerning offering financial products, it demonstrates the companies' focus on customer service, business diversification, and additional revenue generation.

4.1.2 In-house and Outsourced Activities

However, the research was not limited to the categories of services and their respective activities (core business and peripheral activities). It sought to broaden the characterization of the services provided by exploring the forms of coordination adopted by companies to provide services. To this purpose, we verified whether the activities are: (i) developed in-house, as shown in Figure 9, and (ii) outsourced, as presented in Figure 10.

Figure 9 – Activities developed in-house



Administrative Services categories are predominant in all three groups. It can be inferred that companies have chosen to keep activities related to finance, accounting, sales, legal, and product design in-house under the understanding that these activities are either strategic, critical or meet specific company objectives for the business, not justifying the need for outsourcing when developing expertise through internalization. This may be justified because companies can reduce transaction costs by using internal information unavailable to other companies in the market (Alchian & Demsetz, 1972).

In this sense, the marketing & sales activity has a strong presence in all clusters. Combined with administrative and product design activities, it demonstrates companies'

involvement in creating new services that run through the journey that unfolds from testing and analysis to its launch and capturing the value attributed by the customer (Gallouj & Weinstein, 1997).

Financial Services are highlighted in Clusters 1 and 3, with 25% and 26% participation, respectively. Cluster 1 is focused on financial activities, either to develop them as core business or to develop them internally through digital technology (Digital Services). Indeed, this information justifies the high internalization of accounting activity, considering that financial activities (payments, insurance, brokerage, for example) are subject to regulatory agencies and have specific accounting standards, where understanding and compliance are necessary to develop other financial products (product design activity). Next, digital services are second in the most in-house developed service activities.

However, it was noted that some activities are carried out simultaneously, internally, and through outsourcing. This phenomenon demonstrates the multifaceted nature of service operations in some companies, where the coordination strategy can encompass in-house management and outsourcing of these activities.

4.1.3 Motivations (and incentives) to Internalize

Considering the decision to internalize the activities within the company, the research sought to collect information about their motivations for this type of hiring and coordination of activities. A sequence of motivations adapted from the GT International Business Report (2014)[6] was made available so that participants could indicate which best suited their choice to internalize the activities:

(i) developing in-house expertise;

(ii) reduce costs;

(iii) reduce uncertainties;

(iv) control and reduce risk;

(v) strategic planning;

(vi) ensure information security; and

(vii) comply with regulators or legislation.

[6] The criteria and approaches mentioned in the GT International Business Report 2014, initially developed for outsourcing analysis, have been adapted and applied to the context of insourcing. This adaptation is based on the understanding that most motivations and underlying factors may be similar in both outsourcing and insourcing approaches.

Although the strategic choice to develop activities internally may originate from these motivations, separately or together, the companies in Cluster 1 indicated the reduction of uncertainties as the main incentive to internalize activities. Regarding the reduction of uncertainties, the literature on Transaction Costs explains the use of contracts and agreements that define the responsibilities and obligations of the parties involved to reduce losses and uncertainties (Williamson, 1985).

Furthermore, Williamson (1975) understands that the hierarchical governance structure is adequate to transact specific assets, considering that this structure offers ample administrative control. The presence of incentives is scarce, although target contracts and performance bonuses can be considered a form of incentive to contract compliance. In this context, the application of the Service Level Agreement (SLA) is considered an instrument to measure what the company has contracted and what it has received as a service provided between the internal areas of the company (Perez, 2023).

One point of investigation worth exploring is if the main reason for internalizing service activities is to control and reduce risks, it is essential to understand what risks are involved in these activities and how internalization can help mitigate them. This may include operational, security, and regulatory compliance risks, which will be addressed in this research as pre-cybernetic risk management.

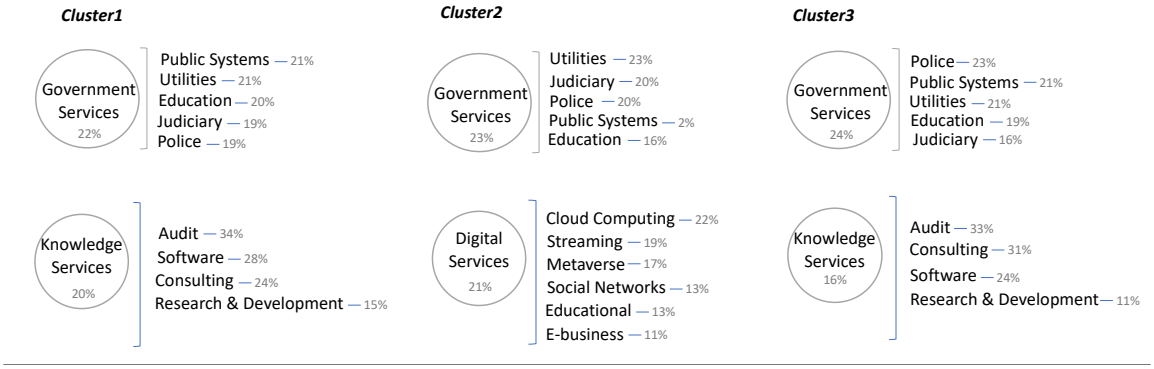
Thus, we questioned through the question: “Concerning the company's internal environment, in the definition of goals, as well as in the bonus policy, employee profit sharing, and description of SLAs, is there a direction for risk management, including cyber risks?” Most companies in Cluster 1 indicated the application of contracts through the SLA in mitigating cyber risks. Concerning Clusters 2 and 3, the presence of companies adopting this practice is 34% and 25%, respectively. By adopting internal contracts and agreements, the companies in Cluster 1 aim to avoid negative impacts related to deadlines, efficiency, and conflicts between areas.

In addition, it is essential to note that verticalization aims to expand the production of services and products, be they structure, equipment, information, technology, and people training. In addition, the verticalization of operations seeks to avoid coordination and negotiation problems arising from dependence on third parties (Alchian & Demsetz, 1972).

4.1.4 Outsourced Services

Outsourcing or "via market" is a concept that contrasts with verticalization. Instead of expanding its operations internally, the company chooses to transact in the market, seeking external partners to supply the goods, services, or resources necessary for its activity (Christensen, 1997; Williamson, 1985). The results of the companies that pursue this management model are shown in Figure 10.

Figure 10 – Outsourcing



The results in Figure 10 reveal that companies "outsource" more Government Services than others services, which may indicate that these companies has a broader outsourcing policy, whether compulsory or facultative. One of the most explicit motivations is services involving financial activities (financial institutions and the provision of financial services), a regulated segment considering its impact on the economy and society in general, which makes companies depend on government regulations to operate. The government, in addition to the recognized role of oversight, can act as an encourager and even accelerator of innovation, involving partners interested in sharing initiatives that will expedite solutions for society (OECD, 2018), bringing in financing agents such as entrepreneurial capital, crowdfunding, start-up, labs, and angel investors.

A subtle percentage difference was identified between the groups concerning the outsourcing of public services. As a result of this observation, a question related to this topic was incorporated into the interview script during the qualitative phase of this research. This inclusion made it possible to investigate the underlying factors influencing decisions to outsource government services.

4.1.5 Motivations (and incentives) to Outsource

Furthermore, as well as exploring the motivations to internalize activities (GT International Business Report, 2014), the research sought to identify the justifications that led to the choice of outsourcing. For this, the following were listed:

(i) Get expertise in the market;

(ii) Reduce costs;

(iii) Reduce uncertainties;

(iv) Outsource the risk;

(v) Strategic planning;

(vi) Expand partnerships; and

(vii) Comply with regulators or legislation.

In a general analysis of this result among clusters, the research revealed that reducing costs, obtaining expertise in the market and, complying with legislation are the main criteria for deciding for outsourcing activities, as shown in Table 3.

Table 3 – Outsourcing Motivations

	Cluster 1	**Cluster 2**	**Cluster 3**
Reduce costs	96%	97%	93%
Get expertise in the market	75%	84%	76%
Comply with regulators or legislation	75%	84%	73%

These motivations received positive responses and proved to be valued by the companies when outsourcing. The high percentage of positive responses in all the clusters suggests a consistent trend across the entire sample of companies, highlighting the importance of these criteria when making outsourcing decisions.

Each company has its priorities and objectives, and the decision to outsource activities can be based on a combination of factors, including financial gains, operational efficiency, and competitive advantage. Many managers seek to optimize activities and reduce costs they consider unnecessary through SG&A (Selling, General, and Administrative expenses) management, which encompasses the operating costs related to the company's management and administration (Oyadomari et al., 2018).

On the other hand, uncertainty reduction, risk outsourcing, strategic planning, and the expansion of partnerships received lower scores. Although they were considered, their lower position indicates that the companies did not prioritize them for outsourcing.

In this sense, if the reduction of uncertainty is considered a minor factor for the decision to outsource, it may be because companies are more interested in other advantages such as cost reduction, access to specific expertise, and even transferring risk to third parties.

In the literature, it is widely discussed that if the transaction costs associated with outsourcing (such as negotiation, monitoring, and coordination costs) are lower than the costs of in-house production, outsourcing may be a more efficient choice for the company. This does not mean that outsourcing is a universal solution for cost reduction. A joint analysis of the benefits, information asymmetries, uncertainties, and risks inherent to business, including those involving the value chain, is linked to strategic decision-making and the company's long-term sustainability (Gallouj, 1997). Even outsourcing can be a strategy for these companies to seek to establish institutional mechanisms (such as reputation and certification) due to the pressure from customers seeking reliable signals of quality (Gallouj, 2023).

So much so that, Ménard (2004) reinforces Williamson's (1996) understanding by stating that in market economies, legal contracts represent one of the most important safeguards for the parties, whose importance is based on five causes: (i) bilateral dependence, (ii) measurement problems, (iii) changes in time terms, (iv) inability to respond in real-time, long latency, and strategy abuse, and (v) risks that lead to weaknesses in the institutional environment.

Moreover, when we return to the results of outsourcing (Table 3) and evaluate the justifications for doing so, in which government services were the most outsourced by all the companies in the sample, it can be inferred that outsourcing of these services may be limited when evaluating cost reduction strategies. In this sense, the service activities that ranked second may provide specific perspectives, although they may not be the majority of activities adopted by companies seeking solutions via the market.

In this sense, auditing activities are a specialized service provided by an independent auditing firm and have adherence in the companies that make up Cluster 1 (34%) and Cluster 3 (33%). This makes sense when these companies have core activities related to specific regulations and standards, such as financial activities, e-commerce, logistics, and software development, and need to rely on expertise through the market, including independent audits, to meet accounting and financial standards and internal control systems.

4.1.6 Professionals from Diverse Areas: The Wealth of Multidisciplinarity

Identifying the position of professionals and departments involved in cyber risk mitigation supports understanding how risk mitigation processes and activities are organized and executed. Understanding the differences in each area can contribute to understanding their specific needs and adaptations.

The survey considered technical and leadership positions to assess cyber risk perception. The choice of professionals from different areas aims to obtain a multidisciplinary and broad view, considering that many professionals need to not only plan but also execute. The indicators presented show the percentage distribution of positions in each cluster and the percentage of professionals who work in specific areas such as finance, compliance, information security, and IT.

Table 4 – Position and Area of Operation

	Cluster 1	**Cluster 2**	**Cluster 3**
% Leadership	93%	81%	76%
% Controllership	25%	19%	13%
% Information Security and IT	19%	27%	20%
% Compliance	16%	N/D	20%
% Financial Area	13%	11%	16%

In an organization, different positions may have different levels of knowledge about risks and inherent risk management strategies. For this, an overview was sought to reveal the profile of the participating professionals.

The data in Table 4 reveals that most companies across all clusters have leadership positions in critical areas, including Controlling, Information Security and IT, Compliance, and Finance. Leadership roles account for 93% in Cluster 1, 81% in Cluster 2, and 76% in Cluster 3. Conversely, technical positions are less common in all clusters, with Cluster 1 having the lowest representation at only 7%. Examining specific functional areas within these companies, it becomes evident that the Controlling function is prevalent across all clusters. However, Information Security and IT also exhibit prominence, with Cluster 2 boasting the highest percentage at 27%.

In Cluster 1, where most of the core business and peripheral activities are Digital, there is a significant proportion of in-house development, and a high proportion of leadership positions are observed. This suggests that the company invests in a management team that

focuses on allocating resources and making decisions that affect company performance (Foss & Klein, 2022), including cyber risk management. However, the percentage of leadership and technical positions does not necessarily mean that the companies surveyed have more or fewer managers. Moreover, the presence of specialized information security and IT professionals indicates that the company is taking specific measures to mitigate cyber risks.

Interestingly, Cluster 2 did not produce any measurable results in Compliance positions, suggesting that these companies do not prioritize a Compliance area in their business or may have a different approach to managing risks.

In addition, the data obtained suggest that Information Security is present in several areas of the companies, with leadership roles dedicated to these areas. While there are differences in specific areas, such as Compliance and Finance, the prevalence of leadership roles in Information Security and Controllership indicates an awareness of the importance of this area in business.

4.1.7 Technology Investment

In addition to obtaining an overview of the professionals' areas of activity, we sought to understand how they and their respective companies are investing in technology based on the following question: "In the last three years, has your company acquired machines, equipment, patents or software based on: (i) the same technology used before and /or (ii) new technology not used before?" Considering the raw result of the obtained data, it was identified that 34.3% of the companies kept their original strategies by acquiring machines, equipment, patents, or software based on the technology already used. In comparison, 65.7% of the companies acquired new technologies not previously used. In Table 5, we bring these results segregated by clusters, which will allow us a more comprehensive description:

Table 5 – Technology Investment

	Cluster 1	**Cluster 2**	**Cluster 3**
New Technology	63%	66%	70%
Same Technology	37%	34%	30%

While Table 5 illustrates a high adherence rate to new technologies among companies in Cluster 3, with an impressive 70% adoption rate, it is essential to note that the difference in new technology adoption between the three clusters is relatively modest. This suggests that, despite the slightly higher adoption rate in Cluster 3, there is a general trend among all clusters

towards incorporating new technologies into their operations. This uniformity in technology adoption underscores the industry's recognition of the importance of staying up-to-date with technological advancements, with only subtle variations between clusters. This result suggests that these companies have assumed an early adopter position by seeking innovation cycles to compete better and advance their business models in the market (Rogers, 2003).

Moreover, Cluster 3 has Commercial Services as its dominant core business, with its own and outsourced wholesale and retail activities also significant. Therefore, this result may reflect the effort of these companies to gain a more significant position in the market, either by meeting customers' demands or by seeking greater competitiveness. While Cluster 1, in which the companies in this group are predominantly active in Digital Services, indicated less investment in new technologies.

Concerning investment in technologies already in use, two aspects must be considered: maintenance and inertia in investing in new technological initiatives. Technological maintenance can bring benefits, making these companies specialized or knowledgeable in specific technologies, which can improve their business model. However, when considering the aspect of disadvantage, i.e., technological inertia, depending on the context of the companies, they may suffer greater vulnerability to new technologies leading them to be reputed as "laggards" (Moore, 2014; Rogers, 2003), i.e., slow to adopt new technologies.

4.1.8 Technology Dependence and Mediation

The exploration of this research involved not only identifying whether companies have been investing in technology in their business, but they also rely on this resource to provide or contract services. A company's dependence on technology allows for a better understanding of the impact of technological changes on its business model and the ecosystem in which it is embedded.

Therefore, addressing examples such as data protection, cybersecurity, cloud computing, and blockchain, the following question was asked: In which process does your company depend on technology services to develop and provide services?

The companies in Cluster 1 demonstrate a high dependence on technology to hire and provide services concerning the other groups. The unrepresentative return (close to zero) of Clusters 2 and 3 demonstrates that the companies in these groups may not have a dependence on technology in these business stages, or a possible lack of knowledge of the use of the

functionalities available in their systems leads them to underutilization of resources, including technological resources.

4.1.9 Economic Growth Initiatives: Physical and Digital Environment

Cyber-attacks have become significant risks in the service economy since the confidentiality and integrity of data and information in cyberspace have received insufficient attention (Norton, 2020). Technological evolution and the growing dependence on digital systems have highlighted cyber risks as critical factors in strategically analyzing companies' economic performance (Chainalys, 2023; CISA, 2021). In this context, participants were asked which initiatives had been prioritized to improve economic performance over the last three years and in which physical, digital, or both environments these initiatives had been implemented. These initiatives were adapted from the CIS European Commission (2018).

In order to present the results, we adopted two methods: first, an analysis of the initiatives' adherence within each cluster, and second, the organization of all growth initiatives, categorizing them as physical, digital, or both, as illustrated in Figure 11.

Figure 11 – Economic Growth Initiatives

Environment	**Iniciatives**
Digital	-Software development, database work, and data analysis. -Investments to expand or enhance reputation in marketing, brand building, and advertising -Reduce lack of knowledge or incomplete information in decision making in my company and/or its value chain - Optimize/reduce the processing time of activities, whether internal or with third parties (suppliers, customers, or government) - Introducing new products or services -Development of specific products and services to meet your customer's needs -Reach out to new customer groups - Development of products and services in partnership with companies (including competitors) -Development of specific products and services by the supplier to meet your company's needs. -Offer to the customers an active role in the creation of the idea and design of products and services -Focus on one or a small number of key goods or services - Reconfiguration of physical spaces, as well as flexibility to adopt new ways of working
Physical	-Reconfiguration of physical spaces, as well as flexibility to adopt new ways of working -Investments directed to better localization (to reduce logistics and space costs) -Reach out to new customer groups.

Among the 175 companies that comprise the three clusters, there was a limited overlap between environments. In other words, classifications were primarily distinct for the physical and digital environments, except for the initiative 'Reconfiguration of physical spaces, as well as flexibility to adopt new ways of working'.

An emphasis on technology and data analysis comes into view. Companies seem to invest in software development, database management, and data analysis, signifying their recognition of the digital landscape's reliance on technology and data-driven insights. Moreover, the significant investments made to expand and enhance the company's reputation in marketing, brand building, and advertising indicate a dedication to establishing a digital presence and reputation. Such investments suggest a recognition of the importance of standing out in the competitive digital landscape to attract a more extensive customer base.

Furthermore, the significance of software development, database work, and data analysis initiatives suggests a commitment to addressing the challenge of incomplete information. By investing in these areas, companies can signal their intent to overcome knowledge gaps, employing tools and data analysis to bolster their decision-making processes.

In addition, the proactive quest for new customer groups within the digital sphere suggests an expansion strategy. This approach subtly hints at identifying previously untapped niches or demographic segments. Introducing new products or services suggests that companies aim to expand their market presence and revenue sources by offering new digital services to attract and retain customers, promoting business growth. Furthermore, alongside the introduction of new products or services, it is implied that companies are strategically adapting to the digital landscape. The reconfiguration of physical spaces and embracing flexible working practices allude to their unwavering commitment to digital competitiveness.

Developing specific products and services in partnership with companies (including competitors) is an initiative that reflects a collaborative approach in the digital environment and may even encompass coopetition. Companies are demonstrating that they are seeking strategic partnerships, including with competitors, to co-develop products or services that meet the needs of the digital market. This can promote innovation and the joint creation of solutions.

Furthermore, the customer-centric approach, actively involving customers in co-creating products and services, suggests a commitment to personalized offerings and building a loyal customer base. This approach hints at the potential benefits of customer engagement in shaping the company's offerings. Additionally, focusing on a limited set of key goods or services would imply a strategic direction in which companies allocate their resources and

efforts. This strategic emphasis alludes to their confidence in competitive advantages and potential growth in those specific areas.

Concerning business strategies in the physical environment, the participants focus on optimizing physical resources, flexibility in adopting new working approaches, and expanding the customer base through traditional methods.

Targeted investments in better locations to reduce logistics and space costs suggests that companies focus on optimizing their physical presence. Investments in strategic locations can help reduce logistics costs and minimize expenses related to physical space. This may reflect a cost-efficiency strategy, where companies seek to improve their physical presence for operational effectiveness.

Reaching new customer groups suggests that companies are adopting traditional or physically present-based approaches to expand their customer base. This may involve strategies such as opening new branches, in-person events, advertising in physical locations, or any actions involving face-to-face interactions with customers.

Two initiatives in common in both the physical and digital environments are 'Reach out to new customer groups' and 'Investments directed to better localization' to reduce logistics and space costs.' These strategies demonstrate the convergence between the two environments, indicating a hybrid approach.

Therefore, a descriptive analysis of the results obtained for the groups of companies identified Cluster 1 as the group with the most significant involvement in initiatives that seek business growth through the economic performance strategy, with 91% adherence. Initiatives in the digital environment indicate an emphasis on innovation, collaboration, market expansion, and technology adoption. Companies are concerned about keeping up to date and adapting to the digital environment in order to remain competitive. As far as Cluster 2 is concerned, the companies showed low adherence to the economic performance strategy, with results close to zero in most of the initiatives aimed at economic growth. In percentage terms, Cluster 2 companies achieved less than 3%. This may indicate that these companies have a conservative approach to growth, seeking to stand out from competitors and possibly focusing on ensuring operational and cyber security in all areas of their business. This may explain the greater focus on administrative activities concerning their core business and peripheral activities (Figure 6; Figure 7, Section 4.1.1). Cluster 3 achieved a percentage of 6%, although it also shows a timid approach from a strategic point of view, similar to the companies in Cluster 2. Therefore, after the descriptive analysis of the results obtained with the characterization of the companies, the organization of activities, and the choice of governance structure, the next topic will address

how the company manages and implements measures (actions or strategies) and mechanisms (systems and devices) that contribute to cyber risk mitigation.

4.1.10 Risk Mitigation

Organizations worldwide have been challenged by confronting and demanding knowledge about cyber security (Strupczewski, 2021). So much so that cyber risk has been cited as one of the most important sources of operational risk in organizations, and discussions have increasingly grown about the events in which this type of risk is embedded (Egan et al., 2019).

As it holds market attention from the accelerated digitization of the economy and social relations, cyber risk is amplified by the combination of probabilities that exist in an organization, from "assets, resources and communication between these that trigger and impact an organization's processes", Bank for International Settlements – BIS (2016). Cyber risk cannot be limited to the Internet, hardware, software, and information systems (Silva & Nogueira, 2019). However, it should be considered the integration of people and their social interactions in computer networks (Klimburg, 2012).

Therefore, the goal of this section is to explore how companies manage cyber risks in their business and to do so, and we will use the survey results as the basis of analysis. Through the question: What are the means used to manage and mitigate risks (in general) in your company*?* For this, management, monitoring, and mitigation methods were listed (Fox, 2018; Strupczewski, 2021) so that participants could point out which of them are part of the risk management process in their companies:

(i) Standardization and integration of operational processes
(ii) Use of technology
(iii) Training and cross-functional teams to identify risks
(iv) Job rotation planned in different functional areas
(v) Control and monitoring of indicators
(vi) Hiring companies specialized in risk management

This question aims to understand the company's risk management culture and how it relates to cyber risks. It is important to note that many companies may not use the specific terminology 'cyber risk' but may still be involved in practices related to information security and operational protection against digital threats. The results showed that companies in Cluster

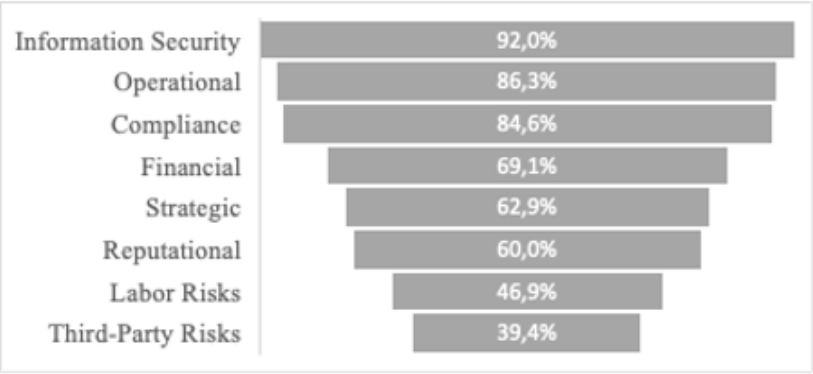
1 adhere highly to the following risk management and mitigation methods: standardization and integration of operational processes, use of technology, and planned rotation of functions in different functional areas.

Regarding Clusters 2 and 3, the results were not very expressive. All clusters showed low adherence to risk identification measures such as training and cross-functional teams to identify risks and hiring specialized risk management companies. It is possible that the companies in these clusters are not fully engaged in bringing in security measures, even though the absence of these methods can expose them to risks, and instability of work processes, besides negatively affecting the reputation and the financial results of these organizations.

Another point of investigation in the questionnaire was to analyze whether companies are adapting their internal norms and policies to keep up with evolving business models. Policies, procedures, and internal rules become relevant in this scenario since the cyber environment profoundly changes how companies operate and adapt to new business models. The results indicated that the companies that makeup Cluster 1 stand out from the others in this aspect. This corroborates the results obtained by the economic growth initiatives adopted by the companies in this cluster (Figure 10 above).

Based on the premise that risks are interconnected and should be managed in an integrated manner to ensure the organization's security (Rossi, 2017), the question was asked: “What risks does the company manage that positively contribute to mitigating cyber risk?”. The participants were presented with a list of risks to select the options that best matched their practices, such as operational, strategic, compliance, financial, reputational, information security, labor, and third-party risks. The results presented in Figure 12 represent the data set's overall analysis without differentiation or cluster stratification.

Figure 12 – Cyber Risk pre-management



The survey found that Information Security risk management (92%), Operational Risk (86.3%), and Compliance risk management (84.6%) most support companies in addressing their most critical business risks and protecting themselves against the rapidly evolving threats in the cyber environment.

The Information Security risk scored significantly high, with 92% adherence. The management of this risk covers the protection of a company's systems, networks, and data against internal and external threats. This involves implementing appropriate security measures to mitigate security risks, such as encryption, access control, authentication, monitoring, and threat detection. In this sense, Information Security risk management supports identifying and assessing process and systemic vulnerabilities in a cyber threat mitigation implementation.

Regardless of the strategies adopted to manage failures in controls, systems, as well as other uncertain events that impact the bottom line, it is recognized that Operational Risk is connected to how the organization operates, acts, and decides its internal control process (Fox, 2018; King, 2001; Lam, 2014). In identifying risks that support cyber risk management, Operational Risk scored 86.3%. Operational Risk is distinguished as the Risk arising from external events, failures, deficiencies, and internal processes, people, and systems inadequacies. So much so that operational risks have been recognized and originated from numerous operational failures. They have grown exponentially with the advent of global connectivity and its extension through value chains (Brazil Central Bank [BACEN], 2017; Cebula & Young, 2010).

With 84.6% participation, the risk of non-compliance refers to non-compliance with the laws, rules, and regulations applicable to an organization. This may make sense when combining this type of management with mitigating information security risks and operational risks after implementing the GDPR (General Data Protection Regulation, 2018) and LGPD (Brazilian General Data Protection Law, 2020), whose legislations value information security standards. Moreover, an ineffective compliance program tends to negatively impact the mitigation of legal and reputational risks and other standards linked to compliance, such as labor, environmental, and financial regulations.

Interestingly, Strategic Risk obtained 62.9% of participation. This type refers to the risk that the organization's strategy does not work as expected, which can lead to market loss and loss of revenue, but can also have impacts in other areas, such as reputation, human resources, and relationships with customers and suppliers, in addition to decisions and strategies linked to the cyber environment, considering that most companies have expanded their operations in

Digital Services, more specifically in operations in the cyber environment, launching products and services that meet the growing demand.

Furthermore, the result revealed that 60% of the sample companies consider that reputation risk can become a measure that supports shielding in the cyber environment where misinformation, false information, and fraud can spread rapidly through the digital environment.

Labor Risks can become more complex in a cyber environment, especially when protecting people who work in remote or hybrid working mode. In this type of risk management, the participation was 46.9%. Furthermore, we can combine the analysis of this result with that of Risks with Third Parties, which took the last position in the sample, with a score of 39,4%.

Both Labor Risk and Third-Party Risk can manifest even more significantly in a digital environment, where organizations often rely on various suppliers and partners to deliver essential products and services. This can include exposure to security vulnerabilities in third-party systems, regulatory compliance risks associated with third-party contractors, and the potential for third parties to misuse confidential company data, even in the company's physical or digital environment.

In relation to the implementation of mechanisms to mitigate cyber risks, as outlined by Böhme et al. (2018) and Strupczewski (2021), participants were asked the following question: 'Does your company have measures to identify and mitigate risks in its activities?' To facilitate their response, a list of mechanisms was provided in the questionnaire, sourced directly from the works of these authors. This allowed participants to indicate which of these mechanisms are implemented by their respective companies:

(i) Phishing[7]

(ii) Data Leakage and Data Theft

(iii) Vulnerability of third-party employees

(iv) Vulnerability in third-party software

(v) Internal fraud

(vi) Violation of privacy

(vii) Use and access to systems and information by your company's employees

(viii) Hacker attacks

[7] *Physhing* it is one of the types of social engineering. Phishing is an attack using e-mail or malicious websites to solicit personal information by posing as a trusted organization (IBM, 2021).

The results showed that only Cluster 1 presented mitigating measures, such as vulnerability monitoring regarding social engineering techniques[8], phishing, fighting internal fraud, privacy violations, and hacker attacks. The other clusters showed non-representative results.

Although Cluster 1 companies invest in cybersecurity mechanisms and measures, they are vulnerable to attack attempts, so much so that they obtained the second highest percentage among the other clusters, when 25% of these companies assumed that they have already suffered financial, reputational, and operational losses arising from these cyber-attacks (Table 6). This may indicate that criminals target the business model of these companies or that their processes can generate vulnerabilities if they are not properly structured and monitored.

Table 6 – Losses and Financial Capability

	Cluster 1	**Cluster 2**	**Cluster 3**
Core Business Services #1	Digital	Digital	Commercial
Core Business Services #2	Knowledge	Administrative	Digital
Losses due cyber attacks (%)	25,00%	12,90%	31,10%
Annual revenues in U$ (%)	11,80%	5,90%	14,70%
1 billion – 10 billions	2,90%	2,90%	10,30%
> 10 billions	8,80%	2,90%	4,40%

Organizations worldwide have been challenged by coping situations such as the abovementioned risk mitigating actions. However, this survey also revealed the financial, reputational, and operational losses related to cyber threats: 22.3% of the companies claimed to have suffered financial, reputational, and operational losses due to fraud, data leakage, disruptions, and hacker attacks.

Notably, companies that reported having suffered financial, reputational, and operational losses due to cyber-attacks account for 16.2% of those with annual revenues between $1 billion and $10 billion, while another 16.2% have revenues above $10 billion. Still, most companies penalized by attacks belong to Cluster 3 (31.1%) and Cluster 1 (25%). These data indicate that even companies with significant economic capacity can be subject to challenges related to fraud, data leakage, disruptions, and hacker attacks.

[8] *Social Engineering*: the attacker uses human interaction to obtain or compromise information about an organization or its systems. Cybercriminals try to exploit the individual's lack of knowledge (Cybersecurity & Infrastrucutre Secururity Agency [CISA], 2021)

Another essential point is that Cluster 2 companies offer Digital Services as their core business, just like Cluster 1 companies. However, they scored the lowest incidence of cyber-attacks (12.9%). This difference can be justified by the second largest core business of Cluster 1 companies: Knowledge Services. Companies that develop Knowledge Services such as auditing, IT development, and consulting, for example, can be particularly attractive for cyber-attacks because they can access their customers' confidential and valuable information.

Table 7 presents other questions to understand how companies address cyber risks comprehensively. The questions sought to explore strategies, practices, and policies adopted by companies to address the challenges arising from the growing threat of attacks on their systems and data security. In this result, a general analysis of the dataset is presented without differentiation or cluster stratification.

Table 7 – Risk Approach

	Cluster 1	**Cluster 2**	**Cluster 3**
Does the company disclose investments related to cybersecurity?	39,7%	35,5%	31,1%
Have cyber risks grown at the same level as opportunities for businesses?	77,9%	74,2%	84,4%
Does your company promote and encourage discussion about cyber risks among your company's employees?	91,2%	75,8%	71,1%
Are the legal mechanisms (legislation and inspections) able to satisfactorily cover and protect the management of cyber risks in organizations?	94,1%	88,7%	84,4%

Most companies do not disclose investments in cybersecurity. Considering the percentages obtained (Table 7), it leads to the inference that companies that adopt the practice of non-disclosure do not invest enough to protect their data and systems or that there is a lack of transparency and communication regarding the actions to manage cyber risks. Alternatively, a possibility could be that companies adopt the lack of transparency as a strategic decision. They may be targeted by malicious people, including third parties and their employees, who may opportunistically exploit the disclosed information.

Whether it is a strategic decision or not, the issue of non-disclosure to third parties can be corroborated by 6.8% of the participants who justified their non-participation in the survey by considering that the questions in the questionnaire were too related to the strategic positioning adopted by their companies. This percentage is likely higher since not all professionals justified their refusal to participate in the survey.

Furthermore, financial, reputational, or operational losses due to cyberattacks may be more frequent than be aware of, as not all companies provide information about their investments in security measures and protocols. Including the percentage of losses presented in this research (Table 6 above).

When asking the participants: “Does your company promote and encourage discussion about cyber risks among employees in your company?”. Companies mentioned promoting and encouraging discussion about cyber risks among their employees (Cluster 1 stood out with 91,2%), which suggests that they are concerned about increasing awareness and knowledge about risk management among their teams. Regarding those companies that do not adopt this practice, one can lead to infer several motivations, such as:

(i) that these companies may not be communicating adequately about the relevance of cyber threats due to a lack of knowledge among executives.

(ii) lack of knowledge of standards and policies that refer to the lack of communication; and

(iii) need for employee interest and engagement.

In addition, most participants recognize that legal mechanisms can satisfactorily cover and protect cyber risk management and support the initiation of innovation activities. This answer resulted from the question: “Are the legal mechanisms (legislation and inspections) able to satisfactorily cover and protect the management of cyber risks in organizations?” It may be a point of investigation to explore the audience that does not share this perception:

(i) have limited or erroneous knowledge of laws and regulations.

(ii) believe that the enforcement of laws is insufficient for cybersecurity and hinders the initiation of innovation in their processes.

(iii) may be skeptical of the capability of the government or regulatory authorities to create, monitor and enforce laws related to cyber threats.

Based on the results presented, cyber management is still a critical issue for many companies. Although most respondents believe that their company promotes discussions about cyber risks among its employees and that legal mechanisms can satisfactorily protect cyber risk management in organizations, there is still a percentage to be explored that does not share these opinions. This suggests that companies may be underestimating the seriousness of cyber threats or that they are not implementing sufficient measures to protect their information and systems from these threats.

4.1.11 Contractual Governance

Contracts are incomplete. Contractual incompleteness in Transaction Cost Theory can be considered "strong" because it has another source: institutional failure (Hart & Moore, 1988; Williamson 1985). An economic implication of contractual incompleteness is that parties are vulnerable to calculated efforts by others to errors, doubts, deceive, renege, cheat, or take advantage of the vulnerabilities of their trading partners. In this regard, the institutions that are ultimately responsible for ensuring the enforcement of contracts cannot enforce those clauses that concern unverifiable variables (Brousseau & Glachant, 2002; Macher & Richman, 2008; Williamson, 1991).

The results obtained in this research have indicated the high level of transactions involving digital technologies and systems and by understanding how companies have conducted contractual terms, it is possible to identify the risks involved in using digital technologies and the obligations of the parties involved in managing these risks. To this end, questions directed at this theme were conducted to complement a view of the characterization, mitigation mechanisms for cyber risks, and the contractual forms for managing these risks.

To enable the participants to answer this topic in an organized manner, it was recommended that they indicate the service categories previously indicated in their characterization (Figure 6): commercial, digital, financial, government, in-company (administrative), knowledge, and personal. The questions were asked to identify the types of services established in contracts and to determine with which party the contract was signed - whether with the client or the supplier. In addition, the periodicity of the contracts was also investigated through these questions (spot, short, medium, and long-term, as well as indefinite contracts and situations in which the contractual relationship is not established).

Table 8 – Types and Contract Periodicity

Cluster	Services	Public	Frequency
1	Digital, commercial and financial	Suppliers	Medium-long term
2a	Knowledge, in-house, commercial, and financial	Suppliers	Medium-long term
2b	Digital, knowledge, commercial and financial	Clients	Medium-long term
3	N/A		

The Digital, Commercial, and Financial services formalized by medium and long-term contracts with suppliers (outsourcing) are characteristics and contractual formats most commonly practiced by the companies that make up Cluster 1. It is important to note that this

cluster stands out for the central practice of Digital Services as a core business (financial and e-business activities) and Government Services as outsourcing (public systems activity), whose financial market is regulated by its laws and institutions.

Cluster 2 companies establish formal medium to long term contracts for their Knowledge and Administrative services. While Knowledge services are outsourced, seeking expertise via the market, administrative services are developed internally, via the hierarchy. This reinforces the understanding that the companies in this group recognize the importance of internal administrative services as a risk management tool through well-defined contracts, in addition to considering these services as part of their core business and peripheral activities (Figures 6 and 7).

The results obtained in Cluster 3 were insufficient to justify adopting contractual governance practices, so the Table is N/A. This shows that there are opportunities to investigate the causes of the low adherence of companies in this cluster concerning risk management through contractual means.

Furthermore, among the contractual phases, the participants pointed out which one with clients and suppliers, they consider to be the most vulnerable and which requires the most attention from the company. The contractual phases presented were:

- Pre-contracting: identification of objectives and clarifications for entering a contract.

- Contracting: formalization of the contract, as well as the negotiation and agreement between the parties.

- Pre-implementation: technical preparation for the treaty to be received, interpreted, and applied.

- Implementation: period of the contract, there will be a need to follow up on its fulfillment).

- Contractual closure: the planned term comes to an end, and thereafter the obligations are ceased.

The companies in Cluster 2 brought significant results indicating that the pre-contracting phase is the most vulnerable and requires the most caution. It is inferred that the vulnerabilities perceived by the professionals in the pre-contracting phase are linked to the management and sharing of risks between the parties, expectations of achieving the object of the contract, in addition to the preparation of the design or well-defined scope of the production and delivery of the object of the contract.

Contractual weakness in this first contractual phase can lead to a "cascading effect" of problems and conflicts between the parties during the execution of the contract. *Ex-ante*

vulnerabilities can arise when the parties do not have complete information about the possibilities (Brousseau & Glachant, 2002; Milgrom & Roberts, 1992). Jiang et al. (2020) revealed that obtaining stocks with higher cyber risk affects and raises the *ex-ante* (before actually performing the transaction or contract) cost of equity capital.

Pre-contractual phase caution can be justified when the parties involved cannot foresee all possible contingencies and situations that may arise during contract execution or even whose incomplete contractual scope or specification may create malicious incentives for the parties, encouraging opportunistic behavior and leading to monitoring and enforcement problems (Klein, 2002).

Another essential aspect investigated was the question about sharing cyber risk responsibility among partner companies to get a broader picture of the contractual governance of the participating companies. Most of the clusters (average 77.6%) claimed that the company shares cyber risk with partner companies, such as contracting for space and project sharing, dedicated vendors, economies of scale, and open innovation; such examples connect with Digital Services.

In addition, another dimension of risks to be explored was to obtain the aspect of trust in hiring third parties - which involves behavioral assumptions, suppliers, customers, and partners. For this, the following question was asked: “Can the trust developed by social and cultural proximity be considered essential in selecting and hiring third parties in your company” The answers obtained were organized in Table 9.

Table 9 – Trust and Social Proximity

Cluster	Yes	No	Partially
1	34%	29%	37%
2	52%	15%	34%
3	44%	27%	29%

This question is introductory to a classic question in social theory: whether human behavior and institutions are affected by social relationships. In this sense, Granovetter (1985) criticizes Williamson's view that economic transactions are strictly driven by contractual and market logic, arguing that social ties and personal relationships strongly influence economic behavior.

Considering the cumulative results of the "Yes" column (Table 9 above), trust developed between parties through cultural proximity is essential to follow in contracting. However, in an

isolated analysis, companies in Cluster 1 tend not only to represent the lowest number (34%) but also partially believe in this element of contracting (37%).

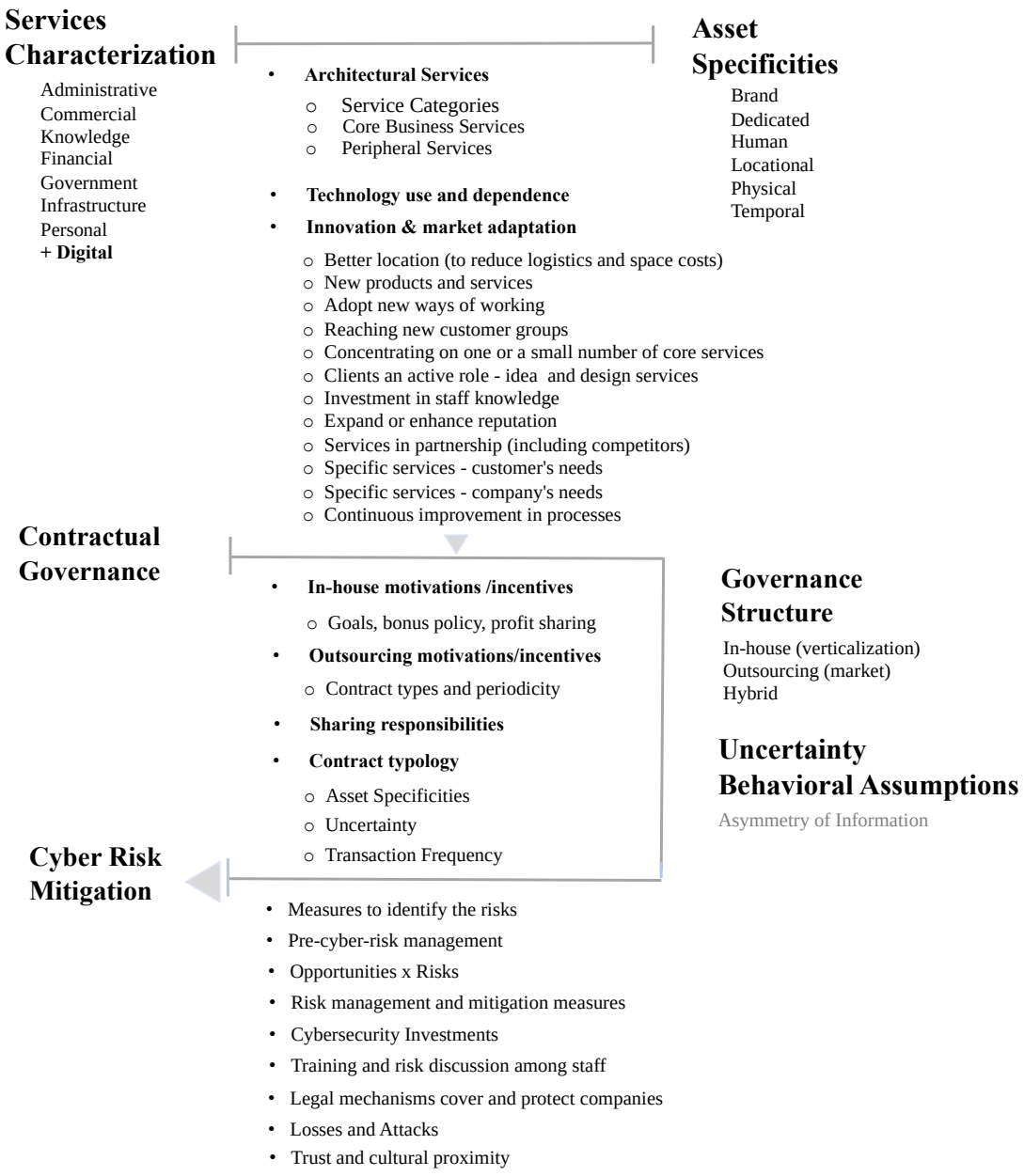
The results of Table 9 (trust and social proximity to perform contracts) combined with Table 8 (types and periodicity of contracts signed); it can be inferred that there is an impact on the contract hold-up[9]. However, it is not guaranteed only by establishing medium-long-term contracts and the trust criterion. Other factors must also be considered, such as contractual complexity involving high asset specificity, the possibility of changing circumstances, and information asymmetries (Akerlof, 1970; Brousseau & Glachant, 2002; Hart & Moore, 1988; Williamson, 1985).

4.1.12 Services Cyber Risk Mitigation Framework – CyberFrame

The construction of the Services Cyber Risk Mitigation Framework (CyberFrame) enables a broad and systematic understanding of the challenges faced in the cyber environment. This initiative aims to support the identification and categorization of the various elements and aspects that make up cyber risk, providing a comprehensive view of the phenomenon. In this way, understanding the causes, impacts, and mechanisms underlying this type of risk can be gained, supporting companies in making strategic decisions related to economic growth and fostering innovation. Furthermore, this framework can be a tool to analyze the choice of governance structure and risk mitigation and identify market opportunities that can be safely and effectively exploited.

[9] Hold-up can occur when one party takes advantage of long-term investments made by another party in a contractual relationship (Brousseau, & Glachant, 2002)

Figure 13 – Services Cyber Risk Mitigation Framework



Source: author (2023)

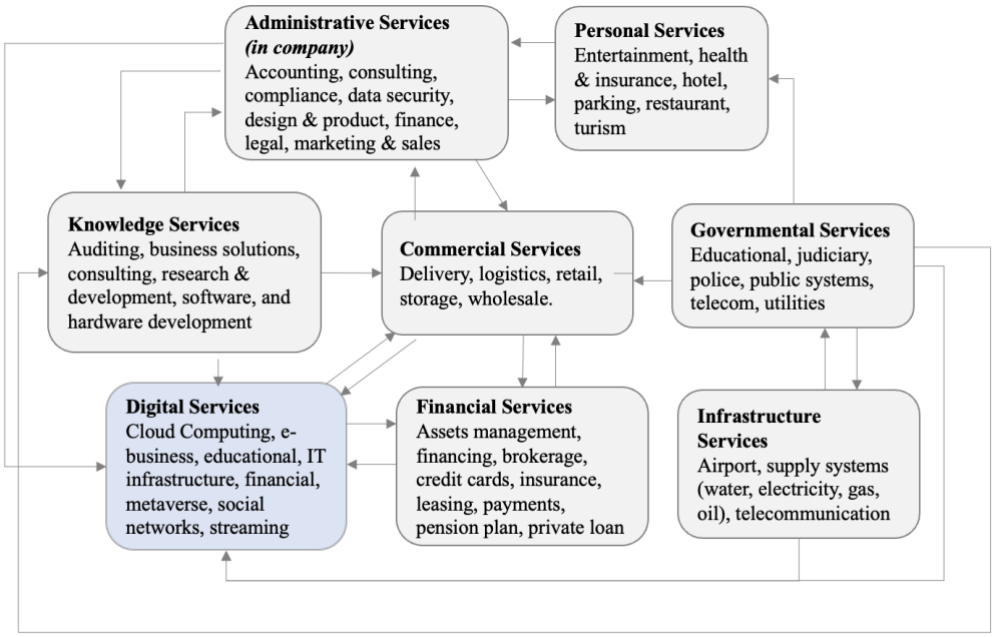
Using the framework, it was possible to outline the thematic axes that will guide the construction of the interview script for the qualitative approach: (Thematic Axe 1) Organization of Activities and Flow of Knowledge in Services (Services characterization and Asset Specificities), (Thematic Axe 2) Contractual Governance, and (Thematic Axe 3) Cyber Risks Mitigation.

Understand the characteristics of the organization of activities, the flow of knowledge, and the characteristics of the services transacted: Companies have explored

expansion opportunities in search of competitiveness, aiming to increase their market share and achieve the desired financial result. However, it is crucial to recognize that this growth brings several risks related to business activities. Therefore, identifying and managing cyber risks is critical for those operating powerfully in the digital environment. In addition, the appropriate choice of governance strategies, whether by internalization or outsourcing, plays a crucial role in mitigating these risks.

Based on the results of this phase of the study, and also considering the contributions provided by the participating companies, it was possible to develop the structure (Figure 14) to provide a comprehensive view of the service sectors that exist in the economy, as well as their connections and interdependencies.

Figure 14 – Service Economy



Note: A service modality has been introduced in the Service Economy literature, which offers a new approach to Digital Services. This addition was included in the model proposed by Soete & Miozzo's Taxonomy (2001)[10] and Bordoloi, Fitzsimmons, and Fitzsimmons (2019) [11].

During the first phase of the research, a significant update was made with the inclusion of the Digital Services category and other service activities related to different categories in the model proposed by Bordoloi et al. (2019), aligning it with the 'Typical Core Sectors' of the Sectoral Technological Taxonomy of Services by Miozzo & Soete (2001). This update provides information on the characterization of digital services traded in the economy. In the second

[10] Miozzo, M., & Soete, L. (2001). *Internationalization of services: a technologic perspective*. Technological Forecasting and Social Change, 67(2-3), 159-185.
[11] Bordoloi, S., Fitzsimmons, J., & Fitzsimmons, M. (2019). *Service management: Operations, strategy, information technology, 9e*. New York: McGraw-Hill Education.

stage of this study, it will be possible to delve deeper into the information provided by the interviewees, allowing us to better explore these digital services in the context of service innovation.

Cyber Risk Mitigation: Cyber risk is a relatively recent scientific debate, while its plurality and complexity have marked a colossal shift in the scenarios of cybersecurity and cyber threats. Under a broad concept used of multidisciplinary action, cyber risk covers areas such as computer science, engineering, business management, economics, and social sciences (Strupczewski, 2021). Therefore, it is necessary to start from the risk framework so that cyber risk can then be better explored.

Interactions with partners, suppliers, and even among employees can also introduce cyber risks. This includes assessing the cyber security of business partners and implementing security measures when sharing information and accessing shared systems. It is essential to consider the legal and regulatory requirements related to cybersecurity, and depending on the industry and country, there may be specific obligations that companies must meet.

In this sense, Hayek (1945) argued that the ability to anticipate is critical in this process. Moreover, the paragraph suggests that anticipation can play a central role in governance structures, which receive input information and produce results even when the available knowledge is imperfect.

Thus, cyber risk can be understood as that which causes physical damage or impacts financial losses when malfunctions occur in the digital environment or when data is overlooked or shared illegally (Nieuwesteeg & Waard, 2018) in situations of cyber threats (Geer et al., 2020; Smidt & Botzen, 2017), the occurrence of financial crimes and fraud (Dudin et al., 2018; Santucci, 2018), amplification of cybersecurity (Böhme, 2005), the incidence of informational disadvantages (Shetty, 2018), and economic risks in Internet of Things (Lee, 2020; Radanliev et al., 2019) constituting as significant risk factors that have permeated the cyberspace.

Contractual Governance: Situations of confrontation require knowledge about cyber security and protection that make it possible to identify the governance structures bringing understanding to the relationship between their management and the cyber risk mitigation mechanisms.

Williamson (1985, 1996) proposed to discuss the particularities of governance structures from the analysis of their properties by making three distinctions among them: market governance (classical, simple spot market exchange), hybrid governance (contracting, long-term type), and hierarchical governance (internalized in firms and offices). A governance structure must offer conditions enabling innovation and mitigating cyber risks. Innovations

have modified the particularities of transactions, leading to the challenge of evaluating and measuring the impacts on asset specificity and its dimensions while generating value for the actors involved (Guimarães et al., 2021).

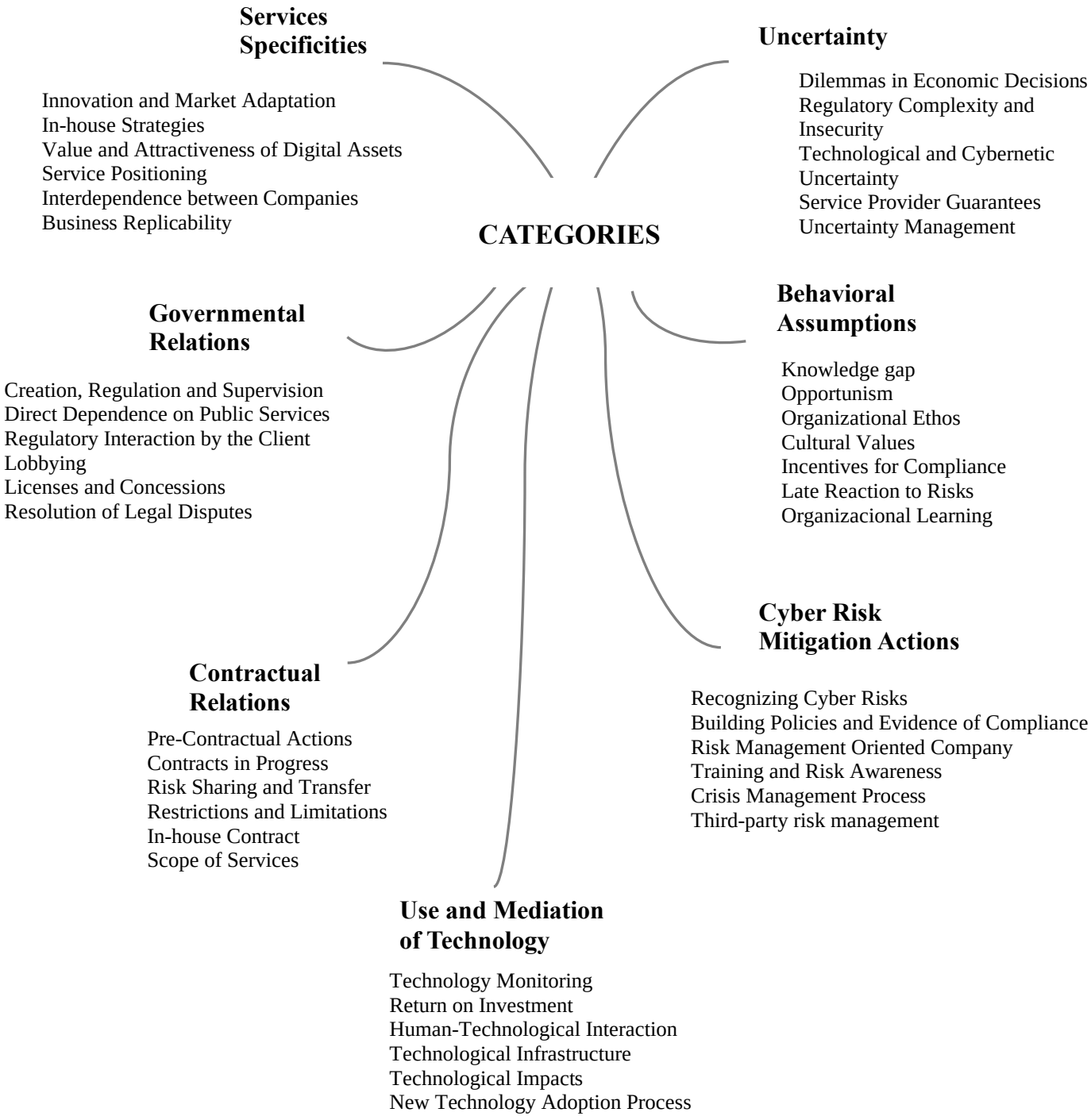
The presence of specific assets in the contractual relationship between companies produces a reciprocal dependence between the parties involved (Williamson, 1975) and still be highly asymmetric. The relationship and development with the customer are considered a highly relevant asset, even when it is related to the services market, whose efficiency and quality are essential to the maintenance and duration of this relationship.

4.2 PRESENTATION OF QUALITATIVE RESULTS

This section will present and analyze the qualitative results concerning the thematic axes defined in the quantitative stage. The thematic axis related to the Organization of Activities and Flow of Knowledge in Services will be explored, covering specific categories such as the particularities of service categories, the use and mediation of technology, and government relations. Next, the Contractual Governance thematic axis will be addressed, detailing categories that include contractual relationships, behavioral assumptions, and associated uncertainty. Finally, the thematic axis related to Cyber Risk Mitigation in Contractual Governance Actions will be examined, focusing on the cyber risk mitigation actions category. Each topic will be dealt with in detail, offering a complete understanding of the qualitative results obtained in this research. Each category will be enriched with the participants' voices.

Supporting the discussion of this study, an integrated analysis was made of the thematic axes obtained in the quantitative stage, together with the categories identified as *a posteriori* (Fig. 15), their respective subcategories, and the recording units acquired in the qualitative stage.

Figure 15 – Cyber Risk in Services Categories



This approach sought a comprehensive perspective, unifying the data and information collected in this mixed research. This combination allowed for a detailed analysis, laying the foundation for discussing the results obtained.

When analyzing the results presented here, specific passages were highlighted by including direct quotes. This approach allowed for a more vivid presentation of the results. It

made it possible to include the participants' voices, which lent greater authenticity to the discussion that will be developed in the subsequent Section.

During the analysis of the categories and subcategories, the range of emerging classifications demonstrated a broad scope of themes related to the study. In order to optimize the effectiveness of the analysis and keep the focus on the most relevant information, an approach aided by IBM SPSS 29 software was used to obtain the frequencies of these classifications. This made it possible to identify the items that exhibited the most excellent recurrence and, therefore, the most significant relevance in understanding the main aspects addressed.

It is important to emphasize that the importance of the categories was not determined exclusively by their frequency of occurrence. To this end, it was recognized that some of the subcategories might have been observed in limited quantities. However, the selection criteria are not restricted to frequency alone but also consider the wealth of discoveries each subcategory can bring to the research. This approach sought to ensure that the analysis encompasses the most prominent trends and captures information underlying the less frequent classifications.

4.2.1 Thematic Axe 1: Organization of Activities and Knowledge Flow in Services

The thematic axes of the "Organization of Activities and the Flow of Knowledge in Services" represents an essential stage in understanding the mechanisms underlying the characteristics and flows of work and knowledge in companies that transact services. This section explores the categories that emerged after analyzing the thematic axes. This made it possible to analyze the 'Service Specificities,' the 'Use and Mediation of Technology,' and 'Governmental Relations'. Each of these categories plays a fundamental role in the governance structure of the companies surveyed, as organized in Table 10.

Table 10 – Description of categories and subcategories related to the thematic axes "Organization of Activities and Flow of Knowledge

Thematic Axes	*A Posteriori Categories*	Subcategories
1. Organization of Activities and Knowledge Flow in Services	**1. Services Specificities**	1.1 Innovation and Market Adaptation 1.2 In-house Strategies 1.3 Value and Attractiveness of Digital Assets 1.4 Service Positioning 1.5 Interdependence between Companies 1.6 Business Replicability
	2. Use and Mediation of Technology	2.1 Technological Monitoring 2.2 Return on Investment 2.3 Human-Technological Interaction 2.4 Technological Infrastructure 2.5 Technological Impacts 2.6 NewTechnology Adoption Process
	3. Governmental Relations	3.1 Creation, Regulation and Supervision of Standards 3.2 Direct Dependence on Public Services 3.3 Regulatory Interaction by the Client 3.4 Lobbying 3.5 Licences and Concessions 3.6 Resolution of Legal Disputes

4.2.1.1 Services Specificities Category

The 'Service Specificities' category is an integral part of the 'Organization of Activities and Flow of Knowledge in Services' thematic axes. This category encompasses factors that make it possible to understand the particularities of the services transacted by organizations. These subcategories provide a framework for analyzing the nuances of services in different sectors and how these specificities impact the organization of activities, flow of knowledge, and the mitigation of cyber risks in services.

Table 11 Description of Service Specificities and subcategories related to the thematic axes "Organization of Activities and Flow of Knowledge"

Thematic Axes	*A Posteriori* Categories	Subcategories
1. Organization of Activities and Knowledge Flow in Services	**1. Service Specificities**	1.1 Innovation and Market Adaptation 1.2 In-house Strategies 1.3 Value and Attractiveness of Digital Assets 1.4 Service Positioning 1.5 Interdependence between Companies 1.6 Business Replicability

Service specificities are specific characteristics or specific specialized or customized assets so that, if they are discontinued, they cannot be recovered or put back on the market. Essentially, they are services or assets that cannot be easily replaced or reverted to their original

state if the relationship or provision of services is interrupted (Klein et al., 1978; Williamson, 1985; Saes, 2008).

Innovation and Market Adaptation. In this context, the 'Specificities of Services' category stood out in the content analysis compared to the other categories. This distinction can be attributed to the diversification of services and business models, giving it a distinctive characteristic.

Driven by the search for competitiveness and expansion, companies are focusing on the strategy of innovation and market adaptation. In this sense, the need to remain relevant and respond to market demands has become a central concern for many managers.

The results obtained in the quantitative phase made it possible to identify the businesses' main activities: 'digital services,' which stood out most in the sample. This finding proved to be of great importance for analyzing the theme of this research.

Digital services, arising from the characteristics of the companies surveyed, connect with the category of ‘Innovation and Market Adaptation.' This category represents an approach that embraces evolving customer needs and directs the development of products and services that meet these ever-changing demands.

In their endeavors to innovate and adapt to the market, companies have turned to digital technology to provide services to customers, such as cloud computing and streaming, which many participants considered peripheral activities of their business. For example, dependence on a cloud or streaming service provider can make the service offered a form of asset specificity. This is because, depending on a cloud or streaming service provider, the company invests in resources, integrations, and configurations that are shaped to work with these providers. If this relationship is discontinued, these specific investments may not be easily transferable to another provider, resulting in a situation where these assets cannot be recovered or reused as effectively.

> "we have a service line for business operations, where we can operate part of the client's business through cloud computing" [E9]

> “Our company has different characteristics, and we need different systems... why did the company think of the digital platform? Precisely to consolidate all these variables, *systems* [emphasis added] within a single platform” [E16]

In addition, it was noted that the number of activities carried out peripherally is more significant than those considered core activities. After analyzing the interviews, it became clear that the diversification of activities and even the verticalization of businesses in the value chain, such as creating a digital bank, logistics, and consultancy services, complement the original business model.

Generating revenue by diversifying services or products has been interpreted as an indicator of business continuity. It is so important that managers and investors have been adopting this metric when assessing a company's financial performance and sustainability.

> "the partners, they really demand that the business be diversified because they say... "you're 100% hostage to [product x], and if something goes wrong in this market, what will guarantee the continuity of the company?" [E14]

However, investments in technology and infrastructure and the quest to diversify services and businesses are factors that accentuate the specificities of these companies' services as they move towards digital services (Fig. 4), which are increasingly being offered and considered as viable business alternatives.

> “[...] and that's what I suggest that companies, in order to protect themselves, hire the services of specialized companies that will always be at the cutting edge of technology” [E9]

> “So, we have a laboratory... testing other businesses” [E14]

By mentioning that the company must be protected by other companies that "will always be at the top of their game," E9 refers to specific and advanced technologies relevant to its business or sector. The excerpt from this speech, as well as that of other participants, points to investments in the search for technologies that are appropriate and specific solutions for the demands and challenges faced by the company. This scenario tends to maximize and enhance the specificities of the services companies offer.

This phase of the study identified companies that establish partnerships with third parties to provide additional services to customers, known as tripartite services. As explained by one of the participants, a practical example of a tripartite service occurs when offering an insurance service in which the customer acquires an intangible cover that protects them against financial risks and losses. However, insurers have often offered additional benefits by partnering with other companies to make the service more tangible and attractive.

> "... it's a company that offers insurance. And in order to make the service more tangible, the insured is offered, let's put it this way, a series of benefits" [E4].

Promoting the 'tripartite service' would aim not only to meet the primary needs of customers but also to enhance the customer experience by providing benefits that add value to the primary service.

In-house strategies. Despite the contracting and dependence on third parties, such as cloud computing and streaming examples, there are simultaneous strategies or even separate decisions. In these cases, companies internalize operations and activities to avoid possible coordination and negotiation problems resulting from dependence on third parties (Williamson, 1985, 1993a).

The sample companies have sought to develop systems, tools, and internal controls as a business strategy. This means that companies have total control over the design, implementation, and customization of these systems according to their specific needs, where solutions are tailor-made to meet the business's unique requirements.

According to E1, “is now tending to develop tools, controls and monitoring internally”, this is a trend in which the decision to develop in-house is based on situations in which the company would need to deal with sensitive information, such as financial data, customer information, intellectual property, and strategies (Biener et al., 2015; Liu et al., 2022). In this sense, the development of internal systems shows that companies seek greater control over the security of this information, reducing the risk of leaks or security breaches.

In line with the decision to internalize, the development of systems and controls leads to results demonstrating that managers are concerned about investing in attracting and training professionals regarding in-house strategies. This includes hiring and retaining people on the team with technical knowledge relevant to the business model.

> "So this is a concrete example that internal development, internal integrations... and obviously a lot of investment that we've made, is bringing good results" [E1]

This internal development strategy is would be guided by the following objectives: reducing costs and, consequently, reducing business uncertainties (Williamson, 1991). This last indicator, which emerged in the quantitative phase, became more evident during the content analysis. Internal teams, in general, tend to have greater knowledge of operations, company

processes, and the complexities of the business. These factors pointed out by the participants can result in the development of more significant commitment to the flows and resources available in the company, according to the narrative of participants E1 and E17:

> “... we end up hiring professionals in the area, training them and they end up developing the functions, or they end up doing the work that I could have relied on a third party to do” [E1]

> “... they have their own interesting line of development that generates efficiency, speed... and commitment” [E17]

According to participant E17, the results show that investing in internal expertise has enabled the company to adapt and adjust systems to be more agile as needs and objectives evolve. In addition, "and generates commitment" refers to building trust between the parties (Foss & Klein, 2022), which is the opposite of opportunistic behavior, which will be explored in the Behavioral Assumptions category.

In this sense, the internal development of systems, controls, and tools, combined with investment in attracting and training professionals, refers to strategic security, in which the participants point to trust in keeping commercial and technological secrets, reducing the risk of confidential information being leaked to competitors.

In highly competitive sectors, protecting strategic plans is central to risk management. Strategic security is based on the protection of assets, information, knowledge, and resources critical to an organization's functioning, competitive advantage, and long-term success (Peteraf, 1993; Wernerfelt, 1984). These assets can include intellectual property, trade secrets, key technologies, and exclusive processes, among others, a concern highlighted by participant E6:

> "so… the information doesn't remain in the hands of third parties, even to have a better way of safeguarding some aspects of strategy" [E6]

E6's statement that 'information does not remain in the hands of third parties' leads to the understanding that by managing sensitive and strategic information internally, the organization can protect its trade secrets, strategic plans, and other assets essential to its competitive advantage. This is especially relevant for those services where competition is fierce and cyber risks constantly evolve.

Value and Attractiveness of Digital Assets. Companies that lead in innovation and technology often have a series of competitive advantages that can result in greater growth, profitability, and market value. This differential is reflected in the maturity of risk management across segments.

> “but the perception is that some are much more advanced, and others are still in the stone age” [E13]

> “as we've raised the level of maturity, we've been creating these mechanisms, reviewing antivirus, reviewing the information security team. We have an external company today that takes care of it, it's dedicated to this work” [E16]

Risk management maturity across segments involves considering how different sectors or areas of the company deal with cyber risk management and how well-developed they are regarding approaches and practices for dealing with risks (Amanowictz & Kamola, 2022; Ani te al., 2019; Böhme & Riek, 2019), whether via internalization or outsourcing. In addition, maturity would be linked to the degree of sophistication and effectiveness of the strategies, processes, and procedures implemented to identify, assess, mitigate, and manage organizational risks (Section 4.1.10, Risk Mitigation).

When analyzing the results of this category combined with the results from the quantitative stage, the value and attractiveness of digital assets, the customer's digital journey became evident as the participants detailed the digital services accessed and used by their customers. This would be the journey a customer takes when interacting with the company through digital means, such as websites, mobile applications, social networks, emails, and other online channels. According to participants E6 and E5, this journey covers all the stages a customer goes through, from discovering and considering a product or service to post-purchase and possible loyalty.

> "so we're talking about technological innovations in digital services, we're talking about social networks, we're talking about technologies, for example, blockchain, Web 3.0.... and the customer themselves, today they're looking for this digital journey [E5]

> "So, financial institutions as a whole, what I see, what I perceive in the market, is moving towards a digital journey" [E6]

Technology, as a facilitator of this digital customer journey in its various forms, is becoming a digital asset in the cyber environment to meet the needs of both the company and its customers.

> "the world is increasingly moving towards digital journeys" [E6]

> "I don't know if you've seen these generative technologies for text, images and so on, this is going to affect the world a lot, both positively and perhaps negatively, but you can use these technologies to suggest developments in the fight against cyber-attacks" [E9]

Value, for the company and the customer, is being translated into their perception their perception of the benefits they receive, either by providing or acquiring and using a specific product or service. The value of the service can be subjective, although it can include a combination of tangible and intangible factors that fulfill the needs, desires, and expectations of both (Benoit, 2023), especially in the digital customer journey, according to participant E5:

> "Portability, convenience, cost, ease. And it's excellent value for money for the customer" [E5]

Moreover, some companies, including traditional ones operating in sectors seemingly far removed from technology, have shown that they are redefining their identity as a business model. Substantial investments in technology, from developing advanced digital platforms to integrating algorithms and automation systems, demonstrate an active search for innovation in their services and products.

> "... we're definitely a technology company" [E1]
> "…but anyway, just so you understand, from our point of view, we're a technology company" [E2]

> "The company I work for is an insurance company, but they are starting to communicate to the market we are a technology company" [E4]

> "... we are a technology and consulting company" [E12]

It can be inferred that those companies that identify their business model as technology companies do so not just out of perception but because their actions and investments reflect a

commitment to technology as a core strategic asset of their business. This can lead to the understanding that these companies are willing to allocate resources and continuous efforts to ensure technology is at the heart of their operations.

Therefore, when applied to the digital asset context, customer value is also intrinsically linked to the technology's and digital resources' perceived usefulness. On the company side, economic value emerges through these digital assets' ability to generate financial returns, optimize processes, and increase operational efficiency.

4.2.1.2 Use and Mediation of Technology Category

Category 2, 'Use and Mediation of Technology,' is a crucial part of the thematic axes 'Organization of Activities and Flow of Knowledge'. This category highlights how organizations employ technologies in services to monitor, prevent, and respond to threats and the return on investment in technologies. In addition, there is a focus on the interaction between people and technology in services, addressing people's skills, understanding, and experience compared to the potential of technological capabilities.

Table 12 – Description of Use and Mediation of Technology and subcategories related to the thematic axes "Organization of Activities and Flow of Knowledge"

Thematic Axes	***A Posteriori* Categories**	**Subcategories**
1. Organization of Activities and Knowledge Flow	**2. Use and Mediation of Technology**	2.1 Technological Monitoring 2.2 Return on Investment 2.3 Human-Technological Interaction 2.4 Technological Infrastructure 2.5 Technological Impacts 2.6 NewTechnology Adoption Process

Technological Monitoring. One risk mitigation practice adopted by most participants is technological monitoring. This type of monitoring is made possible by the use and mediation of technology, which allows for the construction of a database to identify trends and anticipate possible risks.

Furthermore, in situations where an attack is underway, this approach would make it possible to obtain knowledge more quickly, facilitating decision-making by the areas responsible, as mentioned by E2 and E6:

> "There are also cameras on the floors [...] We have 24-hour monitoring... it doesn't stop on holidays; it doesn't stop on Sundays" [E2]

> "this monitoring and information security area already has a guideline for activating the internal cybersecurity audit" [E6]

Also, considering the main lines of use and technology mediation, a relevant technique is observed in the interview analysis results. This technique is based on verifying identity based on the user's access behavior, whether internal or external to the company, as illustrated by participants E6 and E17:

> “you can hire tools that assess behavior, that have artificial intelligence, algorithms and that can help you” [E6]

> “we are dependent on technology, because without technology you can't do this monitoring” [E17]

This type of monitoring is considered a security technique that analyses the specific patterns and characteristics of a person's behavior when accessing systems, applications, or online platforms. Rather than relying solely on traditional passwords or authentication, this approach examines how the person interacts with the cyber environment, creating a specific behavior profile.

> “we have a specific proxy, so all our communication ends up going through content that is monitored and controlled by our company” [E12]

The creation of behavioral patterns through technological monitoring is a technique that is particularly effective in detecting suspicious activity, such as invasions by third parties, fraud attempts, or even insider situations (people with legitimate access to systems and information). The elements that make it possible to create this behavioral pattern include typing patterns, mouse movements, geolocation, and access times, among other ways:

> "It is not just the user password; in addition to the user and password, there's all the behavior that the person has behind it, the location, which is logged in, which is mapped, so you know who's logging in, if it's you, you know?" [E8]

Another important point when building behavior patterns is that this mapping is transformed into input for the consumption of information for machine learning. Internal data

is used as a basis for strategic decisions, operational improvements, and quality of service, as reported by E17:

> "we use it much more with a focus on continuous improvement, what can I automate, how long does it take my employee to carry out that routine, but you're monitoring it online all the time. So the tool has cloud access to all the data you're transacting" [E17]

Mapping ranges from anticipating business risks, mapping, and obtaining internal data, and identifying customer preferences and habits. Based on this information, companies are developing new interests and opportunities for purchasing and hiring. These companies collect and analyze data to create personalized experiences for their customers. This includes targeted product recommendations, suggestions, and interactions more tailored to individual preferences, in line with what E4 and E17 said:

> "Imagine the one whose son has just been born and all. There's an opportunity there. How do we find out? By doing all this analysis, cross-referencing information, then you say, "This guy has the potential to buy this service" [E4]

> "From the moment I provide a card, I already have a bit of buying behavior" [E17]

In the meantime, regarding internal and external data mapping, the companies have mentioned that ethical and data privacy issues have been considered and that this type of practice is transparent and complies with laws and regulations.

Return on Investment. In the dynamics of a company, investments can take many forms, from acquiring assets and developing services and products to improving processes or implementing strategies (Damodaran, 2006; Titman & Martin, 2010). Furthermore, many managers consider and use return on investment as a metric to guide investment decisions. It is a way of measuring the efficiency and profitability of an investment concerning the initial cost (Oyadomari et al., 2018).

In this context, the research showed that managers are concerned about cyber and information security investments. These investments could directly impact the company's financial results and availability of capital, as they could involve costs associated with acquiring technologies, training teams, and third parties, as well as other protection measures.

Most of the time, the financial impact is uncertain, and fines for data leaks and fraud have emerged as one of the determining factors to guide investment decisions in technology (Chanalysis, 2023; IBM, 2021). This is because reputational risk was an element that was frequently addressed in the participants' responses in both stages of this survey:

> “Because the way it is, we could be worried about reputational risk” [E4]

> “And the question of reputation is an essential pillar when making decisions and analyzing the cost of implementing a security tool” [E6].

These factors emphasize the importance of adopting cyber and technological security measures to protect the company against threats. They, therefore, end up influencing the choice to invest in technological solutions that guarantee data integrity and business continuity.

> "Oh, security will help me avoid a R$50 million fine from the ANPD. Security will help me avoid a data leak and that data leak will cost me a lot of money per data leak. Besides, it could hurt my company's image, right?" [E12].

> “if you don't have a technology today that can normalize this data, filter it, and then be able to deal with what you need to deal with and not everything that appears, today it's practically... impossible for us to work with technology. And technology is increasingly the heart of the company, isn't it?” [E12]

The analogy of the heart as the pulsating center of technological solutions highlights how technology can interconnect the company's areas. In addition, it also seeks to extend its "vitality" to the outside or the market.

In this sense, participants emphasized that some companies are allocating part of their turnover, or at least holding comprehensive discussions, on feasibility studies related to new technologies and solutions, according to respondent E16. This approach is not only restricted to business growth, as seen in the specifics of services but also aims to mitigate risks, especially in the digital context.

> “… companies in the logistics segment, they invest around 1.5 to 2% of their revenue in IT protection” [E16]

For participants E16 and E6, the return on investment is perceived as a strategic measure to close gaps, strengthen the company's position in the face of market demands, and respond to the points raised by the audit. This helps ensure that the company is aligned with emerging trends in cyber risks, even if the cost is considered high, as mentioned by E16 and E6.

> “it brought investment, because we managed to close some gaps that we wouldn't have been able to close if we'd come in like we did in the last two years, for example. They weren't serious things [E16]

> “but the board looked inwards and said, "hang on, the audit is charging more, the market is charging more" [E16].

> “but companies are looking for increasingly up-to-date tools and training to mitigate and minimize these risks. So there is a need for this investment. There is, of course, a high cost, right?” [E6]

It was not the purpose of this research to discuss what high or low costs are. However, it is essential to emphasize the dynamics of fixed and variable costs in the context of continuous investment and its return. This dynamic intrinsically relates to how the spending structure is configured concerning the decisions made. While fixed costs remain stable regardless of changes in adaptations, variable costs can fluctuate based on the specific actions taken to optimize returns over time (Frezatti, 2008).

> “...maintain continuous investments, because it's not something... it's not like implementing a SAP there, you do a project, it's difficult and it's implemented” [E15]

Investment in technology in companies that transacts in services, whether to offer or to hire, is continuous, and consequently, the expected return is also continuous. This becomes clear in Table 9, when a descriptive analysis is made of companies more involved in initiatives seeking business growth through the economic performance strategy.

Another critical point noted is the presence of industries that challenge the traditional status quo by adopting new approaches, such as the use of digital catalogs and apps, as examples, which can be considered a form of digital services for offering products and services, according to participants E5, E7, and E17:

> “we have a huge catalogue…that we have commercial rights” [E5]

> “We have four business channels… one of them is trading beauty products, via door-to-door and the online catalogue” [E7].

> “serve the customer, it works in an environment that the market commonly calls omnichannel” [E17]

These respondents, in particular, are companies that are recognized as leaders in their respective sectors and have vertical businesses that range from digital banking, financial services, and property management to Shared Services Centers (SSC). The verticalization process is even emphasized by E1, who points out that this initiative can lead to greater operational efficiency and cost savings since the company has more control over the inputs, quality, and time dedicated to an activity.

> “...and the cost to us is much lower. So that's why some verticals we end up opting to do in-house" [E1]

This observation is relevant as it highlights the ability of traditional sectors to adapt to technological changes by incorporating digital services into their business models. This phenomenon exemplifies current market dynamics, in which innovation has become essential for competitiveness even in historically more conservative industries.

Human-Technological Interaction. Human-technological interaction would comprise contexts and implications when people interact with automated devices or systems, such as computers, robots, artificial intelligence, and other technologies. In practice, social engineering tests and simulations exemplify how interactivity between humans and technologies can be exploited to improve security and awareness in a cyber environment (CISA, 2021).

Social engineering tests and simulations have been one of the strategies companies adopt to train their teams to recognize and protect against malicious or criminal attacks. Meanwhile, people are often considered the main vulnerability to fraud and cyber-attacks (IBRASPD, 2023).

> “Users are the main vulnerability to data leaks” [E3]

> “The door is technology, but I believe that it's the people behind it who open the door to risk, you know” [E13]

Considering the position of participants E3 and E13, it can be inferred that the interaction between people and technology must be accompanied by an assessment of the risks, considering that companies' increased dependence on technology can expose them to cyber vulnerabilities and threats.

On the other hand, this interaction is compromised or practically null in companies with manual processes or predictable and stable activities based on routines (Alavi & Leidner, 2001; Becker et al. 2005; March & Simon, 1993), even though they can be automated. This is because manual execution would be subject to human errors, such as execution failures, misinterpretations, and forgetfulness, which can lead to undesired results, according to E13:

> “... because everything that is very manual ends up having a greater margin of risk” [E13]

In this scenario, technology unfolds in the understanding that business and business models carried out exclusively by humans tend to be reduced in the future - being operationalized through human interaction and technologies, in which business ecosystems will be integrated by physical, digital, and virtual means (Valter et al., 2018).

Although bringing people and technology closer together is a beneficial interaction considered by most participants, there is the consideration that not all people have equal access or skills to use new technologies. The same understanding applies to companies regarding investments in this area.

> “…it's practically a chasm between the knowledge that people have and what technology can deliver” [E16]

The perspective presented by E16 and other respondents highlights the challenges of human knowledge interacting with new technologies. People's skills, understanding, and experience are emphasized when compared to the vast potential of technological capabilities. At the same time, the importance of educating, raising awareness, and bringing people closer to new technologies is emphasized to reduce this gap. This aspect will also be explored in the subsequent categories: Behavioral Assumptions and Mitigating Actions for Cyber Risk.

4.2.1.3 Governmental Relations Category

The 'Governmental Relations' category is part of the 'Organization of Activities and Knowledge Flow' thematic axes. This category covers aspects related to interaction between organizations and government entities. Within this category, the subcategories expose specific aspects of these relationships and provide subsidies for analyzing the complex relationship between organizations and the government regarding cyber risk management, highlighting how government decisions and regulations can affect organizations.

Table 13 – Description of Governmental Relations and subcategories related to the thematic axes "Organization of Activities and Flow of Knowledge"

Thematic Axes	***A Posteriori* Categories**	**Sub-categories**
1. Organization of Activities and Knowledge Flow	**3. Governmental Relations**	3.1 Creation, Regulation and Supervision of Standards 3.2 Direct Dependence on Public Services 3.3 Regulatory Interaction by the Client 3.4 Lobbying 3.5 Licences and Concessions 3.6 Resolution of Legal Disputes

Creation, Regulation and Supervision of Standards. Institutions have been created to bring order and organization to human interactions. This includes reducing uncertainty in commercial and social exchanges by providing a regulated environment where people can reliably conduct their activities (Acemoglu et al., 2020; North, 1991). E12's statement highlights this:

> “I think that today we have a much more regulated world, but it's good, because I think that today the regulatory agencies are starting to tighten up, and that makes us live in a safer environment “[E12]

The combination of formal and informal rules shapes relations between companies and government (North, 1990; Williamson, 1998). Even when considering the role of institutions in the market, each company is responsible for establishing and implementing its own specific internal rules, seeking to create value through its processes and interactions with other companies. Not only do companies establish their own internal rules, but they must also be aware of the regulations imposed by regulatory bodies and government agencies. These excerpts highlight the regulatory weight in their business operations:

> “The regulatory issue has a very big impact in the operation” [E4]

> “So, you see, the regulatory framework is very heavy” [E6]

> “the Mexican government blocked it, it created rules against it (outsourcing), so that companies would have to hire professionals directly” [E18]

Although the company focuses on informal rules, it is essential to note that the government can influence how it organizes its governance structure. In this latest information from E18, the Mexican government has prevented the practice of outsourcing by creating rules that oblige companies to hire professionals directly.

In this context, it is important to highlight two figures: macro-institutions, run by the government, which create and enforce broad regulations, and meso-institutions, whose purpose is to act as a bridge between the government and companies, facilitating the implementation of regulations in the business context, through translation, monitoring and enforcement (Ménard et al., 2022):

> “you can't let the television go off air, that's regulated” [E7]

> “And I think the LGPD[12] has highlighted this even more, the extent to which we have to prove something to the state that we are looking after our customers' information with care and managing it well” [E3]

> “So our company has a product line that is 100 per cent dependent on these communities and it's seasonal. It depends on the seasonality of production and the government carries out various inspections” [E17]

Therefore, the creation, regulation and inspection of standards proves to act as a link that brings institutions together, covering both the macro, meso and micro levels. Together,

[12] The General Data Protection Law (LGPD) is a Brazilian law that deals with the protection of personal data and the privacy of citizens (Law n. 13.709/2018). It was inspired by the European Union's General Data Protection Regulation (GDPR) and came into force in Brazil in September 2020. The LGPD's main objective is to regulate public and private organizations' processing of personal information, ensuring greater control for data subjects over how their information is collected, used, stored, and shared.

these dimensions make up a system that interconnects the government sphere and the business sector, impacting on economic development and business dynamics.

Direct Dependence on Public Services. Regarding dependence on public services, the state's primary interaction could be related to validating and guaranteeing the conformity of these services (Schwartz, 2002). Specifically, when it comes to information security, data protection, governance, and insecurity issues in public services, the state's presence is most evident at the stage of verifying and proving these practices and processes, as discussed by participants E3, E4, and E14:

> "if we're talking about information security, data security, governance, insecurity specifically, I think the only dependence on the state is actually verification" [E3]

> "But because it's linked to the bank, it's much heavier" [E4]

> "…and then there's this other financial part, which will now be answerable to BACEN (the Brazilian Central Bank) [E14]

The unanimity in citing the financial institution as the most regulated sector can be justified by the wide-ranging nature of financial activity in the companies taking part in the survey. As well as being a cross-cutting activity in various organizations, many companies analyzed are dedicated to payment services, credit cards, and benefits, either as their primary or secondary activity. Given this broad presence and the intrinsic relevance of financial services, it is understandable that financial activity stands out as the point of greatest regulatory dependence. Given this context, it is undeniable that financial institutions often become risk management models since managing risks is ingrained in their DNA, covering not only their risks but also those of third parties (Rossi, 2017).

This predominance of financial activity in the survey can also shed light on the results of the quantitative stage, in which the most outsourced services are often linked to the government. The substantial presence of financial activities in the companies surveyed may explain why the most frequently outsourced services are related to government regulations.

In addition, companies are subject to depending on the positions and actions of the government, such as the Public Prosecutor's Office. These institutions perform a regulatory and supervisory role, and depending on the direction they take, they can impact the operational and legal environment of companies:

> "… the staff of the Public Prosecutor's Office take the necessary steps to curb any kind of legal avant-gardism within the judiciary" [E11].

The excerpt highlighted by E11 expresses that the Public Prosecutor's Office's actions can significantly impact disputes and legal discussions promoted by companies. The measures taken by these institutions can influence the legal and regulatory environment in which companies operate, affecting judicial decisions, legal proceedings, and even interpretations of the law (Coase, 2022; Williamson, 1999).

Another factor highlighted in the results was companies' direct dependence on the government in obtaining permits and licenses from town halls. In order to operate legally and in compliance with municipal regulations, companies often need to obtain government approvals, such as operating permits and specific licenses, as participant E16 explained:

> "So we have this dependence on these public services. On the other hand, we depend a lot on licenses, we have several branches throughout Brazil, so we will depend a lot on city halls to get permits" [E16]

There are specific legal requirements and impositions for digital services, and companies must deal with specific regulations and charges that can not only impact their costs but also shape the dynamics of the market in which they operate, as participant E2 explained:

> "so much so that we use public pricing in several cities, including São Paulo, because of the issue of using the road network" [E2]

Finally, in this scenario, utility companies also need concessions to operate in the sanitation, water, and sewage sectors. Specific regulations govern these services, and the companies' dependence on concessions is often directed through Public-Private Partnerships (PPPs). In this business model, a municipality implementing a new sanitation framework law often opens the possibility of tendering for services related to these areas.

Regulatory Interaction by the Client. Certain concessions and regulations may have specific applications in some situations, and others do not affect specific sectors. However, for companies that offer knowledge services, such as consultancy or auditing, or even digital services, including the adaptation of providers and platforms, understanding the application and limits of the regulations that affect their clients' businesses is mandatory for the proper provision

of services. It was noted that these companies are aware of this concern since it implies efforts and costs that influence business coordination.

> "then the others will depend on the focus of our client, if our client is a bank, then we have to follow financial legislation together with the client" [E9].

> "So the concern is data security, financial transactional security, compliance with regulatory requirements [E6]

> "In fact, a large part of our portfolio is helping companies make the adjustments that they need to make" [E12]

Therefore, the results lead to the understanding that government relations can affect the economy and companies directly and indirectly regarding regulatory aspects. It can be seen that the participants were more concerned about operational aspects, costs, and commercial interactions than the actual impact of mitigating cyber risks.

Lobbying. The majority of participants emphasized dependence on and direct discussions with the government. This is evident, for example, in tax discussions that require considerable effort and resources, as companies resort to lobbying to deal with this scenario:

"And now with the new government there is also this discussion to regulate and try to make the social security contribution more beneficial to app workers" [E2]

"So we have to do this well with these entities" [E16]

"the Minister of Finance released a note talking about carbon credits and the issue of green initiatives, our company is also part of this pool of companies that has been discussing it" [E17]

Another aspect emphasized by just over half of the participants concerns the discussions and lobbying practices carried out with the Ministry of Finance and the Brazilian tax authorities, according to E2 and E17:

> "sometimes we also have difficulty looking at the contract, to even define how we're going to tax this when we're taking on that service, there's a difficulty even in knowing" [E2]

> “Then there are the relationships with the government in the tax area - because, well, we're always there on the front line, so we have a closer look at tax audits, I'd say. So we do have a very close relationship with the government” [E17]

Lobbying initiatives aim to improve the orientation of tax legislation. By exploring the specificities of the services offered by these companies, the challenge of interpreting and applying tax regulations becomes evident. This can widen the contractual gap and the environment of uncertainty, adding to the complexity of mitigating cyber risks.

The lobbying reported by the companies is not only limited to discussing legislation for companies already in operation but also covers those that have not yet been able to start their activities, according to the information obtained from E2:

> “Deliveries by motorbike, by app, we depend 100% on that activity being allowed in the municipality where we are going to provide the service” [E2]

In the broader context, these aspects can conclude that there is an active and constant relationship between the government and companies, with companies seeking to influence government decisions at various stages - from the start of operations to tax issues. Companies seeking to influence regulatory or legislative policies can increase transaction costs, as it requires financial resources, time, and effort to engage in lobbying processes and interactions with the government. In addition, the regulatory uncertainty (Williamson, 1979) resulting from policy changes can impact business decision-making and create additional risks.

4.2.2 Thematic Axe 2: Contractual Governance

The thematic axes of ‘Contractual Governance' also proved to be an essential step in understanding the elements underlying the rules, processes, behaviors, and environment in which uncertainty permeates the management and monitoring of contracts. This made it possible to analyze the 'Contractual Relationships', the 'Behavioral Assumptions', and the 'Uncertainty'. Each of these categories plays a fundamental role in the governance structure of the companies investigated, as organized in Table 14.

Table 14 – Description of the categories and subcategories related to the thematic axes "Contractual Governance"

Thematic Axes	*A Posteriori* Categories	Subcategories
2. Contractual Governance	4. Contractual Relations	4.1 Pre-Contractual Actions 4.2 Contracts in Progress 4.3 Risk Sharing and Transfer 4.4 Restrictions and Limitations 4.5 In-house Contract 4.6 Scope of Services
	5. Behavioral Assumptions	5.1 Knowledge Gap 5.2 Opportunism 5.3 Organizational Ethos 5.4 Cultural Values 5.5 Incentives for Compliance 5.6 Late Reaction to Risks 5.7 Organiational Learning
	6. Uncertainty	6.1 Dilemmas in Economic Decisions 6.2 Regulatory Complexity and Uncertainty 6.3 Technological and Cybernetic Uncertainty 6.4 Service Provider Guarantees 6.5 Uncertainty Management

4.2.2.1 Contractual Relations Category

Category 4, 'Contractual Relations,' is part of the thematic axis 'Contractual Governance.' This category addresses contractual relationships within organizations, examining pre-contractual stages, ongoing contracts, and contractual risk management strategies. Within this category, the subcategories provide a detailed analysis of actions and considerations related to contracts.

Pre-contractual Actions. The vulnerabilities identified by professionals in the pre-contractual phase are linked to the management and sharing of risks between the parties, the expectations of achieving the contractual object, and the preparation of a well-defined project or scope for the production and delivery of the contracted object. This outcome was reinforced by the results obtained from analyzing the content of the interviews, as the discussion of *ex-ante* costs in contracts performs a fundamental role in creating agreements that seek clarity and meet the expectations of the parties involved. This explains why pre-contractual actions were widely emphasized by the respondents, such as E8:

> "... drawing up a contract involving technology services is different to drawing up a contract to build a house" [E8]

Technology services contracts cover various complex technical projects, such as digital services linked to software and systems development. In addition, it involves knowledge services, such as IT consultancies, which require detailed specifications, technical requirements, and clauses relating to intellectual property.

Furthermore, companies need to coordinate to manage cyber risks when discussing the use and mediation of technology as a means of growth and adaptation (Figure 11, Section 4.1.9). In this context, the term *ex-ante* cost refers to the costs incurred before a contract is finalized. Technology contracts can refer to the costs associated with preparing, planning, and drafting the contract before the technology services are provided.

In this context, participants cited both pre-contractual assumptions and the use of frameworks for selecting service providers established by companies. These frameworks range from risk analysis and assessment of the business continuity plan, verification interviews and due diligence on the candidate company, high relevance for ISO certifications, and even the application of Black Box Tests on the service provider. There are situations in which, depending on the outcome of the third party's analysis, the third party undertakes to implement controls based on a risk statement as a condition of providing services.

Participant E11 mentioned that in this contractual phase, "we have to find an acceptable level of risk." This implies that the parties assess the risks involved and make decisions that minimize the possibility of losses.

It is essential to recognize that eliminating risk can be impractical or unfeasible regarding effort and costs. This is because, even when the parties are engaged in negotiations and drawing up contractual terms in line with their objectives, it is during this process that knowledge and information gaps can emerge that play a central role in this contractual phase. This is because the parties may not have access to all the information they need or be unable to understand or analyze all the information available (Arrow, 1971; Coleman, 1990; Gallouj, 1997; Williamson, 1993a). This can lead to decisions that must be informed and reflect relevant aspects of the contract.

Moreover, another point of concern for respondents at this contractual stage is opportunism, the behavioral assumption that the parties would seek to prevent from occurring (Coleman, 1990; Williamson, 1993a). This differs from suppressing opportunism, which can lead to more straightforward contracts but lose the ability to deal with complex situations (Williamson, 1993a). This is because by better understanding the intentions and behavior of the

parties before signing the contract, it is possible to establish clauses that discourage harmful behavior after the contract has been signed. This can include penalties for irresponsible actions.

Contract in Progress. Following the contract's conclusion, most participants stated that their companies employ a combined audit approach, including periodic audits and sample audits, to obtain a comprehensive view of the service provider's performance. However, the periodic audit proves to be more resource-intensive, while the sample audit depends on a selection criterion and the choice of a sample that is considered strategic.

One of the requirements of periodic audits is that the service provider provide certifications. This item is worth highlighting, given that these certifications are often used as a signal in response to client pressure (Gallouj, 1997). Certifications are considered seals of approval or evidence that the service provider meets specific standards, practices, or requirements. This practice is widespread in the context of safety, quality, and regulatory compliance. Furthermore, obtaining these certifications becomes even more relevant when the provider is considered a strategic supplier.

> “So, every year the company has to present that its certifications are correct, and then there are audits on some... some suppliers, which they call strategic suppliers” [E17]

In line with Hölmstrom's (1985) understanding, many institutions aim to reduce asymmetry and informational uncertainty. These institutions include certification, reputation, contractual guarantees, contingent contracts, and various quality signals. These approaches have been presented as mitigating the information disparity between the parties involved to promote transparency and trust in contractual relations.

Risk Sharing and Transfer. In addition to the periodic review of contracts, some companies show that they adopt the practice of reviewing and updating contracts at renewal times, although the participants adopt different time intervals for such renewals, as shown in the results of the quantitative stage in Table 12.

Risk transfer is another mechanism that has been used to help the contracting company protect itself against possible adverse impacts resulting from the actions of the contracted company, whether in bilateral or tripartite contracts, as evidenced by the research. In the results obtained in the quantitative stage, the majority of respondents (77.6%) stated that the company shares cyber risks with partner companies about contracting digital services. However, the

failure to transfer the risk is a concern for the participants, although opinion is divided on this issue:

> "... you want to transfer the operation to a third party, it doesn't mean that you are transferring the risk to them, the risk remains with you" [E4]

The management of third-party risks, i.e., those of contractors or clients, has emerged as a central concern in cyber risk mitigation. This is because companies, whether contractors or clients, face the reality that risks are not limited to internal operations. It has become imperative to be prepared to face the risks introduced by third parties, going beyond the contractual relationships identified in this study.

> "...the company is currently concerned - and even the regulatory body is concerned - about what we call third-party management" [E4]

The analyses emphasized that companies have sought to prepare themselves to deal with the risks that can be introduced by third parties into contractual relationships. This is evident, for example, in the chain of service providers and clients.

> "With the advent of outsourcing, there are a series of information security clauses that we also put in place to protect the company, but the problem is only discovered in the middle of the contract" [E4]

> "How do we protect? With contracts and we say to the other party: "if you outsource the operation to someone else, it doesn't matter, if something goes wrong, you'll be penalized" [E13]

Anticipating risks from third parties interacting with contracting parties can be a complex challenge. Often, these risks only become apparent during the execution of the contract since the parties may not have been aware of these problems beforehand when establishing the initial agreement.

4.2.2.2 Behavioral Assumptions Category

Category 5, 'Behavioral Assumptions,' is integral to the thematic axis 'Contractual Governance.' This category explores behavioral assumptions and aspects that significantly

impact how contracts and agreements are managed within organizations. Within this category, subcategories address the behavioral aasumptions and their implications for contractual governance.

Table 15 – Description of the Behavioral Assumptions and subcategories related to the thematic axes "Contractual Governance"

Thematic Axes	***A Posteriori* Categories**	**Subcategories**
2. Contractual Governance	5. Behavioral Assumptions	5.1 Knowledge Gap 5.2 Opportunism 5.3 Organizational Ethos 5.4 Cultural Values 5.5 Incentives for Compliance 5.6 Late Reaction to Risks 5.7 Organizacional Learning

Knowledge Gap. Bounded rationality assumes a fundamental role in the theory of transaction costs. In theory, this discussion is based on the Information Gap. However, considering the results obtained from the analysis and the context of cyber risks, it opted to refer to this category as the Knowledge Gap.

In an information gap situation, people may not have all the information they need to make optimal decisions in economic transactions (Simon, 1990; Williamson, 1993a). However, the term 'Knowledge Gap' is understood to be beyond the 'Information Gap'. This is because people may have access to a considerable amount of data and information but may lack the specialized knowledge to interpret, evaluate, and respond appropriately to this information.

> “they have no idea of the risk they are exposed to because of this lack of information. They don't see themselves at risk” [E11]

> “we still have very little knowledge of the forms of leaks and cyber risks. They change every day, so I think that this speed of change, the company can't always keep up, I'm very concerned about this” [E17]

This difference is particularly relevant in cyber security and data protection, where the risks can be complex and constantly evolving. Therefore, this subcategory, Knowledge Gap, considers the coexistence of two situations: the absence of information and complete

knowledge, the excess of information, and the absence of specialized knowledge[13]. These concepts highlight the need for comprehensive approaches that consider both access to information and the ability to understand it effectively.

When the issue of technology versus people's digital proficiency is addressed again, the challenge of the knowledge gap and time constraints to manage risks is raised as one of the concerns.

> "... the great truth is this, the people who are working day to day, managing the business, not only don't have the knowledge, they also don't have the time" [E5]

While, on the one hand, the lack of expertise or information emerged from the analysis of this study, on the other hand, the results brought up "overconfidence." Overconfidence is based on a lack of understanding of the complexities and risks involved in work activities. This situation can be justified when there is a knowledge gap concerning a specific context, where people tend to underestimate the complexity of situations, which in turn leads them to feel more confident than they should (Gervais & Odean, 2001; Jordão, 2018; Rubisteins, 2001).

> "... I think the underestimation of cyber risk itself, when they say "cyber-attacks only happen to big companies" [E1]

The association between tolerance and optimism concerning cyber risks leads to overconfidence, which can cause people not to protect themselves adequately against cyber threats due to an unjustified belief that they will not be negatively affected.

> "…often, people don't believe that risk is going to happen, because it's never happened in the past, you know? And it gets a bit into that mode... it gets into airplane mode" [E7]

> "…underestimating the risk, like, "oh no, that's not going to happen to me" ... This is cultural, very strong... this terrible culture we have here, isn't it? "Oh, but it's always like this: after someone has died, they'll paint the crosswalk" [E1]

> "... because there's always that internal person who says: "Oh, but that won't happen to us" [E17]

[13] Nissimoff, P. S. B. N. (2023, August 10). "What is the space for strategy, under different behavioral assumptions?" [Classroom discussion with the author]. Postgraduate Course in Organizational Economics and Strategy, FEA USP, São Paulo

E1's statement that "this is cultural, very strong... this very bad culture..." was also addressed by other participants on the issue of late action to risks, whose delay in taking action before the risk materializes is a cultural aspect of the locality. However, the nature of this reaction raises questions about whether late decision-making results from knowledge gaps not identified in time or opportunistic actions, which will depend on an analysis of the context of the situation.

Finally, a knowledge gap can impact the assessment and planning of contractual relationships (Williamson, 1991). This is due to the possibility of unpredictable behavior or actions that are not strictly aligned with the interests of the parties involved. This scenario makes contract analysis more challenging. However, this dynamic will be explored further in the Contractual Relations Category.

Opportunism. Transaction cost economics is an area of study that focuses on the expenses - not only monetary but also time and effort - associated with carrying out economic transactions, such as buying, selling, or contracting (Williamson, 1993b). These transactions are carried out in environments with incomplete or asymmetric information (Macher & Richman, 2008; Tadelis & Williamson, 2012). As this is common in the global economy, opportunism is a concern in contractual relationships. This is because opportunism is related to contractual incompleteness caused by bounded rationality, i.e., the knowledge and information gap.

When one of the parties does not have complete access to the information in the transaction, it can become vulnerable to the opportunistic practices of the more informed party. This can lead to a breakdown in trust and the need for mechanisms ranging from detailed contracts and guarantees to legal disputes to mitigate the risks associated with information asymmetry. In this sense, opportunism and trust are interconnected concepts in economic theory and are often associated with information asymmetry (Coleman, 1990; Williamson, 1991, 1993a).

> “it's people from inside the company who end up leaking information” [E2]

The excerpt from participant E2's speech represents that in a contract signed between a company and an employee, opportunism, and trust also play a significant role in this

relationship. In addition, participant E17 reported that the team was discussing how certain sensitive and confidential information was made public or available to the financial market.

> “... the other day we [the team] were discussing: "How did this information reach the market?"? And we joked and said: "It wasn't for us, poor mortals, was it?" [E17]

When participant E17 mentions the expression "it wasn't us poor mortals," he is jokingly referring to the fact that the team members were not directly responsible for the information leak. Instead, it is inferred that they are commenting that this information may have been obtained or disseminated through unknown means, suggesting that these means go beyond the understanding of risk management to which they have access. In this context, interview participants highlighted that scams and fraud may involve subtle patterns or interactions that may not be easily detectable, requiring more detailed analysis and specialized technology to identify irregular behavior.

The discussion about the origin of information and privileged sources that result in unfair advantage is relevant to transparency and the integrity of markets. When participant E17 mentions that there is "opportunism in the use of information," he refers to those parties who take advantage of the information they have in a strategic or self-interestedly.

> “there is opportunism in the use of information, especially in our country, when we talk about competition” [E17]

This opportunistic behavior can create unequal competition and damage the market's integrity or the business environment. When it comes to a bilateral relationship that involves the security of information on clients, employees, projects, and other sensitive aspects, information asymmetry can be a challenge in this context since one of the parties can have access to confidential information that the other party does not have, which can lead to advantageous or unfair behavior, especially when it involves sensitive and strategic business information.

Other behaviors obtained in the content analysis results, which also stem from opportunistic behavior, are the comfort zone and resistance to change. These behaviors have been revealed to be common in human dynamics, especially in contexts of accelerated and constant change, such as the cyber environment. In this sense, participant E11 pointed out that

human beings tend to settle into familiar and predictable situations, as this provides a sense of security and stability.

> “that people don't want to change that quickly. Human beings aren't made for that...They want a period of enjoyment that they no longer have” [E11]

In a professional environment, implicit and explicit contracts between the employee and the company exist. Resistance to change can be perceived as opportunistic when people choose not to adopt or comply with the terms of the contract. This behavior can create tension between individual interests and organizational objectives. In an account of company situations, participants E7 (internalization process) and E17 (outsourcing process) exemplified behaviors resistant to implementing new systems in the company.

> "I've always done it this way, right, and it's never been a problem" [E7]

> “even halfway through, the staff said: "ah, let's go back to the other old system, which is simpler" [E17]

Cognitive dissonance in the economic context can lead people to ignore or minimize the risks associated with financial contracts or agreements (Ferreira, 2023). The results revealed cognitive dissonance when employees face internal conflicts between the desire for immediate convenience and the need to invest effort and resources to manage conflicts or problems, including those caused by cyber threats. Therefore, it is a postponement by affective decision-making, which can result in irrationally optimistic decisions, leading to adverse financial consequences (Ferreira, 2023).

The participants reported recurring situations both in their own companies and in their clients' companies. These scenarios were based on valuing or defending a compliance posture; however, their concrete actions consisted of not complying with policies and standards.

So much so that this perception is corroborated by examples such as one-off actions, like adjustments and corrections to systems and processes, which are carried out on the eve of audits or certification renewals, once approved, the teams and managers resume their work routines without worrying about the compliance required for their processes, according to E8 and E11.

> "…sometimes they do this *faking* [emphasis added] companies do this for the external auditor to see, for the ISO auditor to see "oh no, you have management... how nice" [E8]

> "... people don't want to, they think it's too fast, that there's no time for them to adapt. Sometimes that's true. And then you have to pretend that you have adapted" [E11]

In addition, cognitive dissonance is also revealed by simulated knowledge behavior. In the following excerpt, the conflict between action (signing without reading) and knowledge (not knowing about the guidelines and restrictions) is a report of cognitive dissonance.

> "we've had situations where people have been caught in a situation of non-compliance and they've said, "Ops, I didn't know I couldn't do that", "no, buddy, it's written there and you signed it"..."ah, but I signed it without reading it" [E7]

If managers, teams, and service providers do not adhere to policies and standards, the results sought in internal or external contracts can become inconsistent. This is because a lack of adherence to policies and standards can lead to inadequate processes, inconsistent decision-making, and a lack of control, directly impacting the company's ability to fulfill its contractual promises consistently and reliably.

The opportunistic stance concerns many participants, as it reveals that a lack of compliance can worsen the situation by exposing the company to more significant risks, making it more susceptible to cyber-attack fraud compromising trust, reputation, and finances.

Organizational Ethos. The findings in this research indicate that although they recognize opportunistic situations, information asymmetry, and knowledge gaps, participants value creating a company culture rooted in awareness, assessment, and management of business risks, including cyber threats. This suggests that risk management is not just an isolated process, but something incorporated into operations and the team's perspective. Based on the significance attributed by the participants and the reports collected, the term Organizational Ethos was coined to express this appreciation.

Risk management culture was highlighted when the organization's perspective and perceived attitude towards risk identification, assessment, and mitigation was brought to the fore. The participants perceive A risk management culture when awareness is raised about the importance of considering risks in all the company's decisions and processes.

By mentioning that 'so that people are more all ears, right?', participant E5 emphasized the need for people to be more attentive and receptive in order to cultivate a culture of excellence in the company. This implies that this mentality is not restricted to a specific area but permeates all actions and decisions (Damodaran, 2009).

Furthermore, participant E10 emphasized that 'it's a culture of excellence in general. That the place has a culture of excellence and people try to do their best there'. Here, importance is given to a culture of excellence in all areas (Assi, 2021). E10 pointed out that when a company cultivates a culture of excellence, it becomes an environment where people strive to achieve their best.

Moreover, finally, participant E16 highlighted the phrase 'everyone is a risk manager.' From this perspective, every professional becomes a cyber risk manager. This means that everyone would recognize the importance of identifying, assessing, and mitigating cyber threats in their activities. This approach would not be restricted to a specific area or group but would extend to every internal and external professional. So much so that the participant emphasizes the promotion of integrated risk management in the company in a new speech.

> “It's that people understand that risk management is not an area, it doesn't have to be a person, it has to be a process that will be crossed by the whole company [E16]

Up to this point in the analysis, the results have outlined an approach based on building trust, risk culture, and the role of people as cyber risk managers. This perspective has given these people the autonomy to observe and direct actions related to these risks. However, a discovery emerged that initially seemed contradictory: a strategic front to strengthen the Organizational Ethos concerning cyber risks would be the adoption of Zero Trust Governance.

In a Zero Trust Governance environment, security is based on strict authentication, granular authorization, and constant monitoring. Zero Trust architecture represents a cybersecurity paradigm that transcends network boundaries. Zero Trust Network Architecture (ZTNA) seeks to redefine the concept of trust, replacing implicit trust with an adaptive explicit trust approach (Google Cloud, 2023; Microsoft, 2021; NIST, 2023). Unlike traditional security projects based on presumed trust, the "Zero Trust" approach rejects this assumption. In other words, this governance does not assume that certain elements are trustworthy by default. Instead, it requires constant checks, regardless of the origin or history of the item under analysis, in pursuit of a high level of security and mitigation of cyber risks. The adoption of the "Zero

Trust" architecture is presented as a dynamic response to growing threats, as explained by participants E10 and E12:

> “it's that Zero Trust policy, you don't have prior access to anything, you're not induced to have access to anything. Everything you have access to you ask for, you declare its use, you declare the purpose of your request” [E10]

> "... I don't know if you've ever had the chance to go into the bureau. But there's nothing, you know? You leave your cell phone outside, you leave all your stuff, you go through a metal detector, you work in a place where there's no paper, the computer is fixed, things... like, the keys you have to print are all blocked, you know? There are two cameras, one on each side to see what you're doing” [E12]

After understanding the application of this type of governance, it becomes clear that the zero-trust approach should not be interpreted purely as a lack of trust in people. The focal point is to establish resilient systems that consider fallibility and potential risks, regardless of their origin, whether human or technological. This would demonstrate a proactive approach to dealing with the complexities inherent in security and reliability, strengthening resilience and human and technological interaction.

It is perceived that the adoption of the "Zero Trust" approach, coupled with the strengthening of the cyber risk management culture, has been perceived as a demonstration of transparency and commitment to protecting data and the privacy of clients and partners.

The predominant positioning in the results is aimed at directing efforts towards mitigating cyber risks since preserving reputation assumes a role of total relevance. In this context, there was considerable emphasis in the quantitative stage on the reputational value the company undertakes to be perceived by the market, partners, employees, and the government. This point is also widely emphasized in the interviews conducted.

In the results, it became clear that in a market where intangible assets such as reputation are as important as tangible assets such as facilities and equipment, competitiveness requires solid relationships with all the public involved (Fombrun, 2018).

In fact, participants E5 and E9 emphasize the protection of the company's reputation, suggesting that when faced with the choice between taking a risk that could damage the company's reputation and choosing not to take that risk, the preferred decision would be to avoid the risk, even if it means losing opportunities or gains.

> “so if you have to make a decision between taking a risk that could hurt your reputation and not taking the risk, it's better not to take the risk [E5].

> “so internally we take great care of our name, the name and reputation of our company, it's an ethical company, a company that values honesty [E9]

> “the issue of reputation is an indispensable pillar in decision-making and in analyzing the cost of implementing a security tool” [E6]

Reputation is also considered an essential element in decision-making and in analyzing the costs associated with implementing security tools, as E6 explained. One of the motivations put forward to protect the company's reputation can be further intensified by the fear of social rejection, including the phenomenon known as "cancellation." Nowadays, with the growing influence of social networks and online public opinion, social rejection can have a significant impact on a company's business, as explained by E2 and E12:

> “social rejection can have a significant impact on a company's business, as explained by E2 and E12:

> “that today image is everything, the cancellation thing on the web is something like that... it terrifies [E2]

> “we have a verb called "to cancel", you know? So, in the same way that you cancel a person, you often cancel a company, don't you?” [E12]

Dudenhoefer (2020) argues that the culture of cancellation involves taking a public stance against a person or institution due to actions considered objectionable or offensive. Without going into the merits of whether this practice is effective in holding decisions and actions accountable, the point is that this practice, especially for companies that need to have cyber security in their processes, causes the company to incur expenses for monitoring social media, according to the comments made by E1 and E2:

> “we use the services of a third-party company that gives us inputs on everything that is being said on the deep web, on the internet, on social networks, related to the brand and related to our clients” [E1]

> “there is monitoring of social networks to see if there is anything, if there is nothing about my company and possible attacks or possible fraud” [E2]

At the end of the analysis in this section, it becomes apparent that the risks arising from behavioral assumptions, resulting from the knowledge gap and opportunism and the search for Organizational Ethos, can have a decisive impact on the dynamics of contractual relationships and vice versa. This is because it involves organizational efforts and costs regarding time, resources, and investment.

4.2.2.3 Uncertainty Category

Category 6, 'Uncertainty' is part of the thematic axis 'Contractual Governance'. This category addresses issues related to uncertainty that play a critical role in the governance of contracts and agreements in organizations. Within this category, the subcategories provide an analysis of these facets of uncertainty and their implications for contractual governance.

Table 16 – Description of the Uncertainty and subcategories related to the thematic axes "Contractual Governance"

Thematic Axes	***A Posteriori* Categories**	**Subcategories**
2. Contractual Governance	**6. Uncertainty**	6.1 Dilemmas in Economic Decisions 6.2 Regulatory Complexity and Uncertainty 6.3 Technological and Cybernetic Uncertainty 6.4 Service Provider Guarantees 6.6 Uncertainty Management

Accurately predicting the implications of a security incident presents an almost impossible challenge. Since cyber threats are frequent and dynamic, dealing effectively with them requires considerable flexibility and adaptation for companies. Security threats and technologies have been a concern for stakeholders, considering that rapid technological change creates an environment of uncertainty, as it is difficult to predict which approaches, behaviors, and technologies will be used in the future.

Dilemmas in Economic Decisions. Among the measures adopted to manage risk, it is worth noting that companies have opted to include cyber insurance in their contracts to reduce the financial impacts in a materialized risk situation. Participant E15 described that in this type of insurance, the insurer requires the contracting company to implement specific control measures as part of the requirements for obtaining cyber insurance.

> “in order for the insurer to approve cyber insurance, it requires the company to implement a series of controls, and this turned out to be successful” [E15]

According to this participant, in the phrase "and it ended up working," he emphasizes that this approach has been successful, considering that this implementation resulted in introducing a new service to clients. However, the participants stressed that cyber insurance cannot prevent threats such as data leaks, and fraud. It has an *ex post* effect since its protection is perceived after the incidents (Klein, 2002; Williamson, 1985).

In this context, considerable investments and efforts have been made in response to uncertainty. A dilemma has emerged from this situation: the complex reconciliation of investment and security. This impasse revolves around harmonizing the search for growth and efficiency on the one hand and the protection of sensitive assets and information on the other. This dilemma results from constant competition in the allocation of resources aimed at fostering business development and safeguarding (Williamson, 1993a) cyber security measures.

The connection between disbelief and uncertainty can be explored by analyzing the reactions of the parties involved to the possibility of unrealized risks, as explained by participant E8:

> “since the risk hasn't been consummated yet, is the LGPD really going to happen? Does it make sense to make all this investment?" So accept the risk, because if it does happen, you've accepted it here and it's your problem", you know? ((laughs)) Then you tell the guy to sign, and he doesn't sign” [E8]

When there is the prospect that the risks will not materialize, doubts arise about the viability and justification of the investment. Questions turn to the validity of investing significant resources in something whose uncertain outcome has not yet materialized.

In this context, this uncertainty regarding the LGPD or protection measures can be analogously considered as a cascade effect of uncertainty: a situation in which doubts about the implementation and application of the regulation affect decisions from regulators, managers, and, lastly, the market. This dynamic process highlights how uncertainty can influence decisions at various layers of the decision-making hierarchy, affecting the LGPD's overarching objective: data protection. In addition, this uncertainty can also reverberate in contracts and

their completeness, making them potentially more vulnerable to contingencies and revisions in the face of the uncertainties associated with the regulation.

Extending the analysis further, this type of situation is not uncommon in companies - the lack of signature by the professional, despite verbal acceptance, highlights the conflict between recognizing the possibility of an adverse event and the willingness to make this acceptance official through a formal commitment.

In this sense, when we examine the results of the quantitative stage, on average, a third of companies have published or disclosed their investments in cyber security to the market (Table 7). However, this figure may seem timid, given the broad frontiers of digital investments that these companies have made. These investments, listed in Figure 10, range from introducing specific new products and services aimed at meeting customer demands and the requirements of the customer journey itself to strategic alliances and projects aimed at mitigating information gaps in decision-making, both within the company and in its value chain.

Deciding between allocating resources to investments and security often leads to conflicts during planning and deliberation meetings between managers. However, it is possible to identify exemplary situations that illustrate this scenario:

> “he wants to know the following: "Okay, if this happens, the size of the problem is this right here, and this is where we're going to lose money, got it?" "Got it." So, there's no point, therefore, you have to know how to communicate, because if not, the guy doesn't understand, and the guys don't... And they really don't have to understand, you know? It's not their role, *decision makers* [emphasis added] [E8]

The relevance pointed out by the participants regarding the participation of a cybersecurity and risk professional in such meetings emerged as a measure to mitigate these dilemmas.

> “but I notice that especially medium and large companies make quite an effort to have a CISO[14], to have a security professional involved in committees, in the board, right? [E1]

According to the participants, the cyber security specialist plays a key role in translating technical complexities into information accessible to decision-makers, thus contributing to

[14] CISO: Acronym for Chief Information Security Officer. The CISO is responsible for leading and managing an organization's information security strategies

targeted analysis and the appropriate selection of mitigation strategies. In addition, the presence of this professional in planning discussions, according to the participants, can result in the early identification of potentially vulnerable points, providing a more proactive and practical approach to cyber risk management. This is because, although uncertainty about cyber-attacks persists, a comment that came up frequently in this research, according to the excerpt from E1, was:

> “we often say that the incident is not a "if" it will happen, right? It's a "when" the incident will happen. [E1]

Therefore, this scenario leads to a concern to maintain a contingency plan to deal with incidents considered certain to occur but uncertain when they will occur. However, the lack of a contingency plan in some companies is often linked to the uncertainty about whether the risks will materialize (Damodaran, 2009; King, 2001). This dilemma arises when managers face the difficulty of accurately measuring the probability of a risk event occurring and its financial and operational impact on the business. As a result, some organizations may hesitate to allocate time, resources, and effort to draw up this plan, according to the excerpt from E6.

> “medium and small-sized companies often prioritize expenses first before looking at the benefits, mainly because many times if they don't conduct this analysis, they won't survive” [E6]

> “looking even more at small and medium-sized businesses, they don't have [a contingency plan]. So, it's necessary to be more concerned about the issue” [E1]

Focusing on expenses first is deeply rooted in the pragmatic nature of small and medium-sized companies, which often operate by prioritizing cost containment to minimize financial risks and maintain the business's financial health, essential factors for its long-term sustainability (Brazilian Micro and Small Business Support Service [SEBRAE], 2023; Welsh & White, 1981).

Regulatory Complexity and Insecurity. Regulatory ambiguities can lead to uncertainty or lack of clarity in a situation's standards, rules, regulations, or guidelines (Brousseau & Glachant, 2002; Williamson, 1993). At the same time, it creates challenges when trying to assess risks and make decisions. This is because when norms are ambiguous, different

parties can interpret them differently. This can lead to misunderstandings and conflicts about how actions should be conducted.

> “and then the change in legislation came and shook things up [...] they are not clear on how you are going to implement that, and you make inquiries, wait for a response, it really depends in that sense on the regulatory aspect” [E14]

The excerpt from participant E14 emphasizes the generation of uncertainty caused by practical application, leading to consultations and dependence on regulations to properly understand how to follow the new rules.

> “you know, these are extremely complex entities. Some are impossible to deal with, simply like that, we can't obtain a license” [E16]

Although addressed in the 'Government Relations' category, the issue of dependence on public services presents another perspective on this relationship between government and business - participant [E16] communicates the difficulty a professional faces when dealing with public bodies, which can result in uncertainty. In some cases, the inability to issue licenses reflects the bureaucratic or administrative challenges that can be present in certain governmental or regulatory institutions (Williamson, 1993a, 1999). This uncertainty is especially significant given that obtaining the license is for operating the company's activities.

What can be observed is that bureaucracy and slowness in obtaining licenses and concessions can limit the ability to innovate or develop new products and services. This can result in delays in launch plans and the ability to keep up with the competition - such as licenses to operate specific technologies and intellectual property.

In this context, one aspect that stands out in the results is the limitation of the guarantee offered to mitigate cyber risks, according to E11 and E3.

> “when there's an imposition of fines or other measures to curb certain behaviors, they are administrative measures that can be appealed” [E11]

> “how many experiences have we lived through where we know that paperwork isn't the only assurance, right? [...] And the argument, you can see it in a judicial process, a judge and two lawyers resolving their clients' cases, you'll see that the judge decides in favor of whoever presents a stronger argument, who brings more proof, evidence” [E3]

The arguments suggest that in a legal context, decisions are often based on the soundness of the arguments put forward by the parties involved, in addition to the evidence and proof presented. Although this procedure is evident in the legal sphere, the message conveyed is that uncertainty still prevails even in a regulated environment where the rules may be evident. This is because the resolution of legal disputes, including those involving cybercrime, can depend on various factors, such as the skill of argumentation, the weight of evidence, and the judge's perspective.

An additional point is that the legal system tends to prolong the time needed to reach conclusive answers or decisions in appeals processes. The absence of precise answers or well-established time limits can give rise to an environment permeated by questions about the outcome of cases. When this uncertainty involves the possibility of reputational losses, it becomes even more disturbing since the potential impact extends beyond the legal sphere, also affecting public perception and the trust of the parties involved (Williamson, 1993a, 1999).

Technological and Cyber Uncertainty. As companies witness their businesses expanding, they encounter a landscape of challenges and opportunities, often steeped in uncertainties. The rise in demand for products or services may necessitate significant infrastructure, human resources, and operational processes adaptations.

> "Am I prepared for this growth? Do I have a compatible infrastructure?" [E6]

In this context, one of the concerns identified is uncertainty regarding interoperability[15] (Mancini, 2021). This uncertainty is considered critical in cyber risk management, especially in environments where several systems, technologies, or services must operate in sync. This occurs when different systems or components interact, exchange information, and function in a coordinated manner.

This research has shown that interoperability is a factor of great concern and has gained prominence in the segments of the participants in this study, such as the financial sector, transportation and logistics, application-based services, utilities, manufacturing, TV channels,

[15] "The term interoperability in this report refers to the ability of different digital services to work together and communicate with one another. This interconnection can allow users to combine multiple services with complementary functionality, "or example when a photo received by email is automatically added to a user's online photo storage service" (Riley, 2020; Mancini, 2021, p. 12)

e-commerce, and insurance. It is worth noting that, especially in the context of the health sector, interoperability was mentioned as a critical element, considering that a lack of interoperability can lead to significant losses.

In addition, systems that are not interoperable can be more challenging to maintain and update, as changes to one system can unexpectedly affect other interconnected systems, bringing challenges, uncertainties, and costs in terms of maintenance.

The lack of interoperability between systems can have various causes. One of them is criminal attacks and intrusions, exemplified by cyber espionage perpetrated by those who, theoretically, should support companies in their security. This type of espionage involves actions by government agencies that seek to collect confidential and sensitive information from other governments, organizations, or people through cyber activities (Devanny et al., 2021). These actions can be politically, militarily, or economically motivated.

> “There's even espionage... in this sense, and even worse, there's governmental espionage, with countries seeking to digitally invade other countries” [E6]

Participant E9 expressed this concern and reflected on companies' worries when seeking innovation and investing in research and development to create products and services. This apprehension is rooted in the fear that this information could fall into the wrong hands, be it rival governments, competitors, or cyber criminals. This kind of exposure can negatively affect the company's position in the market beyond mere information leakage. Furthermore, even the government itself can become a victim, as participant E6 explained:

> “the other day we saw an action by the Federal Police identifying hackers who infiltrated some public agency. So, we observe this type of concern not only in private companies but also in public entities that provide societal protection services as well” [E6]

In addition to the uncertainty generated by interoperability, another highly relevant issue that emerges as technology advances is the lack of knowledge about the maturity and limits of cyberspace.

> “They are defining this world of the internet, the cyberspace, but I'm not aware of how that definition stands currently. And what's beneath? What is the limit? I don't know” [E8]

In this excerpt, participant E8 has a general perception of the internet and cyberspace, but he doesn't know how this definition is established. He also expresses a sense of uncertainty and raises the question of what is to come, referring to "what is down the road," in other words, what is to come in the future regarding technological developments. However, not in isolation, other participants show the same concern, according to participant E17:

> “What I see about this issue of the online world is that we're not fully accustomed to it yet, right?” [E17]

> “I believe that the biggest threats come from contact with the outside, staying at the organization's borders” [E3]

This last excerpt expresses the belief that the greatest threat originates when encountering external elements located precisely at the border of the organization. This indicates that the predominant source of risk or danger arises when there are interactions or connections with factors outside the organization's internal environment.

Providing a counterpoint to this position is essential, as it is not always exclusively determined by the external environment. Internal factors, such as risk management strategies, the resilience of the governance structure, and the ability to adapt, also perform an important role in reducing the vulnerabilities that amplify the environment of uncertainty:

> “It's like they have very powerful computers, so encryption is much faster. In just five minutes, the intruder encrypts the machine. However, I mentioned that sometimes you can't find the IT person in time to come here and save a machine like this” [E12]

Vulnerabilities can come from both internal and external perspectives. From participant E12's observations, this dynamic can be seen in both environments. Observing that an attacker can encrypt a system in minutes raises concerns about security and incident response. The speed with which the attack can be carried out highlights the need for robust security measures and a well-designed incident response plan. Internally, finding an IT professional in time highlights an additional challenge in managing cybersecurity incidents. A rapid response would be a basic premise for blocking or minimizing a hacker attack.

In this respect, literal interpretations, which can be linked to knowledge gaps, have the potential to lead to inaccurate attributions of the uncertainties faced by companies. It is, therefore, important to emphasize the importance of considering interpretation within a broader context, as E2 explained:

"Because if you take it literally, everything can be blamed on technology" [E2]

The view that technology is the only source of uncertainty can be simplistic. In this understanding, participant E2 reinforces that technology is not the only source of technological or cybernetic uncertainty. Although technology can contribute to uncertainty in specific contexts, it is essential to recognize that uncertainty can be influenced by several factors, including regulations, market changes, unpredictable technological advances, and even behaviors, as discussed in this study.

4.2.3 Thematic Axe 3: Mitigation of Cyber Risks in Services

The thematic axes 'Mitigation of Cyber Risks in Services' supported the analysis of the *a posteriori* category 'Cyber Risk Mitigation Actions', whose direction allowed an in-depth exploration of how the subcategories contribute to a comprehensive understanding of cyber risk management practices in services.

4.2.3.1 Cyber Risk Mitigation Actions

Category 7, 'Cyber Risk Mitigation Actions,' is an exclusive part of the thematic axes ' Mitigation of Cyber Risks in Services'. This category addresses companies' actions and strategies to mitigate cyber risks that could affect their services. These subcategories provide a framework for analyzing organizations' specific actions to mitigate cyber risks in their services, highlighting the importance of preparing for, managing, and responding to cyber threats.

Table 17 – Description of the Cyber Risk Mitigation Actions and subcategories related to the thematic axes " Mitigation of Cyber Risks in Services"

Thematic Axes	***A Posteriori* Categories**	**Subcategories**
3. Mitigation of Cyber Risks in Services	**7. Cyber Risk Mitigation Actions**	7.1 Recognizing Cyber Risks 7.2 Building Policies and Evidence of Compliance 7.3 Risk Management Oriented Company 7.4 Training and Risk Awareness 7.5 Crisis Management Process 7.6 Third-party risk management

Recognizing Cyber Risks. Recognizing protection measures is vital so people can take appropriate precautions, follow security guidelines, and understand the risks associated with a

given environment or situation. This recognition was widely highlighted in the results and is based on implementing security layers by the participants' companies.

Security layers are measures or components implemented in systems, networks, or applications to protect against security threats and risks (Schneier, 2022). According to the participants: “We have to go through several layers of security” [E2], and “there is a whole layer of security here” [E6].

In combating technology with technology, these layers of security would be one of the approaches used as a technological solution to solve or mitigate the problems often associated with the technology itself but are linked to its uninformed or abusive use (ChainAnalysis, 2023; Schneier, 2022).

In this way, recognizing cyber risks is a precursor to using technology (Hopkins, 2014; Strupczewski, 2021). Furthermore, protective measures were pointed out for the physical environment, which may involve understanding security protocols, such as access control systems and emergency procedures (Egan et al., 2019). In addition, preventive action was highlighted as the best way to mitigate risks, considering that many respondents emphasized that a reactive stance is the most common practice found in most companies:

> "In fact, if you consider public and even private security, technologies are developed after something has already happened” [E2]

> “You might have a situation where someone is committing fraud here and there in a process. There comes a point when you catch it, and eventually you catch them. Our challenge is to try to identify this as early as possible” [E7]

In addition to implementing security layers, other ways have been cited to act in predicting and neutralizing cyber threats:

> “So, today we're going to have issues, but why? Because the attackers, they've stepped up, they've improved their attack methods, and they've reached a point now where it's exactly like how we used to create a point of defense” [E12]

Participant E12 is concerned that the tactics used by attackers (cybercriminals hackers, among others) are becoming so sophisticated and effective that they are surpassing the security measures that used to be implemented to protect systems and information. This may indicate a

growing challenge in cyber security, requiring actions to adapt to business and evolve to anticipate new threats (Arachchilage & Love, 2014; Jain & Gupta, 2022).

In this context, the construction of policies and evidence of compliance can provide a series of guidelines and practices that can help implement better protection measures. These measures are applied with the support of technology to face the constantly evolving challenges, as will be discussed in the following subcategory.

Building Policies and Evidence of Compliance. In the view of the participants, it must be ensured that employees understand and identify the security measures and procedures to be followed concerning cyber threats. A dedicated cyber risk management structure would be vital for monitoring, assessing, and mitigating possible threats. This involves proactively identifying vulnerabilities, assessing the potential impact of a data breach, and developing strategies to minimize these risks (Biener et al., 2015; Brewer, 2000; Egan et al., 2019):

> "the governance structure itself, it already has the Corporate one. It already specifies the roles and responsibilities of all areas - there are several policies there for both providing and validating data access." [E6]

> "In a company where the direction is to deliver results at any cost, it becomes much more difficult for a person to stop, evaluate and observe the rules." [E14]

> "When I talk about technology risks, we have well-established roles within the company - meaning, regardless of the risk area." [E14]

The results of this study highlight the importance of Periodic Compliance and Ethics Committees in cyber risk management. Based on the participants' experience, the committees would play a multifaceted role in the organization, ensuring compliance with data security and privacy regulations and contributing to preventing and mitigating risks in the digital landscape.

The analysis points out that the constant review of security-related policies and procedures carried out by these committees allows for the early identification of possible threats, facing challenges such as information asymmetry, technological knowledge gaps, a highly competitive market, and uncertain environments. These challenges, inherent to the constantly evolving cyber environment, would make the committees' work even more relevant in promoting a culture of responsibility concerning cyber security.

In addition, the ability of the company's governance structure to coordinate responses in the event of cyber incidents could be linked to the relevance of these committees in minimizing damage and preserving the company's reputation.

> “So, what is done more often is that internally there are some committees, which are committees that stay in constant communication to understand how to perform this verification. The IT department working together to understand the business's needs, but at the same time, on the other hand, you have the information security department that stays there ensuring if it's compliant or not.” [E7]

Participant E7 highlights a context of internal collaboration in his company, in which committees and multidisciplinary teams maintain regular communication to understand and validate information security. The information technology team works in synergy to provide the necessary support, while the information security team oversees adherence to the stipulated guidelines.

Periodic Compliance and Ethics Committees would emerge as a critical component in safeguarding digital assets and sustaining sound and adaptable business practices in the face of the challenges of the cyber environment.

Risk Management Oriented Company. Concerning the risk management-oriented company, the results of this research highlight that Information Technology (IT) performs an intrinsic and vital role in the strategic decisions of organizations, opening space for the inclusion of security professionals in strategic discussions. In fact, Segate et al., 2019 emphasize this position in the literature by stating that the IT area is part of companies' strategic decisions.

Through automation, data analysis, and personalization, IT can improve operational processes, improve customer experience quality, and support company operations. Including security professionals in strategic discussions is necessary to guarantee the protection of digital assets and resilience against cyber threats. IT ensures the integrity of information, facilitates global expansion, and enables agile adaptation to market fluctuations, making it a key element in companies' strategies.

In addition to including information security and technology professionals in the company's strategic discussions, the issue of layout design to mitigate the risk of conflicts of interest is presented as a proposal for information security management.

However, at the same time, as concerns were raised about the participation of professionals who could be strategic in mitigating cyber risks in the company, another point noted in the analysis was that segregation of roles and responsibilities alone would not be enough. In addition, the design of the layout of the company's areas can support risk management about restricted access, combined with 'zero trust' governance, mentioned in the behavioral assumptions category.

> "…and they are located in a building separate from the back office there, because there is very sensitive information." [E2]

> "our team, for example, always works closely together and away from others who might... use that information..." [E4]

The statements by participants E2 and E4 mention a layout design that emphasizes proximity between people on the same team while keeping them away from others who may have conflicting interests (Akerlof, 1978; Milgrom & Roberts, 1992). This layout is a strategy for minimizing possible conflicts of interest and preserving the confidentiality of sensitive information. Proximity between team members facilitates synergy and internal communication, while distance from others can reduce the possibility of interactions that could result in disputes or unwanted disclosure of strategic data. Although this arrangement can contribute to efficiency and information security, it is also essential to consider the implications of isolation between areas and possible effects on knowledge sharing.

In addition to concerns about the layout of teams, the analysis revealed that participants value the role played by technology but also recognize that effectiveness is enhanced when there are transparent processes and workflows.

The analyses consistently emphasized the importance of investing in solid processes in parallel with adopting technology. Many participants pointed out that an approach centered solely on technology and innovation may be inadequate since processes play a crucial role in operational effectiveness, as E6 and E15 pointed out:

> "There's no point in going out and 'shooting' for evolution without being prepared." [E6]

> "I still think that there is a lot of investment in technology, there needs to be, it can't be any different, but there also needs to be investment in processes." [E15]

The perception is that technology alone is not a complete solution to risk management challenges. This conclusion reflects the growing awareness that the synergy between technology, innovation, and processes is fundamental to optimizing safety and efficiency. In this context, another relevant point raised is the presence of outdated and unreviewed processes, according to E13:

> “I'm going to be quite honest: there are companies - as I told you, I can't disclose names - but like this, you're surprised that it's something quite archaic” [E13]

This can have implications and complications, especially when introducing innovations or technologies. Obsolete processes may not be adequate to deal with the demands of emerging technologies, leading to operational, knowledge, and information gaps, and risks may not be easily identified.

A third important observation between the triad of technology, innovation, and processes is that an excess of restrictions and data control can stifle the agility of processes and underutilize the technologies implemented. Participant E10 highlighted this concern:

> I think this failure due to excess also exists. When you block so many processes for obtaining data, circulating data within the organization, you are in fact reducing the risk of exposure [E10]

Although control proves to be vital for security, it would need to find a balance to allow the effective use of technology. The analysis also made it possible to identify and highlight that an excessively restrictive approach can create obstacles to innovation and the exploitation of the full potential of the technological solutions adopted by the company.

In processes considered by the participants to be the most mature, they highlighted the importance of creating a threat-hunting team. This team would be dedicated to identifying incidents and vulnerabilities, both internal and external. One of the participants pointed out:

> “do we have a threat hunting team? Yes, we do! Internal, who look for vulnerabilities here and abroad” [E1]

The points made in this section emphasize that risk management cannot be a secondary action. It needs to be included in companies' strategies, combining technology, innovation, solid processes and the approaches mentioned in Organizational Ethos.

5 DISCUSSION OF THE COMBINATION OF RESULTS

In this section, the results stemming from the analysis of the two research phases will be presented. The subsections will center on the following subjects: Cyber Risk in Services Categories', with their respective frequency data contributing to the development and examination of ‘Governance Structures and Asset Specificity Degree in Cyber Risk Mitigation Practices’. Drawing upon the outcomes derived from the categories of cyber risks in services, it becomes possible to attain a comprehensive overview of the categories most frequently associated with internalization, the hybrid approach, and outsourcing.

Another factor explored concerns the effects of the governance structure on cyber risks in services. To this end, a theoretical model was developed, starting with consideration of the services specificities, and culminating in actions to mitigate cyber risks in services. In addition, a subsection is dedicated to 'Contribution and Expansion of Knowledge: The cyber environment as an asset specificity in Transaction Cost Theory.' In this context, it is proposed to include cyber services in the sectoral technological taxonomy of services, based on the approach of Miozzo & Soete (2001). This aims to identify the determinants, directions, and measurable characteristics identified throughout this research.

5.1 GOVERNANCE STRUCTURES IDENTIFIED IN CYBER RISK MITIGATION IN SERVICES

In the scope of this research, a analysis was conducted on the statements provided by the participants, which addressed various topics, including activities, decisions, assignments, concerns, and perceptions related to the coordination of these elements. This analysis allowed for the classification of governance structure choices adopted by companies that engage in service transactions regardless of the sector and are involved in activities requiring cyber risk mitigation.

As part of the investigation in this study, through the coding performed in the content analysis stage of the interviews, the frequency with which governance structures, including outsourcing, insourcing, and the hybrid mode, were evidenced by the participants was measured. This frequency was assessed using the IBM SPSS 29 software, as shown in Table 18.

Table 18 – Cyber Risk in Services Categories (Frequency)

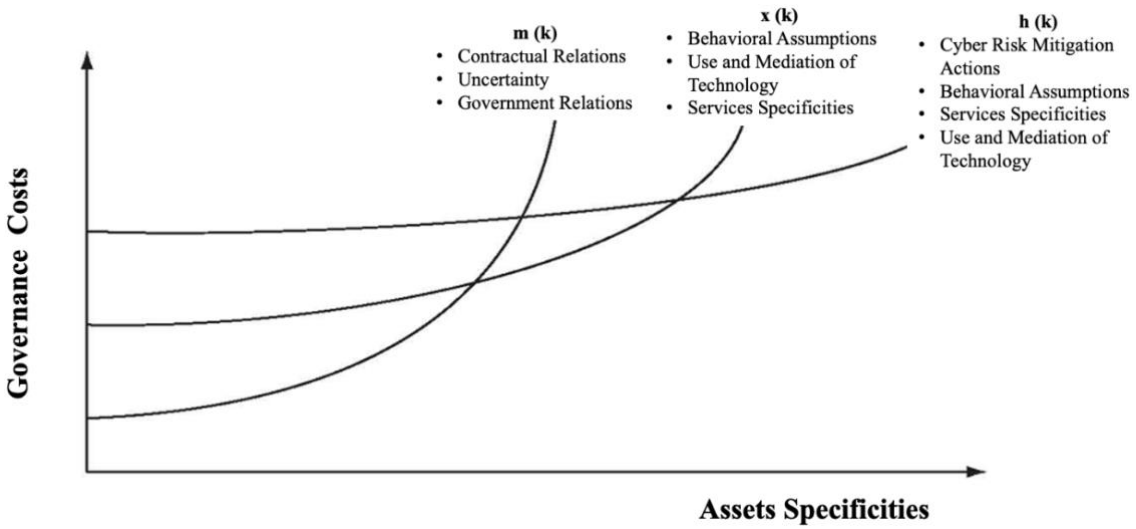
Outsourcing (Market)	**%**
Contratual Relations	82,3%
Uncertainty	57,1%
Governmental Relations	56,3%
Use and Mediation of Technology	18,6%
Services Specifities	15,9%
Behavioral Assumptions	7,9%
Cyber Risk Mitigation Actions	6,5%

Internalization (Hierarchy)	**%**
Cyber Risk Mitigation Actions	87,0%
Behavioral Assumptions	70,4%
Services Specifities	69,6%
Use and Mediation of Technology	64,7%
Uncertainty	28,6%
Governmental Relations	37,5%
Contratual Relations	14,0%

Hybrid Mode	**%**
Behavioral Assumptions	21,7%
Use and Mediation of Technology	16,7%
Services Specifities	14,5%
Uncertainty	14,3%
Cyber Risk Mitigation Actions	6,5%
Governmental Relations	6,2%
Contratual Relations	3,7%

The results presented reflect the analysis of cyber risk management and mitigation practices in services and how the different governance structures perform in this dynamic scenario. The governance structures could be represented graphically to provide a holistic visualization. This is particularly relevant in the context of cyber risks in services, considering that increased asset specificity is associated with higher governance costs, as shown in Figure 16.

Figure 16 – Governance Structures and Asset Specificity Degree in Cyber Risk Mitigation Practices



Source: author, based on Wiliamson (1975)

The Figure 16 presented conceptualizes how economic agents develop mechanisms to deal with transaction or governance costs, as theorized by Williamson (1975, 1996). The figure shows that there are three distinct categories of governance structures. Two are considered extreme: governance through hierarchy (h), which involves an internal organization with centralized control, and governance through the market (m), which involves transactions based on contracts. The third governance category is an intermediate form, i.e., a hybrid combining elements of hierarchical and market governance.

The scenario presented showed that 82.3% of the activities linked to contractual relations are outsourced, 14% are developed internally, and 3.7% have hybrid governance. This result shows that most companies have turned to the market for these services. Furthermore, this result suggests that although contracts may represent the core of transaction cost theory, circumstances may involve the strategic choice of taking advantage of specialized resources available on the market.

A concern became evident when companies decided to outsource related to IT development and project consultancy while also investing in training of these outsourced teams. However, as employees acquired skills, they began considering opportunities in other companies offering more attractive salaries. In this context, a game of expectations emerged, where the subsequent company reaped the rewards of the training investments made by the original company. However, this situation triggers a dilemma known as 'hold-up contracts' (Brousseau, & Glachant, 2002): the original company, aware of this dynamic, is reluctant to invest in outsourcing since it cannot wholly reap the benefits of this investment, resulting in

inefficiencies in the outsourced process. In addition, the high turnover rate among outsourced professionals was criticized by at least 55% of the participating companies.

With regard to the internalized part, it mainly encompasses in-house contracts based service level agreements (SLAs) (Perez, 2023), on targets, and direct contracts between the company and its employees. This indicates that the company chooses to keep these activities within its domains, possibly to have direct control over processes and results. It is important to note that the company is seeking to guarantee performance and quality standards by implementing target contracts and internal SLAs.

The activities associated with the environment of uncertainty are distributed as follows: 57.1% of them are coordinated through the market, 28.6% are managed internally by the company, and the remaining 14.3% are managed through hybrid governance. This strategic differentiation is illustrated by allocating almost a third of the activities conducted internally to specific companies while the remaining two are outsourced. According to the participants' accounts, this approach can be considered a thoughtful strategy. Outsourcing is seen as a practical approach to acquiring this external expertise, conferring a competitive advantage by taking advantage of external solutions (Williamson, 1991). In addition, the company can manage financial, operational, and cyber risks through collaboration with specialized partners while maintaining direct control over the internally managed portion (Bouncken et. al., 2018; Brandenburger & Nalebuff, 2020; Thorgren et al., 2009).

However, considering the perspective of the 28.6% of companies that have chosen to coordinate internally is of the utmost importance. This fact is reinforced by the motivations identified in the quantitative phase of this research, in which the development of internal expertise, the comparative analysis of costs and benefits, and the quest to reduce uncertainty in decision-making continue to be factors that led to the choice of internalization (see Section 4.1.3).

In the context of government relations, the coordination of these activities is distributed as follows: 56.3% through outsourcing, 37.5% are managed internally by the company, and the remaining 6.2% adopt a hybrid approach.

Most government relations were established through interactions with the market, which included hiring consultancy services, a lobbying, or a law firm to represent the company's interests before the government. This approach was consistent with companies seeking expertise in the market to deal with regulatory, judicial, and political aspects. The choice to outsource the mediation of licenses and concessions, rather than dealing directly with these

processes, is often rooted in the search for efficiency, concentration on core business, and access to expertise. These aspects were highlighted in the results of the quantitative stage.

Another relevant point is that outsourcing companies must familiarize themselves with the client's laws and regulations. This perspective also fits into the realm of government relations. This dynamic was perceived during the interviews, in which the presence of different clients from knowledge services companies was identified as participants in this study. This diverse set made it possible to conduct a comprehensive analysis from both the contractor's and the client's point of view. Thus, the profile of these service providers, who require in-depth knowledge and implementation of each client's specific laws and regulations, fits into the context of government relations established both through the market and through internalization.

Actions to mitigate cyber risks in services gained attention, comprising 87% of the mitigation activities performed internally by the company. On the other hand, companies that opted to look for solutions on the market or adopted a hybrid governance approach accounted for 6.5% each. These results reflect the prioritization of internal risk mitigation activities, showing that activities are highly specific to the company and cannot be easily transferred to third parties without losing knowledge or quality, where internalization is preferable.

Opportunistic behavior, conflicts of interest, and information asymmetry can arise in both the internalization and externalization of activities; in addition, can information and knowledge gaps (North, 1990; Williamson, 1993a). In this sense, it was identified that the behavioral assumptions resulted in 70.4% of them being managed internally, 7.9% outsourced, and 21.7% hybrid.

The higher tendency for behavioral assumptions to be more present in internalization processes leads to the inference that, internally, the company has the advantage of closer monitoring and an environment that can facilitate the direct exchange of knowledge. The proximity between teams and internal communication and monitoring systems can lead to greater identification of knowledge gaps or opportunism and, consequently, a more favorable environment for finding solutions. On the other hand, the distance and lack of direct access can make detecting and correcting knowledge gaps difficult when activities are outsourced quickly.

Behavioral assumptions, such as opportunism and the knowledge gap, can vary in their manifestation (Dequech, 2001; Simon, 1990; Williamson, 1981). In other situations, these behaviors are perceptible in interactions, while they may be more veiled in other contexts. However, regardless of their intensity or frequency, it is through contractual safeguards that it is possible to coordinate the effects of these assumptions. Consequently, the results suggest that

the safeguard approach is more effectively managed internally. In Miller's (1992) view, hierarchy is an effective structure when it represents the best option that individuals can achieve within a viable institutional arrangement.

Concerning the specificities of the services, 69.6% are internalized service activities. Outsourcing accounted for 15.9%, while the hybrid approach accounted for 14.5%. Internalized activities cover various intangible services, from the development of controls and knowledge services to in-house strategic security consultancy. In fact, in this research, a significant presence of companies have sought vertical integration to oversee their entire value chain. In these cases, verticalization was presented as a specific business strategy, especially when integrating segments such as financial services, IT development, technology-oriented services, and logistics. Additionally, in outsourcing and hybrid governance, various services are covered in the specificities of digital services. Items such as contracting providers, the use of cloud storage, and streaming services comprise a significant portion of the outsourcing contemplated to reduce investments in fixed assets. This leads us to understand that the companies that adopt these outsourced services have chosen not to invest in fixed assets. This approach can be especially advantageous in sectors subject to rapid technological change (Ben-Asher & Gonzalez, 2015)

Concluding hierarchical governance or internalization, the use and mediation of technology resulted in a percentage of 64.7% about internalized activities, while 18.6% correspond to outsourcing and 16.7% represent the hybrid model.

The activities involving the use and mediation of technology were mainly manifested in digital services activities, which have been part of both the core and peripheral business operations of technology-oriented companies, as presented in the quantitative stage of this study (Figure 7 and Figure 8). Also noteworthy within this category is the return versus investment analysis, where governance is defined through cost-benefit analysis.

Understanding how companies coordinate their governance structures about the specificities of services is of great importance in the context of transaction costs, where the balance between the specificities of assets and governance costs can impact the strategy and conduct of these companies' businesses.

The specific and specialized nature of these assets is justified by the decision to internalize cyber risk mitigation actions, the management of behavioral assumptions, specificities services, and the use and mediation of technology.

In an environment where cyber security and technology play a crucial role, internalizing these activities reflects the understanding that these assets are intrinsically linked to the

company's core operations. This is because, as Gallouj (2000) noted, technology is undoubtedly a central element of service innovation, driving efficiency, meeting customer expectations, and providing competitive advantages.

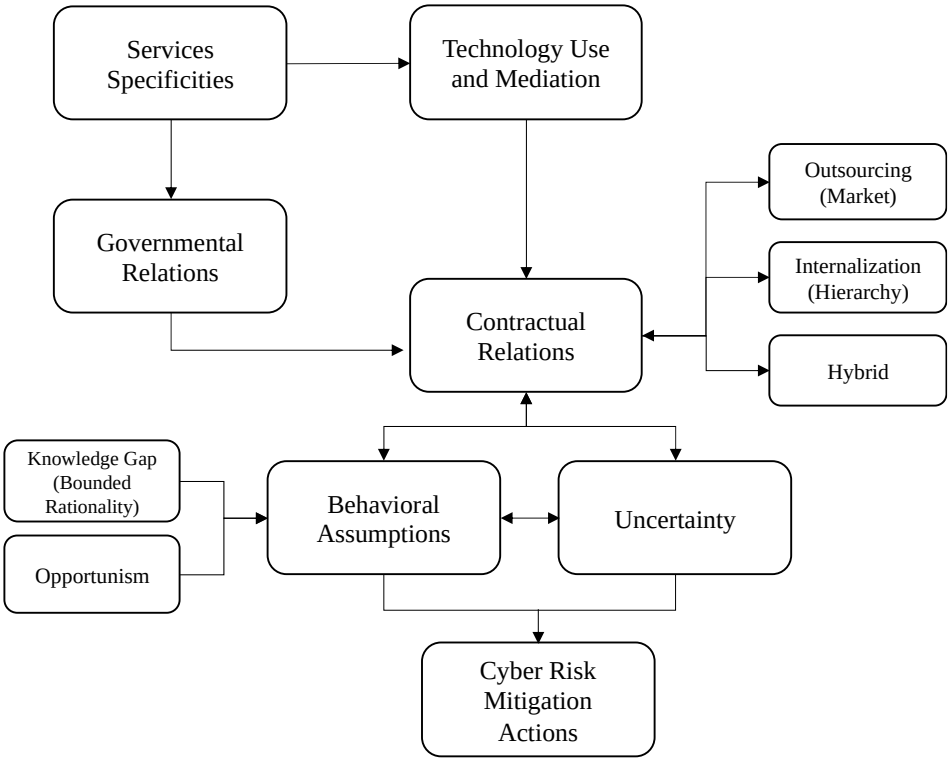
Activities involving cyber risk mitigation and the use and mediation of technology, for example, require specialized knowledge and resources tailored to the company's unique needs (Williamson, 1985). This results in the formation of specificities assets, which have substantial value within the organization's context but may have little value outside that environment. This finding aligns with Transaction Cost Economics, which establishes a relationship between asset specificity and increased governance costs. In the context of cyber risks, the internalization of mitigating actions, along with other categories more suited to this type of governance, contributes to higher governance costs.

The choice of outsourcing, encompassing contractual and governmental relations and the inevitable associated uncertainty, is based on the need to balance complex operational and strategic variables. The decision to outsource part of operations emerges as a response to the dynamics of the business environment, where agility and specialization can be crucial factors for competitiveness.

In this sense, the choice of governance structure affects institutions' efficiency, costs, and competitiveness. It influences how companies mitigate the cyber risks arising from the services and operations they are managing. The results show that the governance structure can enable companies to focus on their core competencies, reduce operating costs, and increase flexibility; this choice allows companies to organize themselves ambidextrously (March, 1991; Popadiuk, 2015), i.e., exploit opportunities and manage cyber risks simultaneously in the services contracted and provided.

This assessment is fundamental, as the choice of governance structure is based on available resources, strategies with growth objectives, and competitive advantage (Barney, 1991; Foss, 2005; Peteraf, 1993; Williamson, 1985). From the point of view of business decision-making, managers analyze and decide through contracts which activities are best allocated internally and which can be conducted through outsourcing or even through the hybrid mode, as shown in Figure 17.

Figure 17 – Theoretical Model of the Effects of the governance structure on the practices of mitigating cyber risks in services



The 'Theoretical Model of the Effects of Governance Structure on Cyber Risk Mitigation Practices in Services' supports exploring the relationship between governance structure (how a company or organization is coordinated and controlled) and cyber risk mitigation practices in services. This research has made it possible to build this model, seeking to understand how governance structures can affect the mitigation of cyber risks in services. The consequences for contract management can be demonstrated based on the specific characteristics of services, their dependence on government regulations, and the use and mediation of technology. Therefore, each topic of this model will be presented to understand these dynamics.

Services Specificities. When analyzing the results of the organization of service activities for the segment indicated by the participating companies, it can be seen that, although there are some industry terminology standards to be followed, a business model can differ "from the standard" to the point of appearing not to belong to an industry. Companies are involved in various services, focusing on digital, financial, knowledge, and commercial. This suggests that they are meeting diverse market needs and demands. Many of these services involve intangible assets, such as intellectual property, proprietary knowledge, algorithms, customized systems, and unique relationships, according to the results obtained in this study. These assets are

difficult to quantify and transfer, making the sale or donation of these services challenging (Farina et al., 2017; Williamson, 1985). Companies operating in financial and digital services areas deal with sensitive and valuable information, like financial data, customers' personal information, and intellectual property. This specialization makes these companies attractive targets for cybercriminals (Bowcut, 2021). In addition, financial services are subject to strict regulations, and transferring services within these sectors requires compliance with these regulations, which can be time-consuming, costly, and effort intensive.

In this sense, the first stage of this study provided relevant data and information that made it possible to identify a company's core and peripheral activities. This identification was fundamental to discerning the specificities of services, differentiating those that make up the core of the business, responsible for generating revenue, from those that complement these activities, known as peripheral activities, as stipulated by Djellal and Gallouj (2023). The data collected during the interviews showed that these central and peripheral activities are crucial to the development of the business. They are relevant to understanding how they are related and interdependent, especially the governance structures adopted. Understanding the interactions between these activities plays a fundamental role in making strategic decisions and optimizing business processes, contributing to the success and efficiency of the organization.

These central and peripheral activities are crucial to the development of the business. They are relevant to understanding how they are related and interdependent, especially concerning the governance structures adopted. Another striking point in this study is that the percentage of activities developed as peripheral is higher than core business activities. This may be related to the diversification of services companies have sought to expand their client offerings.

This makes it clear that the relationship between core and peripheral activities can influence the choice of a company's governance structure. Although transaction cost theory predicts that specialized and complementary peripheral activities can lead to strategic partnerships or outsourcing to optimize these functions while maintaining control over core activities, the results revealed that many companies consider specific services a critical competitive differentiation factor. This is because such services contribute directly to their competitive advantage, deciding to internalize them strategically to maintain more direct control over quality, innovation, and economic rents (Barney & Clarck, 2007).

When evaluating service specificity in this study, the unique characteristics of the service in question can influence how contracts are developed and managed (Brousseau & Glachant, 2002). This is because specialized services require specific security approaches to

protect the digital assets associated with that specific service. This is relevant since all the participating companies are investing in technology, either in maintaining it or developing new initiatives (as shown in Table 5, Section 4.1.7).

Use and Mediation of Technology. Technology is crucial in developing companies, especially those transacting digital services. In this scenario, services are not just a means to interact with customers; they represent the company's core purpose. This is evident in providing knowledge, commercial, administrative, financial, and other areas integral to the service economy.

Depending on the specifics of the services, these will affect how the use and mediation of technology are managed. Furthermore, assessing the technological implications involves understanding how technology is used and mediated to provide the service and identifying potential vulnerabilities and threats, which is fundamental to protecting digital assets and the integrity of services.

Companies that offer digital services as a core and peripheral activity and tend to internalize their activities to a greater extent than seeking outsourcing in the market have a high dependence on technology for contracting and providing services. The cloud services market is growing and has created opportunities for traditional companies to diversify their businesses and enter new markets. This study observed that two-thirds of the companies have incorporated new technologies into their operations that were not previously used. This finding reinforces the need for companies to invest in innovation and adaptation to the market (Birkinshaw et al., 2016), aspects that predominated in the perceptions shared by the participants throughout the two phases of the survey.

Investment in technology may not necessarily be oriented towards market growth, as indicated by the results obtained from companies in Cluster 3, where 70% of investments are directed towards new technologies. However, only 3% of these companies have pursued intensive growth strategies. Companies that handle administrative and digital activities internally often prioritize operational efficiency and internal control. Through technology investments, they aim to automate processes, enhance operational accuracy, and reduce internal costs. This strategy can serve as a means to uphold transaction quality and efficiency. Additionally, traditional sectors are demonstrating their capacity to adapt to technological changes by integrating digital services into their business models. This underscores the growing significance of innovation for competitiveness, even within historically conservative industries.

An important point to highlight is the concern that emerges before deciding to invest in technology. Managers have shown concern about investments in cyber and information

security, given the direct impact of these investments on companies' financial results and availability of capital. The costs related to the acquisition of technologies, team training (human-technology interaction), and protection measures are key factors in this context. Uncertainty about the financial impact is often associated with fines resulting from data leaks and fraud, influencing technology investment decisions. In addition, the risk of reputational damage is emphasized in the participants' responses. Therefore, continuous investment in technology is a characteristic of the sector, reflecting its dynamic nature, especially in companies that offer or contract services.

This data highlights how the services specificities, business objectives, and economic growth strategies shape approaches to technology and cyber security in different business sectors. Each company encounters distinct challenges and opportunities, which mirror the ever-changing dynamics of the market. Consequently, comprehending key technologies and their susceptibility to cyber threats empowers companies to pinpoint specific risks that could impact their digital assets and operations as they strive for growth in the market. his approach enables them to allocate security resources and implement strategies to address the most critical risks effectively. This analysis assists in the assessment of cyber risks.

A consideration of technology in cyber risk mitigation strategies has been found to help avoid costly litigation. In a security breach, the presence of contracts can play a crucial role in determining liability and remedial measures (Dixit, 2002; Frant, 1996; Klein, 2002; Miller, 1992). This can reduce the potential for protracted legal disputes and their associated costs.

In addition, the use and mediation of technology in modern business operations have a significant impact on interactions with the government. Technology permeates many aspects of business, from data collection and storage to cyber security and regulatory compliance.

Governmental Relations. Alongside the Use and Mediation of Services, Government Relations is essential, especially in regulated sectors or where the government has direct involvement. If the service is subject to strict government regulations, the governance structure may need to incorporate legal requirements and regulatory compliance. Dependence on the government may require a greater emphasis on transparency, accountability, and reporting since the government and the general public may be interested in the activities and results.

When a company has a direct dependence on public services, this means that its operational processes or even the viability of its business depend heavily on the services provided by the government or public sector entities. Outsourcing government intermediation activities can be mandatory to ensure compliance with regulations and compliance in

government matters. This can include obtaining licenses, regulatory compliance, and influencing policy formulation or lobbying practices.

The interaction between formal and informal rules (Klein, 1999; North, 1990) plays a fundamental role in shaping relations between companies and government. In this context, how the government relates to and operates profoundly impacts the company's business model and activities. Depending on the specific nature of the service offered, as well as the company's degree of emphasis on innovation, adaptation, and investments in technology, interactions with the government in line with the company's business model, economic activity, and location can influence or even determine the choice of the most appropriate governance structure to manage its operations.

Cyber risk management, including cyber security and data protection, are critical aspects of modern business operations. The relationship with the government involves the need to report security breaches and take measures to prevent and mitigate cyber threats. This is also reflected in drafting contracts that include specific cyber security and data protection clauses.

Government relations can influence the choice of governance structure. The actions of institutions at macro and meso levels can directly impact companies' transaction costs. This impact can increase or decrease these costs, depending on how the translation of business models is carried out. This context highlights the complexity of operating in the digital economy, where digital services have to deal with specific regulations and charges that can not only impact their costs but also shape the dynamics of the market in which they operate. The reference to 'public pricing,' for example, as a fee related to the intensive use of urban roads, indicates that digital services, such as app-based deliveries, face specific costs associated with using urban roads for their activities.

Contractual relations. What is a firm? A set of contracts, all incomplete. This definition reveals that the firm is an organizational structure comprising a series of contractual agreements between its constituent parts, such as employees, managers, service providers, and partners. However, these contracts cannot foresee all the possible situations that may arise over time, making them incomplete by nature.

The decision on the contractual approach was presented as one based on an analysis of the requirements, available resources, costs, urgency, and complexity, all of which are linked to the specificities of the service. It is also essential to consider the long-term implications, such as the ability to maintain a qualified in-house team or depend on specific suppliers in the long term. These decisions relate to the choice between internalizing the activity, outsourcing, or a

hybrid approach, and the units of analysis stem from the individual transactions associated with each choice.

In an effort to mitigate this inevitable incompleteness (Llewellyn, 1931; Williamson, 1985), some parties resort to generic "force majeure" or "unforeseen circumstances" clauses in the hope of covering a more comprehensive range of unexpected scenarios. However, even these clauses can be interpreted differently, often leading to further debate and uncertainty rather than swift resolution.

In this perspective, the technologies and diversification of services promoted by the cyber environment have changed the particularities of transactions, leading to the challenge of assessing and measuring the impacts on the specificity of assets and their dimensions, generating value for the parties involved. By incorporating the contractual decision into a cyber risk management process, the organization tends to ensure that resources are allocated in a targeted manner to meet cyber security challenges and mitigate the risks associated with the service in question. If the service requires specific assets or valuable resources, this can impact how governance is established to ensure the efficient use of these resources. However, several factors in addition to the characteristics of the assets must be taken into account when assessing the cost of governance and the organizational structure of a company, including the context of the markets in which it operates, the interactions between these markets, and the levels of uncertainty involved (Silva & Saes, 2007).

Depending on the organization's choice of governance structure, how it acquires and manages the resources needed to deal with cyber risks can have a significant impact on both the effectiveness of cyber security and the mitigation of these risks:

- Outsourcing: the option of hiring external companies with expertise in cyber security was seen as a viable approach to harnessing specialized knowledge without the need to invest in internal training. This strategy can be advantageous when the organization is looking for a quick and effective solution or does not have the necessary resources to develop in-house, such as "assets, capabilities, organizational processes and competencies" (Barney, 1991, p. 101).

A relevant observation that emerged in the second phase of the research is that small and medium-sized companies often resort to outsourcing to a greater extent. However, this is not a premise, as the different costs involved in transactions can influence the choice between market and hierarchical governance structures.

The decision by smaller companies to outsource has proven to be a response to the scarcity of resources and the search for efficiency. However, this choice has challenges, especially when maintaining the desired quality standards. Decreased direct control over

outsourced activities and the need to manage contractual relationships can be more pronounced in these scenarios. In addition, when transactions or agreements cannot be adequately defined in detailed contracts due to complexity, uncertainty, or other limitations, a company is established to manage and coordinate these activities more directly and effectively (Hart, 1995).

This leads to the point that while large companies have the resources and financial capacity to decide on and maintain a broader range of operations in-house, smaller companies often turn to outsourcing as a strategy to access specialized resources and focus their efforts on what matters to their business. The literature highlights that in situations where the long-term relationship involves greater trust and interdependence, the governance structure can evolve into something more collaborative, based on partnership (Ménard, 2004; Williamson, 1985).

- Internalization: developing in-house means investing in resources, which has proven appropriate when the service requires specialized expertise. The company has the resources and capabilities to train and retain talent in-house, as practiced by some participating companies. It is worth noting that, according to the data, 52.6% of the companies analyzed have an annual turnover of more than 1 billion dollars, indicating that the majority of them consider insourcing to meet security requirements aligned with their organizational culture and strategic objectives. In addition, internal contracting aims to provide a level of control over security operations and customization to the organization's needs.

Internal organization (vertical integration) is a more attractive choice when the assets involved are highly specific (Zylbersztajn, 1995). In addition, companies have adopted verticalization to broaden the scope of services and products, covering various aspects such as infrastructure, equipment, information, technology, and personnel training. Verticalization has been adopted to avoid coordination and negotiation challenges when the company relies heavily on third parties, such as logistics, financial services, and IT development. This means that by internalizing more processes, the company can reduce its dependence on contracts and external agreements and manage its operations more closely.

Drawing a parallel between the diversification of the supply of services, which is highlighted in the results of peripheral activities and the verticalization, Klein and Alchian (1978) define "quasi-rent" as the difference between the value that a specific asset has in its best use and its subsequent best alternative use. Thus, the greater the specificity of the traded asset, the greater the likelihood of "quasi-income transactions." In other words, the adoption of vertical integration as a form of transaction governance can occur when "quasi-income transactions" are present. Vertical integration can support reducing governance costs, such as monitoring costs and ensuring contract compliance. This means that companies can use vertical

integration to capture "quasi-rents" and increase their profits, which can be the primary justification for having more peripheral activities than core business (Saes, 2007, 2009).

Furthermore, this is confirmed by the reports of at least 20% of the participants that their companies have adopted the process of verticalizing previously outsourced services and have obtained additional income by offering these services as third-party services. In this sense, Transaction Cost Economics has generated much empirical research focused on quasi-income from specific assets (Zylbersztanjn, 2019). However, internal organization can be effective in terms of adaptation when assets are highly specific. However, this can come with bureaucratic costs and the loss of some competitive incentives the market provides.

In this scenario, Granovetter (1985) challenges Williamson's traditional view of transaction costs, which argues that hierarchical organizational structures (governance through internalization) result in lower transaction costs and are always effective. The author points to the need to examine power relations and decision-making in organizational hierarchies in greater depth. This broader perspective can be explored in greater depth in the context of the verticalization of activities, where the company provides services for itself and exploits opportunities for additional revenue. In any case, choosing between the internal organization and the market involves a continuous balance between efficiency and costs.

- **Hybrid governance**: hybrid approach involves combining internal and external resources. This can include hiring service providers to work alongside the internal team or for specific projects. This approach can be flexible and effective in meeting different demands and resources.

The recognition of administrative activities and the expanded vision of core business are points of emphasis in the results of this study. This result is relevant due to the observation that much of the literature does not usually consider administrative activities to be intrinsic components of companies' core operations (Johnston & Clark, 2005; Matteis, 1979), focusing instead on the efficiency of internal processes and constant cost reduction. This recognition emphasizes that companies in Cluster 2, especially those operating in sectors with strong regulation, value internal administrative services as a risk management tool, established through contracts.

However, the results of this survey also reveal a trend towards outsourcing and internalizing administrative activities, especially in medium-sized and large companies with less regulation. This practice has become even more prominent. The results indicated that companies have chosen to outsource and internalize administrative services and knowledge

services, including accounting, project consulting, and IT development, in order to reduce operating costs and focus more on their core competencies (Prahalad & Hamel, 2009).

Uncertainty. When examining the impact of governance structures in companies regarding cyber risk mitigation, the research results unveil a complex relationship. The quantitative research underscores the primary motivation for internalizing activities as the reduction of perceived uncertainty, indicating that companies seek greater control in critical areas like cybersecurity and compliance, where uncertainty is a significant component. However, the literature suggests that uncertainty often stems from the company's boundaries, "where market governance is sustained by standardized transactions of all kinds, regardless of the degree of uncertainty" (Williamson, 1985, p. 79).

Internalizing these activities is a viable strategy if the company deals with sensitive and strategic information that can be affected by opportunistic behavior and knowledge gaps (Alchian & Demsetz, 1972). Internalization would allow the company to reduce the risk of exposure to behavioral assumptions, such as opportunistic behavior by third parties or knowledge gaps. This would align with the idea that internalization is a way of seeking greater control over critical areas, especially when uncertainty is a concern (Williamson, 1979; 1991). In addition, the results revealed that in outsourcing, contractual relationships, governmental issues, and uncertainty were attributes frequently mentioned in the interviews, highlighting the importance of these elements in the decision to outsource. These factors played a role in the companies' cyber risk management and governance strategy. In this context, both results lead to the understanding that companies seek security and management of these activities, which can be essential for mitigating cyber risks and protecting strategic information.

The analysis showed that governance structures have implications for cyber risk mitigation. In a general analysis, although the motivation for internalization is often expressed as reducing uncertainty (Section 4.1.3), this uncertainty is specifically related to those areas where the company seeks greater control, even when external uncertainty, from the company's borders, persists. The frequency of these attributes in the interviews underscores their relevance to companies' decision-making in relation to cyber risk management.

The choice between insourcing and outsourcing should be informed by the specific nature of the uncertainty and each company's cyber risk mitigation goals, considering the uncertainty arising from external boundaries, whether complex (Keynes, 1937) or estimated (Knight, 1921). Services that involve high risks or uncertainties may require a certain governance structure, including the implementation of adequate safeguards, to ensure that the results meet expectations and protect the parties involved.

Behavioral Assumptions. The results revealed that the choice to internalize and adopt a hybrid governance model can be justified by a company's aim to deal more closely with behavioral assumptions, taking advantage of the internal environment to mitigate the contractual risks associated with these behaviors. However, the decision to adopt a governance structure, whether through internalization or externalization of activities related to cyber risk mitigation, can be influenced by several considerations. Internalization has been shown to offer advantages such as greater control, direct monitoring, agility in responding to cyber threats, and the development of technical expertise within the organization. On the other hand, outsourcing can be a viable strategic option, reducing operating costs in some circumstances and providing access to specialized expertise.

The final decision must result from a careful analysis that considers both the organization's cyber security strategy and financial capacity. In addition, if the company with sensitive information or if strategic issues play a critical role in its transactions, these factors must be carefully pondered in the decision to adopt a governance structure. For this reason, the inclusion of bidirectional arrows between contractual relations and the options of internalization, outsourcing, and hybrid mode is justified by the central role that contract monitoring plays in evaluating and adjusting governance structures. The analysis of results and contract execution is a critical factor in optimizing the choice of governance structure.

The purpose of this model is to provide a conceptual framework for understanding and analyzing the interaction between factors and how they influence cyber risk mitigation in services. In addition, it seeks to direct research, develop cyber security strategies and decisions regarding the choice of governance structure.

5.2 CONTRIBUTION AND EXPANSION OF KNOWLEDGE: THE CYBER ENVIRONMENT AS AN ASSET SPECIFICITY IN TRANSACTION COST THEORY

This study undertook a detailed investigation to meet this research's objectives. However, along the way, opportunities were identified to advance theoretical understanding. The research results revealed an interesting addition to the existing knowledge in the Transaction Cost literature.

When evaluating the results of the '*a posteriori* categories', two of them, referred to as "Use and Mediation of Technology" and "Cyber Risk Mitigation Actions," stood out with greater frequency among the survey participants. It was noted that these elements had not

previously been addressed in governance structures, specifically in the asset's specificities in the Transaction Cost Theory.

It should be emphasized that, from the 1970s to the mid-2000s, Williamson pioneered Transaction Cost Theory's paths. This theory is constantly concerned with discovering efficient ways of obtaining and allocating economic and financial resources in companies through governance structures. Since then, business models, markets, and the economy have evolved significantly, and various studies have highlighted the importance of governance structures.

Although the theory continues to generate various scientific fruits and relevant research and remains fully applicable to current business contexts, it is essential to emphasize that incorporating the 'cyber asset' can enrich and update it, opening doors to future research opportunities.

Although the addition of cyber assets was not the objective of this study, it is relevant to contribute to the evolution of theory and literature, providing new findings that can enrich the field's understanding.

In this context, the service economy emerges as a critical element, playing roles both as an end activity and a middle activity; in other words, roles focused on the contribution, influence, or value that the service brings to the dynamics of the cyber environment and its operations.

This study also revealed that digital services emerge as central elements of cyber assets. Digital services permeate various operations, and business dynamics highlight the importance of understanding and analyzing this comprehensive integration. Digital services can be considered asset specificities, influencing transaction costs between different parties involved in an economic transaction. In addition, several elements can be observed that highlight asset specificity within the digital services and cyber environment scenario. Specific investments, adaptations, and the creation of new processes, the importance of expertise covering activities related to the cyber environment, and the interactive nature of these assets emerged as critical factors. The intrinsic interdependence of these assets manifests itself through interactions with other elements of the digital ecosystem. These interactions occur through interconnected platforms and services, where each component contributes to the functionality and connectivity of the ecosystem.

In this sense, just as Transaction Cost Theory analyzes organizational and contractual choices and the efficiency of markets, the Service Economy in contemporary economies, the latter exerts a significant influence on a wide range of socio-economic aspects, such as job creation, contribution to Gross Domestic Product (GDP), fostering innovation and

technological advances (Gallouj, 2002). In the context of the digital economy and the growing importance of technology, digital services play a crucial role in service innovation, often transforming how services are designed and delivered (Bryson, 2023; Witell, 2023).

Regardless of the treatment for temporal specificity attributed by Williamson, the opportunity is given to explore new ways of operationalizing it, including through the cyber environment:

> While the relation of technology to organization remains important, it is scarcely determinative […] Rather I contend that transactional considerations, not technology, are typically decisive in determining which mode of organization will obtain in what circumstances and why. Central to the analysis is what I refer to as the "organizational failures framework." (Williamson, 1975, p. 2)

Cyberspace is a digital ecosystem that encompasses the technological infrastructure, communications, information systems, and human-machine interactions that take place on the internet and other digital media; it is a space where information is created, transmitted, stored, and shared (Beer, 2002), and gives dynamism to business, communication, education, research and many other aspects of modern society (Lévy, 2000, 2003).

Cyber risks have become a critical concern for companies in an increasingly digital and connected world. Rosati, Gogolin, and Lynn (2022) point out deficiencies in internal organizational control, whose cybersecurity failures are considered significant risk factors that negatively affect companies' financial results.

Furthermore, in the constantly evolving cyber environment, specificities assets in the contractual relationship between organizations can result in a relationship of mutual dependence, as noted by Williamson (1975), even if this relationship is highly asymmetrical. This can translate into shared cyber risks, as organizations depend on each other to protect their systems and data in the ever-changing digital environment.

In addition, some research describes ambidextrous organizations as those able to simultaneously exploit existing competencies and new opportunities (Raisch & Birkinshaw, 2008). An organization that focuses solely on exploitation can generate an endless cycle of searches and failures regarding security and data protection. While those that focus exclusively on exploitation obtain returns, but with shaky sustainability, facing increasing risks of cyber-attacks (Levinthal & March, 1993).

Gallouj and Djellal (2011) complement this perspective by arguing that the concept of transaction costs in the context of service innovation can be particularly significant due to the uncertainty and ambiguity associated with service innovations and the need to negotiate new and often complex service contracts. Service innovation often requires drafting new contracts or modifying existing contracts to accommodate the changes introduced by the innovation. These contracts can be complex due to the innovative nature of the services and the need to address specific related issues. Negotiating and drafting complex contracts also contributes to transaction costs.

Although technological innovations may originate in the manufacturing industry, their service use shapes their nature. Based on the approaches of Pavitt (1984) and Miozzo & Soete (2001), this technological sector taxonomy was expanded to incorporate digital platform services and digital services to provide a comprehensive categorization structure for companies operating in the cyber environment. This taxonomy takes into account criteria such as companies' typical core activities, sources of technology used, the nature of manufacturing or service provision, the profile of users served, value appropriation strategies, technological trajectory, sources of technological knowledge, and the relative size of innovative companies.

Figure 18 – Cyber Services Taxonomy: Digital Services and Platforms

Firm category	Typical Core Activities	Sources of Technology Manuf/Serv	Type of User	Means of Appropriation	Technological Trajectory	Source of Technology	Relative size of innovating firms
Specialized Suppliers	**Digital Platforms**	**Services**	**Price Sensitive Performance Sensitive**	R & D, Know-how, Skills, Copyright, Product Differentiation, Communities, Brand reputation, Patent, Process Certifications and Regulatory Compliance	**System Design**	**In house Outsourcing or hybrid**	**Large**
	Authentication and Authorization in Online Systems Cloud Server Hosting E-commerce or e-business Educational Identity and Access Management Intrusion Testing (Pentest) and BlackBox IT Infrastructure Metaverse Network Monitoring Other Cyber Services Security Incident Response Social Networks Streaming Others Plataforms						
Science Based	**Digital Services**	**Services**	**Price Sensitive Performance Sensitive**	R & D, Know-how, Dynamic learning economics, Copyright, Product Differentiation, Certifications and	**Cost-Cutting Networking**	**In house Outsourcing or hybrid**	**Medium Large**
	Application and Website Development Blockchain and Cryptography Solutions Cyber Security Consulting Cyber Threat Intelligence Services Data Management and Cloud Storage Disaster Recovery and Online Data Backup IaaS (Infrastructure as a service) Online Content Management Services PaaS (Plataform as a service) SaaS (Software as a Service) Threat Detection Services Other Digital Services						

Source: author, based on Pavitt (1984) and Miozzo & Soete (2001)

Within this taxonomy, the "Specialized Suppliers" category has been expanded to include companies that provide digital platform services. These companies cater to an audience sensitive to the price and performance of the services offered. Evaluating the costs and benefits still represents a significant decision-making challenge for many companies when using and mediating technology in their business, especially in mitigating cyber risks. However, these companies drive innovation through various sources of technology, which include research and development (R&D), specialized knowledge, technical skills, copyright, product differentiation, community collaborations, brand reputation, patents, process certifications, and regulatory compliance, which supports the validity of the criteria addressed by Pavitt in 1984.

The appropriation of value in this context is carried out through the design of systems and other sources of technology, adapting to the company's business model. However, it is important to note that, in some cases, companies may adopt approaches that focus more on non-technological aspects, such as reputational value and strategies, to appropriate value, as discussed in the results of this research (in return on investiment, Section 4.2.1.2). This diversity of strategies can reflect the multifaceted nature of innovation in the digital platform services landscape.

“Specificity plays a crucial role in the value appropriation strategies of innovative companies”; when a company develops specific innovations, it can create a substantial competitive advantage (Pavitt, 1984, p. 348). This advantage not only manifests itself in the ability to apply this innovation effectively, but also in the ability to maintain control over the underlying knowledge for a period of time, making it difficult for competitors to replicate. Specificity can be shaped strategically to align with the company's business model, ensuring that the economic benefits are directed towards it.

Therefore, digital services can be characterized by providing functionalities or benefits through digital interfaces, often accessible via the Internet. They can cover various services, from video streaming applications to cloud storage solutions and online payment systems. These services can cater to a global audience and offer convenience and accessibility to an unprecedented degree.

On the other hand, digital platforms serve as infrastructures that enable the creation and development of digital ecosystems. They act as technological foundations on which other digital services can be built. Platforms like social networks, online marketplaces, and content management systems enable collaboration and interaction between players, such as users, developers, and companies. This ability to create

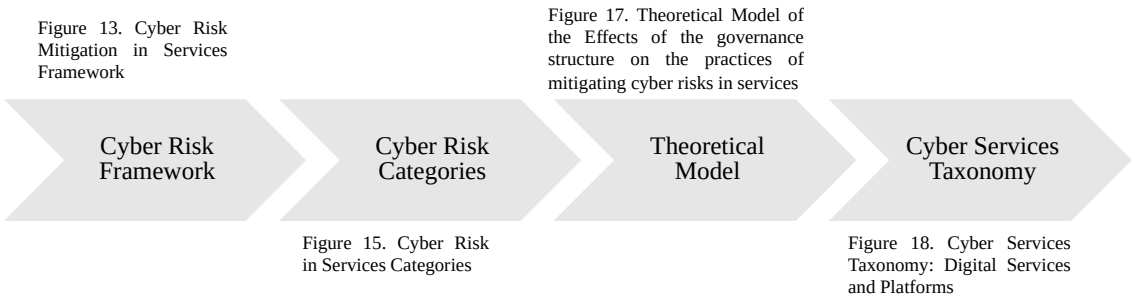
interconnected and dynamic ecosystems is one of the main characteristics that sets digital platforms apart.

In addition, in the context of digital platforms and digital services, certifications and regulatory compliance can act as means of appropriation with various implications (Pavitt, 1984). These certifications can be both optional and mandatory. In addition to demonstrating a commitment to quality and safety standards, compliance can impact a company's reputation. This, in turn, contributes to building the trust of customers, investors, and business partners.

In the research, new companies have begun to transact through digital services with the incorporation of new technologies in recent years (Table 5, Section 4.1.7), whether through verticalization, internal contracting, or external contracting. This transformation highlights the importance of strategic adaptation for digital services companies. According to Pavitt (1984, p. 362), these "dynamics learning economies" not only affect the entry of imitators in process technologies but have also caused a "paradigm shift" due to rapid and significant technological changes.

Therefore, the discussion sought to trace a trajectory that illustrates the evolution of the research results, demonstrating the dynamic nature of the investigation in question, s organized in Figure 19.

Figure 19 – Progression of Research Results



The findings of this research began with the development of a framework dedicated to companies seeking growth and needing to manage cyber risks in services. From this framework, thematic axes unfolded that made it possible to identify categories of cyber risks, contributing to answering the research question. In addition, this journey led to the constructing of a theoretical model that illustrates how the governance structure can affect cyber risks in services. In addition, a proposal is presented to contribute to and

expand knowledge, highlighting the cyber environment as an asset specificity in the Transaction Cost Theory. Finally, the research created a Cyber Services taxonomy covering both digital services and platforms. This sequential flow not only aligns with the progression of the research but also seeks to visually enhance the clarity of understanding by highlighting how each aspect emerged as the analysis went deeper.

6 CONCLUDING REMARKS

This research aimed to explore how the governance structure affects the mitigation of cyber risks in services. This question was conducted considering that the service industry represents a global contributor fostering innovation and technological advances (Gallouj, 2002, 2023). Companies in the sector are increasingly adopting technological solutions to improve efficiency, meet customer expectations, and gain a competitive advantage. However, these initiatives also come with risks related to cybersecurity, data privacy, and cyber resilience.

To this end, a methodological approach was adopted based on the premises of Creswell (2010), Bardin (2020), and Bryman (2021), whose mixed research was divided into two distinct phases. Firstly, an attempt was made to analyze the specificities of the services transacted in the cyber environment. Companies are involved in a variety of services, with focuses covering sectors such as digital, financial, knowledge, and commercial. In many cases, the business model can differ considerably from the standard to the point where they appear not to belong to a specific industry, especially when they adopt verticalization of their business chain. In addition, the aim was to identify the role of technology in governance structures in companies facing cyber risks. This indicates that these companies are responding to diverse market needs and demands.

In addition, managing risks from third parties, i.e., contractors or clients, has emerged as a concern in cyber risk mitigation. This is because companies are not only subject to the risks inherent in their internal operations. Analyses emphasize the need for companies to be prepared to face the risks that can be introduced by third parties, going beyond formalized bilateral and tripartite contractual relationships. Anticipating risks from third parties is the challenge, further underscoring the uncertainty environment.

Illuminating the thematic objective of this research, while digital operations and services can reduce information search costs, allowing service takers and providers to find each other more easily, they can also bring with them new challenges, such as increased costs associated with measures against contractual incompleteness, fraud, and risk. This is an example of how institutional changes can generate trade-offs, where reducing one type of cost can lead to another increase.

In this sense, it also sought to identify the measures companies use to deal with the challenges associated with cyber threats. In the first phase of this research, by developing a framework of cyber risk in services, the factors involved in risk mitigation

were explored as companies seek growth and innovation. This Services Cyber Risk Mitigation Framework was used as a priori categories, resulting in three thematic axes: 'Organization of Activities and Flow of Knowledge,' 'Mitigation of Cyber Risks in Services,' and 'Contractual Governance,' which became the thematic pillars of this research.

In addition, efforts were made to ensure that the thematic axes were flexible enough to accommodate topics and new information that could emerge during the content analysis of the interviews, following the premises of Bardin (2020). So much so that this process, which was carried out manually, resulted in 2.648 codes, the analysis of which was broken down into categories *a posteriori*. These categories were analyzed within the context of governance costs and the assets specificities (Figure 16). In addition, it was possible to trace the flow of the effects of the governance structure, through theoretical model, starting with the characterization of the specificities of the services and ending with the actions to mitigate cyber risks in services (Figure 16 in section above).

However, the samples showed relatively limited participation from companies in the personal services and infrastructure sectors. However, it is essential to note that this limitation did not restrict the research results. On the contrary, it identifies and supports future opportunities to explore these sectors in subsequent studies.

Understanding that research can go beyond the mere identification of specific questions and objectives, this research also provided an additional contribution to advancing knowledge in the areas of Transaction Cost Theory, Service Innovation, and Strategy. The research contributed by highlighting the 'Cyber Environment as Asset Specificity in Transaction Cost Theory' (Section 5.2). In addition, by focusing on service innovation, the thesis aims to explore the nature of services in activities carried out in the physical-digital environment. In this way, the research aims to improve the understanding of services as fundamental elements in the modern economy. As a practical contribution, understanding how the cyber environment permeates a company's operations is fundamental information for identifying cyber vulnerabilities.

The taxonomy entitled 'Digital Services and Platforms Taxonomy: Cyber Services' emerged from this research. Notably, digital services became the focal point, further emphasizing the growing importance of innovation and technology in the service sector. Digital services and platforms stand out for their ability to harness technology to meet the specific demands of the cyber environment. Their emphasis on connectivity and ability to adapt to rapid change support the characterization of the cyber domain.

The identification and categorization of digital services can support the organization in making more informed decisions about how to approach cyber security and risk management. Therefore, this taxonomy on cyber services can contribute to the Service Innovation literature and a better understanding and management of cyber risks in an organization.

In this way, the research aims to improve the understanding of services as fundamental elements in the modern economy, providing subsidies for the development of more effective and informed policies and strategies in sectors increasingly dependent on this growing economic dimension.

As part of these future investigations, examining how organizations manage cyber risks introduced by third parties in technology contracts is suggested. In addition, the research could focus on the co-creation of secure digital services, investigating how collaboration between customers, suppliers, and other stakeholders can enhance cyber security. The adoption of 'Zero Trust' Governance approaches in technology contracts is also a promising area for analysis, including evaluating case studies of companies that have succeeded with this approach.

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APENDIX A

QUESTIONNAIRE
(free translation from Portuguese)

Universidade Presbiteriana Mackenzie

Université de Lille

Governance Structures and the Effects on Cyber Risk Mitigation Capacity in Services

This research is part of the doctoral project "Governance Structures and the Effects on Cyber Risk Mitigation Capacity in Services", developed by doctoral researcher Marcia Rossi under the guidance of Professor Gilberto Perez (Universidade Presbiteriana Mackenzie/Brazil) and Faïz Gallouj (University of Lille/France).

Cyber Risk is related to a malicious electronic event that can cause business discontinuity and financial loss. It can be considered the one that covers all other risks associated with online activity, such as the storage of personal data and transactions that may result in image damage, financial loss, and adversities in business (NAIC, 2018).

With this research, it is intended to understand how choices of governance structures can influence the cyber risk mitigation capability of companies providing and contracting services.

This theme gains importance due to the increasing complexity and expansion of markets, in which managers have come to understand that sustainability is not an isolated action, but necessary to manage and mitigate any risks, be they of the business itself or of its value chain.

There are 24 questions dedicated to the organization of a**ctivities and knowledge flow, cyber risk mitigation, and contract governance**. It is estimated to take about 30 minutes to answer them.

Your contribution will be of great relevance to this study. Thank you for your interest in our research.

1. Name (optional):

2. Job title (position in the company): *

3. Area of expertise in the organization: *

4. Country (where you work): *

5. Companies' name (optional):

6. Company Segment/main activity: *

7. Number of employees:

8. Company's home country: *

9. **Total company's annual income (US$)** *

Marcar apenas uma oval.

- ◯ < 5 milhões
- ◯ < 10 million
- ◯ 10 million – 100 million
- ◯ 100 million – 500 million
- ◯ 500 million – 1 billion
- ◯ 1 billion – 10 billion
- ◯ > 10 billion

ORGANIZATION OF ACTIVITIES AND KNOWLEDGE FLOWS

Core activities generate operating revenue for the company and are considered its core business.
Peripheral activities are the production of goods or services that complement the company's core activities.

1. Please, answer only the service activities below about your enterprise:

- What are your company's primary business activities;
- Of these activities, which ones are developed or provided by your enterprise;
- Point out what service activities your company hires from third parties.

1.1 ***DIGITAL SERVICES*** activities

	Core Activity	Peripheral Activity	Activity developed by your company	Outsourced Activity
E-business	☐	☐	☐	☐
Digital streaming	☐	☐	☐	☐
Financial	☐	☐	☐	☐
Educational	☐	☐	☐	☐
Cloud computing	☐	☐	☐	☐
Social Networks	☐	☐	☐	☐
Metaverse	☐	☐	☐	☐

1.2 **KNOWLEDGE SERVICES** activities

	Core Activity	Peripheral Activity	Activity developed by your company	Outsourced Activity
Consulting	☐	☐	☐	☐
Auditing	☐	☐	☐	☐
Software development	☐	☐	☐	☐
Research and Development (R&D)	☐	☐	☐	☐

1.3 **COMMERCIAL SERVICES** activities

	Core Activity	Peripheral Activity	Activity developed by your company	Outsourced Activity
Wholesale	☐	☐	☐	☐
Retail	☐	☐	☐	☐
Delivery	☐	☐	☐	☐
Logistics	☐	☐	☐	☐

1.4 **PERSONAL SERVICES** activities

	Core Activity	Peripheral Activity	Activity developed by your company	Outsourced Activity
Health	☐	☐	☐	☐
Hotel	☐	☐	☐	☐
Restaurant	☐	☐	☐	☐
Entertainment	☐	☐	☐	☐

1.5 **GOVERNMENT SERVICES** activities

	Core Activity	Peripheral Activity	Activity developed by your company	Outsourced Activity
Educational	☐	☐	☐	☐
Judiciary	☐	☐	☐	☐
Police	☐	☐	☐	☐
Utilities (water, energy, transportation)	☐	☐	☐	☐
Public Systems	☐	☐	☐	☐

1.6 **FINANCIAL SERVICES** activities

	Core Activity	Peripheral Activity	Activity developed by your company	Outsourced Activity
Financing	☐	☐	☐	☐
Insurance	☐	☐	☐	☐
Payments (in general)	☐	☐	☐	☐

1.7 **IN COMPANY SERVICE** activities

	Core Activity	Peripheral Activity	Activity developed by your company	Outsourced Activity
Finance	☐	☐	☐	☐
Accounting	☐	☐	☐	☐
Legal	☐	☐	☐	☐
Design and Production (D&P)	☐	☐	☐	☐
Marketing and Sales	☐	☐	☐	☐

1.8 Other activity not specified above:

2.What are your company's expected goals when it chooses to outsource services activities? *

	High	Medium	Low
Get expertise in the market	◯	◯	◯
Reduce costs	◯	◯	◯
Reduce uncertainties	◯	◯	◯
Outsource the risk	◯	◯	◯
Strategic planning	◯	◯	◯
Expand partnerships	◯	◯	◯
Comply with regulators or legislation	◯	◯	◯

3. What are expected goals for your company to internalize any type of service's activity? *

	High	Medium	Low
Developing in-house expertise	◯	◯	◯
Reduce costs	◯	◯	◯
Reduce uncertainties	◯	◯	◯
Control and reduce risk	◯	◯	◯
Strategic planning	◯	◯	◯
Ensure Information security	◯	◯	◯
Comply with regulators or legislation	◯	◯	◯

4. In the last three years, your company has acquired machines, equipment, patents, or software based on:

- ☐ The same technology used before
- ☐ New technology not previously used

5.Which mechanisms is your company dependent on to provide services? *

	High	Medium	Low	Not dependent
Use of technologies	◯	◯	◯	◯
Use of patents, copyrights, or trademark	◯	◯	◯	◯
Specific knowledge from professional associations	◯	◯	◯	◯
License out its own intellectual property rights	◯	◯	◯	◯
Use of trade secrets	◯	◯	◯	◯
Continuous search for innovation	◯	◯	◯	◯

6. In which process does your company depend on technology services? (examples: data protection and cybersecurity, cloud computing, blockchain, among others) *

- ◯ To provide services
- ◯ To hire services
- ◯ To provide and hire services
- ◯ Investments in technology are under analysis by the company
- ◯ There is no dependent on technology

7. Regarding the company's internal environment, in the definition of goals, as well as in the bonus policy, employee profit sharing, and description of SLAs, is there a direction for risk management, including cyber risks? *

* SLA is the agreement established about the deadlines and parameters of the service, that is, it is the document where the minimum conditions such as quality and deadline are specified for a service to be considered satisfactory.

- ◯ Yes, there is this type of direction
- ◯ This proposal is under study
- ◯ Partially (each leader decides his team's goals)
- ◯ No, there is no direction
- ◯ N/A

8. During the three last years, how important were the following initiatives to the economic performance of your company? Please indicate whether these initiatives occur in the digital/online environment, physical environment, or both. (Part 1/4) *

	High	Medium	Low	Not important	Digital	Physical
Investments directed to better localization (to reduce logistics and space costs)	☐	☐	☐	☐	☐	☐
Introducing new products or services	☐	☐	☐	☐	☐	☐
Reconfiguration of physical spaces, as well as flexibility to adopt new ways of working	☐	☐	☐	☐	☐	☐
Reach out to new customer groups	☐	☐	☐	☐	☐	☐

Continuation (Part 2/4) *

	High	Medium	Low	Not important	Digital	Physical
Focus on one or a small number of key goods or services.	☐	☐	☐	☐	☐	☐
Offer to the customers an active role in the creation of the idea and design of products and services.	☐	☐	☐	☐	☐	☐
Investment in the teams' knowledge	☐	☐	☐	☐	☐	☐
Investments to expand or enhance reputation in marketing, brand building, and advertising	☐	☐	☐	☐	☐	☐

Continuation (Part 3/4) *

	High	Medium	Low	Not important	Digital	Physical
Development of products and services in partnership with companies (including competitors)	☐	☐	☐	☐	☐	☐
Development of specific products and services to meet your customer's needs	☐	☐	☐	☐	☐	☐
Development of specific products and services by the supplier to meet your company's needs.	☐	☐	☐	☐	☐	☐

Continuation (Part 4/4) *

	High	Medium	Low	Not important	Digital	Physical
Optimize/reduce the processing time of activities, whether internal or with third parties (suppliers, customers, or government).	☐	☐	☐	☐	☐	☐
Software development, database work, and data analysis.	☐	☐	☐	☐	☐	☐
Reduce lack of knowledge or incomplete information in decision making in my company and/or its value chain.	☐	☐	☐	☐	☐	☐

MITIGATION CYBER RISKS

9. What are the means used for risk management and mitigation (in general) in your company? *

	Very Weak	Weak	Moderate	Strong	Very Strong
Standardization and integration of operational processes	◯	◯	◯	◯	◯
Use of technology	◯	◯	◯	◯	◯
Training and cross-functional teams to identify risks	◯	◯	◯	◯	◯
Planned job rotation of staff across different functional areas	◯	◯	◯	◯	◯
Indicator control and monitoring	◯	◯	◯	◯	◯
Hiring companies specialized in risk management	◯	◯	◯	◯	◯

10. What type of risk management does your company manage that can positively contribute to mitigating cyber risk? (multiple answers) *

- ☐ Operational
- ☐ Strategic
- ☐ Compliance
- ☐ Financial
- ☐ Reputational
- ☐ Information security
- ☐ Labor risks
- ☐ Third-party risks
- ☐ Outro: _____

11. Has your company ever suffered losses (financial, reputational, and operational) from cyber threats? *

- ◯ Yes
- ◯ No

11.1 If yes, how did this happen?

_____

_____

12. Does your company have measures in place to identify and mitigate the risks in its activities?

* *Phishing:* e-mail or a text message that impersonates a person or organization they trust, i.e., a co-worker, a bank, or a government agency.

	Very Weak	Weak	Moderate	Strong	Very Strong
Phishing*	◯	◯	◯	◯	◯
Data Leakage and Theft	◯	◯	◯	◯	◯
Vulnerability of third-party employees	◯	◯	◯	◯	◯
Vulnerability in third-party software	◯	◯	◯	◯	◯
Internal frauds	◯	◯	◯	◯	◯
Violation of privacy	◯	◯	◯	◯	◯
Use and access to systems and information by your company's employees	◯	◯	◯	◯	◯
Hacker Attacks	◯	◯	◯	◯	◯

13. Does the company disclose investments related to cybersecurity? *

◯ Yes
◯ No

13.1 Why? *

14. Does your company promote and encourage discussion about cyber risks among your company's employees? *

- ◯ Yes
- ◯ No

14.1 Why? *

_____

_____

15. Do your company's internal policies, procedures, and rules keep up with the new business models that permeate cyberspace? (examples: e-business, social networks, marketplaces, among others) *

- ◯ Yes
- ◯ No

15.1 Why? *

_____

_____

16. Are the legal mechanisms (legislation and inspections) able to satisfactorily cover and protect the management of cyber risks in organizations?

- ◯ Yes. Facilitates and supports the initiation of innovation activities
- ◯ No. Hinders or increases the costs of innovation activities
- ◯ Outro: _____

17. Have cyber risks grown at the same level as opportunities for businesses? *

- ◯ Yes
- ◯ No

17.1 Why? *

_____

_____

CONTRACT GOVERNANCE

18. What are the most frequent types of contracts formalized by your company with suppliers?

- Short-term = up to 1 year
- Medium/Long term = from 3 years onward

	Spot Contract	Legal Person	Short-term	Medium/Long term	Indefinite	No formalization
Digital Services	◯	◯	◯	◯	◯	◯
Knowledge Services	◯	◯	◯	◯	◯	◯
Services by the Government	◯	◯	◯	◯	◯	◯
In Company Services	◯	◯	◯	◯	◯	◯
Commercial Services	◯	◯	◯	◯	◯	◯
Personal Services	◯	◯	◯	◯	◯	◯
Financial Services	◯	◯	◯	◯	◯	◯

19. In the formalization of contracts with suppliers, are there contractual clauses related to risk management, including cyber risk, inherent to the activities contracted by your company? *

- ◯ Yes, there are clauses pertinent to these risks
- ◯ This proposal is under study by the company
- ◯ No, there are no clauses directed to this
- ◯ N/A

20. What are the most frequent types of contracts formalized by your company with costumers/clients?

	Spot Contract	Legal Person	Short-Term	Medium/Long-Term	Indefinite	No formalization
Digital Services	◯	◯	◯	◯	◯	◯
Knowledge Services	◯	◯	◯	◯	◯	◯
Services by the Government	◯	◯	◯	◯	◯	◯
In Company Services	◯	◯	◯	◯	◯	◯
Commercial Services	◯	◯	◯	◯	◯	◯
Personal Services	◯	◯	◯	◯	◯	◯
Financial Services	◯	◯	◯	◯	◯	◯

21. In the formalization of contracts with costumers/clients, are there contractual clauses related to risk management, including cyber risk, inherent to the activities contracted by your company? *

◯ Yes, there are clauses pertinent to these risks
◯ This proposal is under study by the company
◯ No, there are no clauses directed to this
◯ N/A

22. In which contractual phase with consumers/clients, do you perceive to be the most vulnerable that requires the most attention from the company? *

Vulnerability is related to the weakness that leads the company to its own risk or the risk to third parties.

	High	Medium	Low
Pre-contracting (identification of objectives and clarifications for entering a contract)	◯	◯	◯
Contracting (formalization of the contract, as well as the negotiation and agreement between the parties)	◯	◯	◯
Pre-implementation (technical preparation for the treaty to be received, interpreted, and applied)	◯	◯	◯
Implementation (period of the contract – there will be a need to follow up on its fulfillment)	◯	◯	◯
Contractual closure (the planned term comes to an end, and thereafter the obligations are ceased)	◯	◯	◯

23. Is there shared responsibility for cyber risks when involving partner companies? (examples: sharing space and projects, dedicated suppliers, economy of scale and open innovation) *

◯ Yes
◯ No

23.1 Why? *

_____

_____

24. Can the trust developed by social and cultural proximity be considered an essential element in the selection and hiring of third parties in your company?

◯ Yes, it is considered an essential element in my company
◯ Partially essential
◯ Not considered an essential element
◯ Outro: _____

Please let us know if you are interested in:

- [ ] Receive the results of this Survey
- [ ] Participate in an interview with the researcher, as the second stage of this Research

If you have answered one of the options above, we kindly ask you to inform the e-mail and/or phone contact.

_____

_____

APENDIX B

Interview Guide
SEMI-STRUCTURED RESEARCH
(free translation from Portuguese)

Topic 1. Specificities of the services provided and contracted

1. What are the main characteristics of the services provided by your business model?

2. What are the main services contracted by your company.

3. What are the main motivations for your clients to hire your company's services?

4. What are the activities that your company chose to develop internally and not hire third parties?

Topic 2. Use of mechanisms (information, processes, knowledge and consequences) to deal with problems associated with cyber risks (Whether to hire a third party or to be hired)

5. What measures or mechanisms does your company use to identify and mitigate risks related to:

- Information asymmetry

- Fraud

- Overconfidence of employees

6. How does your company assess or discuss risks related to knowledge gaps or opportunistic behavior with your colleagues, teams or even related to hackers and/or independent people in your organization?

7. How does your company avoid delays or uncertainty in discovering new threats to your company's business??

8. Does technology play a role in mitigating cyber risks in your company? If yes, what type of technology is used?

9. Considering the business model in which you operate, how do you perceive the challenges and future trends in cyber risk mitigation?

10. Could you comment on whether damage to reputation, fines and penalties have been enough for companies to increase investments (precaution) in risk mitigation?

Topic 3. Contractual forms practiced by the company (or required by the other party) to mitigate cyber risks.

11. When formalizing contracts with third parties, in general, how does your company handle risk management, including cyber risk, inherent to contracted activities?

12. To what extent is the transfer of risk to third parties through contractual means permitted or common?

14. How do companies fail to mitigate cyber risks? What are the most appropriate measures/actions to create a scenario of cyber and contractual resilience in organizations?

15. How can employees be encouraged to manage business risks, through their target contracts?

.....

Additional questions

- Would you like to make a comment that you consider relevant?

- Survey feedback (respondent perception)